Deriving the Particle Zoo from Observer Consistency

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Abstract

This paper carries the Standard Model structure derived by Observer-Patch Holography (OPH) into particle calculations. Its finite source model selects the odd-Weyl Spin typing, the Standard Model charge lattice, and the maximal faithful matter image $(\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1))/\mathbb{Z}_6$ on the declared carrier. Laboratory identification of the current and flux sectors, physical seam and family attachment, four-dimensional topology, scalar dynamics, and continuum quantum-field realization are work in progress.

The selected branch gives exact charge quantization, a dimensionless electroweak hierarchy, a zero Higgs naturality defect in its declared chart, and conditional Higgs and top coordinates. A strict finite-order W/Z pole algebra and two external Standard Model calculation engines test the matching protocol. They do not produce an OPH-native weak-boson pole. A positive Hermitian C_3 face circulant obeys the exact identity $Q = 1/3 + (2/3)(|b|/a)^2$, hence $Q = 2/3$ exactly when $|b|/a = 1/\sqrt{2}$; a finite tracial construction supplies that balance under its stated premises. Conditional on the declared Clebsch pattern, one-loop transport gives $m_s/m_d = (m_\mu/m_e)_{\mu_U}/9 = 22.9743$. Against the two 2024 Flavor Lattice Averaging Group light-quark ratios it is 12.8% to 15.2% high, rejecting that route. The reciprocal quark and weighted-cycle neutrino candidates also fail their tests. The six absolute quark masses are non-identifiable from the stated source data, and absolute lepton and hadron masses are empirical inputs rather than source-only results. No nonzero pole mass is claimed as a source-only prediction.

What This Paper Contributes

The Standard Model gauge paper supplies an exact finite source-model current algebra, the odd-Weyl Spin typing, the hypercharge lattice, a three-color carrier, and the maximal faithful matter image $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$ from the incidence, response, and matter premises alone. The source packet thereby selects the $\mathbb{Z}_6$ quotient on its declared carrier. It also supplies the distinct conditional compact-gauge/Tannaka reconstruction route, a conditional generation window $3 \leq N_g \leq 5$, and the connection and metric carrier roles. Under the single-complete-object, faithful-action, and operational-cost premises, the exact screen-band theorem selects rank three inside that window. The declared finite simulator recovers the same projector at its lowest positive generator frequency. Tensoring that response band with the declared generation table gives a conditional complex rank-45 candidate. Chirality and the diagonal $\mathbb{Z}_6$ action are properties of the table. A separate finite source-domain receipt checks the declared operator $D_\sigma \otimes I_{45}$ and conditional inheritance of a positive dimensionless gap. The source does not select this matter action or bridge the twelve-port Spin packet to the local operator domain. Identification with physical matter poles and continuum fields is open. Classical massless modes require explicit action, background, and phase premises; quantum poles require a stronger particle certificate. This paper studies the downstream particle readout after the local closure coordinate is supplied.

The finite source packet selects the maximal faithful $\mathbb{Z}_6$ quotient on its declared carrier. This does not identify the finite flux sectors with laboratory currents or choose the physical seam action. Two structural routes feed that downstream calculation. Incidence expresses J as a polynomial in adjacency, and target-blind port readback derives the signed response and current algebra. Under the declared fermionic Spin category, anomaly balance and tensor descent give the conditional matter chain. Transportable sectors plus Tannaka reconstruction give a separate conditional compact group. The target-free matter reduct does not select the generation count inside the window. Under separate complete-band and cost-order premises, the screen-band theorem selects rank three. Tensoring that response band with the declared generation table gives a conditional rank-45 candidate. A separate local receipt checks a declared tensor-identity operator and conditional gap inheritance. It neither selects the matter action nor transports the twelve-port Spin packet to the local domain. Matter-pole identification, continuum Spin/locality, the physical seam choice, the third persistence leg, and laboratory identification are open. Completeness for extra light sectors is a separate open attachment. The intertwiner between the two current objects is also open.

A single screen-cell coordinate $P_\star$ feeds the electromagnetic, electroweak, and mass branches. The exterior matter witness fixes four weak doublets. Conditional on the shared physical load carrier, the selected branch emits $v/E_\star$ with Higgs naturality defect $\epsilon_H = 0$; a weak scale in GeV also requires an independently closed $E_\star$. Each declared pixel map has a machine-certified interval proof of a unique fixed point. On the target-anchored empirical-closure branch, the measured charged-lepton triple lies inside every outward-rounded arithmetic enclosure. These are diagnostic intervals rather than prospective prediction intervals. The branch inverts exactly at the witness: one anchor-gap value closes all three masses on the measured triple at once, inside the certified band. The scheme bridge is therefore a declared confirm-or-refute target. Absolute charged-lepton and first-principles hadron masses remain outside the theorem, and empirical endpoint data are identified wherever a transport bridge uses them.

1 Introduction

Observer-Patch Holography (OPH) asks a concrete follow-up question after the gauge structure is fixed. Can one local screen constant organize the particle spectrum, or are the masses and mixings a collection of unrelated inputs?

Conditional on its explicit finite response, rank-15 matter, and descent packets, The Standard Model gauge paper supplies the Standard Model gauge structure

$$\frac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_c = 3, \qquad 3 \le N_g \le 5.$$

The gauge group, hypercharge lattice, and color count come from the stated premises alone, and the window by itself leaves the generation count open. The graph carrying the alternating group on five letters, A_5, together with anomaly cancellation and the target-free matter reduct does not select a count. A strengthened screen interface does: among single complete faithful bands in the window, the exact Laplacian costs $5-\sqrt{5} < 6 < 5+\sqrt{5}$ select the rank-three band uniquely. The declared unitary simulator recovers the same residue at its lowest positive generator frequency. Tensoring this mode with the declared generation table gives a conditional rank-45 candidate.

Chirality and the diagonal $\mathbb{Z}_6$ action come from that table. A separate local-domain receipt checks a declared tensor-identity operator and conditional gap inheritance. The source does not select the matter action or bridge the twelve-port Spin packet to that domain. Physical family interpretation requires matter-pole, continuum, seam-selection, persistence, and laboratory receipts. The paper also supplies the exact hypercharge lattice and a conditional exterior representation witness. If a trace-balanced block carrier $V = C \oplus W$, with dimensions $3 + 2$ and hypercharges $(-1/3, 1/2)$, is physically selected, then $\Lambda^2 V \oplus \Lambda^4 V$ branches exactly as $Q \oplus u^c \oplus e^c \oplus d^c \oplus L$. It has the three one-Higgs invariant lines, cancels all five gauge and mixed anomalies, and contains four weak doublets per generation. The determinant balance, Spin lift, non-vacuum exterior package, and center/deck descent are receipt-checked under the response and matter contracts. The scalar scan fixes compatible charges and Yukawa channels, not $H = W$ or scalar multiplicity. Laboratory identification of the current and flux sectors, exclusion of extra light sectors, attachment of the A_5 face representation to physical families, scalar attachment/dynamics, and continuum QFT are open. The presentation-invariant normal-form framework of Ref. [1] isolates the same-source confluence, cross-source boundary-identification, liveness, and refinement criteria used upstream. It quotients hidden coordinates, labels, worker layout, and ancillas when they leave observer-visible records unchanged. Visible port incidence, topology, response maps, and repair laws remain physical branch data. The framework supplies no particle-sector selector or numerical particle output; every such claim below is conditional on the specialist gauge branch and the particle-specific receipts.

The particle branch is a one-fixed-point forward reconstruction problem with an explicit closure matrix. It propagates one dimensionless pixel ratio P through the spectrum. The construction does not emit photon, gluon, or graviton particle masses from symmetry labels. It records their conditional classical carrier modes and quantum-particle gates that are work in progress, while the Higgs candidate is on its declared quantitative surface. The hierarchy/naturality bridge is closed on its selected branch. The restricted quark source-spread non-identifiability theorem, the common-scale rejection of its reciprocal-ray candidate, and the weighted-cycle neutrino branch are visible as obstruction or comparison surfaces. Target-anchored quark mass textures, the mixed-convention formula diagnostics, and compare-only absolute neutrino attachments are withheld from public prediction tables. The P-closure root, the electroweak W/Z rows, charged leptons, and hadrons obey the sector-specific boundaries stated below.

1.1 Why this particle content is selected

On the declared branch, particles are not inserted by hand as a list of elementary ingredients. They are the stable observer-visible excitations and carrier modes that are left once three layers of structure have been fixed:

1. overlap consistency on the observer-patch network;
2. the explicit finite response and rank-15 matter contracts, anomaly balance, and tensor descent that imply the finite Standard Model gauge chain;
3. the conditional rank-three screen selection, its abstract tensor product with the generation table, and the separate declared local operator context.

The first layer says what kinds of local data can be compared consistently across neighboring patches. The second fixes the finite Lie algebra, charge lattice, color carrier, and global quotient through receipt-checked conditional data. The third separates the finite carrier theorem from its declared physical completion. The result is an exact contract-conditional finite carrier and one-generation matter packet. The selected response band and generation table give a conditional rank-45 candidate. The separate local-domain branch verifies the declared tensor-identity operator without selecting it from the source. A physical particle interpretation requires source binding, laboratory current identification, matter-pole and continuum attachment, interaction, scalar attachment/dynamics, and QFT receipts.

The transportable-sector/Tannaka route is a separate conditional compact-group reconstruction. Physical source binding of the finite current and equality with the Tannaka current are open. The conditional rank-45 candidate does not become a physical family without source-selected local action, cross-domain Spin transport, matter-pole, continuum, seam-selection, persistence, and symmetry-descent receipts.

Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency and Deriving Standard Model Gauge Structure from Observer Overlap Consistency [6, 7] is therefore central to this paper. That paper gives the exact conditional finite structural theorem that identifies the packet

$$\frac{\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1)}{\mathbb{Z}_6}, \qquad N_c=3, \qquad 3\le N_g\le 5,$$

together with one scalar-doublet channel and a rank-15 internal matter witness. The gauge group, charge lattice, and color count use the stated premises alone. The target-free matter reduct leaves the three-generation value unselected. On the strengthened screen interface, exact cost minimization selects the canonical rank-three band and the declared simulator realizes its response residue. Its tensor product with the declared generation table is a conditional rank-45 candidate. A separate finite local-domain receipt checks a declared tensor-identity operator. It does not select that action or identify the local domain with the Spin packet. Physical matter-pole and continuum attachments require separate derivations.

On that selected one-Higgs branch the scalar carrier has a canonical local screen-chart model. Let $C_{\mathrm{EW}}\cong\mathbb{CP}^1$ be the support-visible electroweak chart and fix the positive Hopf line-bundle convention by the neutral component condition $Q(\phi^0)=0$. Borel–Weil gives

$$H_{\mathrm{OPH}}=H^0(C_{\mathrm{EW}},\mathcal{O}(1))\cong\mathbb{C}^2,$$

the first nontrivial holomorphic section space. With OPH's hypercharge and $\mathbb{Z}_6$ normalization this is exactly the $(1,2)_{1/2}$ Higgs carrier. Projectivization classifies a nonzero section direction as a point of $\mathbb{P}(H_{\mathrm{OPH}})\cong\mathbb{CP}^1$, but it forgets the scalar hypercharge phase. For the lower-component vacuum vector

$$\phi_0=\frac{v}{\sqrt{2}}\begin{pmatrix}0\\1\end{pmatrix}, \qquad v\neq 0,$$

one has

$$e^{i\alpha T_3}e^{i\beta Y}\phi_0=e^{i(\beta-\alpha)/2}\phi_0.$$

Consequently $[\phi_0]$ has the projective two-torus stabilizer $\mathrm{U}(1)_{T_3}\times\mathrm{U}(1)_Y$, modulo the inherited finite center, whereas the vector ϕ_0 is fixed only when $\beta=\alpha$, locally. Its connected stabilizer is therefore the electromagnetic diagonal $\mathrm{U}(1)_Q$, generated by $Q=T_3+Y$. Thus the projective geometry explains the carrier-ray classification, while the chosen nonzero vector supplies the symmetry-breaking statement. This explains the representation, charge, and symmetry-breaking geometry of the one-Higgs slot. It does not derive m_H, the quartic, v, or Coleman–Weinberg dynamics; the weak scale, Higgs/top quantitative surface, and $\epsilon_H=0$ hierarchy/naturality closure are the OPH quantitative branch.

This also delimits the phrase "and no others." The finite OPH claim concerns the low-energy packet implied by the stated response, matter, and descent contracts. Additional connected gauge generators, additional light Higgs multiplets, extra light chiral families, or low-energy supersymmetric partners are absent from the declared completion because that completion declares them absent. This is a declaration, not source-derived exhaustion of physical sectors. Propagating gauge-particle existence requires the complete primitive data, action, phase, and quantum-pole premises below.

The individual families then emerge for different reasons. The realized electromagnetic, color, and dynamical-metric branches identify connection and metric carrier roles. Explicit Maxwell, perturbative pure-Yang–Mills, and pure-Einstein quadratic actions then yield classical transverse or TT massless modes on their stated backgrounds and phases. Quantum photon, gluon, or graviton states require the separate physical-Hilbert-space, pole-residue, and asymptotic/phase receipt. The weak bosons arise when the electroweak gauge sector is propagated through its quantitative closure branch. Quarks and leptons arise from the conditional chiral matter packet together with the rank-three screen selection. Their physical three-family interpretation requires the conditional rank-45 tensor candidate to be identified with physical matter-pole residues and continuum fields, with the local matter action and cross-domain Spin transport selected. Their family splittings are then read from the deeper overlap-transport and excitation machinery, not from a second arbitrary postulate that says "copy the family three times and assign masses by hand." Once color is realized and confined, stable hadrons are composite readout channels of the quark/gluon sector.

The role of the pixel constant P is also important to state clearly. Here, P does not decide whether an electromagnetic particle pole exists, whether there are three colors, or whether the candidate family band is physically attached. The electromagnetic representation role, the color carrier, and the declared generation completion belong to the finite packet before the quantitative closure step. What P does is set the shared quantitative scale on which receipt-certified content is read out numerically. So the logic of the paper is:

$$\begin{aligned}\text{observer consistency} &\rightarrow \text{realized gauge/matter branch}\\ &\rightarrow \text{representation roles} \rightarrow \text{conditional action-level carrier modes}\\ &\rightarrow \text{common scale } P \rightarrow \text{receipt-gated quantitative spectrum.}\end{aligned}$$

That is how the derivation explains why a universe with the realized branch and the shared pixel scale exhibits this particle zoo.

1.2 Logical basis and reading rule

The particle derivation uses the same OPH basis as the SM/GR derivation in Ref. [6, 7]. Its three axioms are the oriented twelve-port observer screen, observer agreement, and conditional maximum randomness. On that basis, the SM/GR derivation in Ref. [6, 7] supplies the structural chain used here: the exact finite gauge group, hypercharge lattice, and color triplet $N_c = 3$ under explicit response and matter contracts; a separate receipt-conditional Tannaka classification; and the conditional generation window with its declared three-generation completion. Extra-light-sector completeness, laboratory current identification, and physical family attachment are open. Refs. [3, 4] supply the patch-net, repair, measurement, and regulated screen language for flavor transport and observer-facing readout.

The quantitative particle side adds one local closure variable. The common pixel ratio P, reported by the synthesis paper's incomplete outer/inner declared map [2], feeds the forward electroweak map and its descendants. The public empirical comparison branch displays

$$\alpha^{-1}(0) = 137.035999177(21), \qquad \alpha(0) \simeq 0.00729735256433, \qquad P_C \simeq 1.6309682094.$$

The comparison pixel P_C is defined from the measured endpoint; it is not a source-root coordinate. The computation has a fixed source order: golden-ratio entropy balance gives φ, boundary Gaussian normalization supplies the $\sqrt{\pi}$ width, a trial P feeds the source map through unification, running, and electroweak anchoring, and Ward-projected electromagnetic transport gives the Thomson endpoint used by the outer/inner pixel fixed point. The declared numerical map fixes a unique root on its certified interval. The map is incomplete because no derivation identifies its numerical endpoint with the physical electromagnetic readout. The first-principles diagnostic trunk records the certified coordinates

$$P_{\text{fwd}} = 1.630972095858897\ldots,$$
$$\alpha_{\text{root}}^{-1} = 136.994835177413\ldots.$$

The root is the interval-certified unique fixed point of the declared numerical map. The approximately 0.041 inverse-alpha difference to the public Thomson endpoint is a residual of that incomplete map. It neither derives a QCD/hadronic contribution nor identifies a physical endpoint relation. The QCD-free hierarchy witness is the cleaner first-principles stress test. Combining the source/root value with the finite-screen unified gauge-width contribution evaluated at the CODATA-derived comparison pixel, $\alpha_U(P_C)$, gives the mixed-provenance no-hadron diagnostic $A_{\alpha_U}^{\text{fp}} = 137.0359595136\ldots$, below the Thomson endpoint. It is mixed-provenance comparison bookkeeping rather than a source-only fine-structure prediction. The certified self-consistent gauge-width fixed point is $\alpha^{-1} = 137.035660136946577\ldots$; the mixed diagnostic combines the inner value at P_{fwd} with α_U at the CODATA-derived comparison pixel and is excluded from the physical result. The detailed endpoint table appears in Section 4. The separate proposed cosmic record-closure target $\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}$, with $N_{\text{CRC}} = \log D_\star$, defined through the correctable code of the reachable public record atoms under globally coupled checkpoint kernels, belongs to the cosmological-capacity branch and supplies no particle theorem. The Newton coupling uses the separate selected no-G scale certificate $\gamma_\star = \ell_\star \nu_{\text{Cs}}/c$, equivalently $B_\star = 3\pi/\ell_\star^2$, so the particle branch does not derive G by back-solving the pixel ratio. The map's public checkpoint packet, carrier representation, whole-fiber scalarization, extension/refinement, finite-size slack, and fixed-point receipts are open. Its identification with the electroweak bridge is conditional on a positive, unital, refinement-natural identification of the screen load with the electroweak load; the bridge value $N_{\text{EW}} \simeq 3.532 \times 10^{122}$ is about 6.6 percent above the Planck-Λ central capacity $N_\Lambda \simeq 3.313 \times 10^{122}$. If the missing physical identifications are supplied, the pair $(P_\star, N_{\text{CRC}})$ determines dimensionless curvature products such as $\Lambda_\star a_{\text{cell}} = 3\pi P_\star / N_{\text{CRC}}$, not the SI scale product $\Lambda_\star N_{\text{CRC}}$ by itself. The rounded $N_\Lambda \simeq 3.313 \times 10^{122}$ display is a cosmological central-value label, not the high-precision input for G. Structural carriers, quantitative outputs, conditional calculations, and continuation calculations therefore inherit the declared theorem premises, with particle-facing quantitative burden carried by the local declared-map root $P_\star$. The public fine-structure display row is a comparison endpoint conditioned on the measured Thomson value. The first-principles diagnostic row and empirical hadron closure rows are separate support records. Section 4 records that boundary in full.

2 P-Closure and the Reverse-Engineering Strategy

The reverse-engineering claim is easiest to state plainly. In ordinary phenomenological use, the Standard Model does not derive the observed particle masses and mixings from one common microscopic constant. It is usually presented with 19 free parameters in the minimal massless-neutrino theory, and with at least 26 once neutrino masses and mixing are included, depending on neutrino-sector conventions. OPH organizes that situation around one universal pixel ratio P from the outer/inner closure relation described in the synthesis paper. Whether any parameter compression is realized follows from counting the declared settings against the independent landed basins. The same P must drive all downstream bosonic, quark, lepton, neutrino, and hadronic branches.

One clarification matters. The broader self-closure formulation uses the universe-level equation

$$N = \log M_0(\mathfrak{U}_N),$$

where M_0 is the multiplicative size of the largest correctable public record code. Its selector-free finite form is stable global screen capacity

$$\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}, \qquad N_{\rm CRC} = \log D_\star,$$

for a universe supplied with logarithmic capacity N. The synthesis paper specifies the finite read-back directly by $M_0(q) = \alpha(G_q)$, the independence number of the compound confusability graph of the reachable public record atoms under the globally coupled checkpoint family. Under whole-fiber scalarization and a faithful capacity carrier it is deflationary; under confusability-reflecting capacity extension it is monotone, so top-down iteration reaches the greatest fixed point on a declared finite chain. Execution and the finite-size selector are work in progress. The electroweak identification additionally requires the common screen/electroweak load-carrier hypothesis. The conditional bridge value $N_{\rm EW} \simeq 3.532 \times 10^{122}$ is about 6.6 percent above the Planck-Λ central capacity $N_\Lambda \simeq 3.313 \times 10^{122}$, a propagated 2.4 to 2.5 one-dimensional sigma; read through $\Lambda_\star \ell_\star^2 = 3\pi/N$, the same bridge is the zero-dial relation $\Lambda \ell_P^2 = 3e^{-6\pi/(P\alpha_U(P))}$ only after the independent capacity, horizon, and common-load receipts. No correction term is derived, and the same exponent runs the weak hierarchy of this paper. The particle-spectrum derivation studied here uses the local declared-map root $P_\star$ in place of a long particle-by-particle parameter list.

Observation is allowed to supply a branch hint. That is a consequence of the OPH picture, where the universe is a closed fixed structure and internal observers can read approximate coordinates from experiments. The proof obligation is stricter: the exact $P_\star$ used by the particle branch is computed by the self-referential numerical pixel equation, and the root is unique on the declared interval. A further derivation is required before that root is a physical endpoint. The pixel equation and its uniqueness statement are carried by the shipped interval contraction certificate: the closure map is a self-map with a derivative bound certified by interval arithmetic on the declared interval. The domain-global statement is discharged as well: the companion domain-global certificate bounds $\sup |g'| < 1$ on every piece of a 256-piece subdivision of the declared numerical domain ($\alpha^{-1} \in [100, 200]$, both readout maps, empty exceptional set), so each declared readout map has exactly one fixed point on that numerical domain. This certificate does not provide the missing physical endpoint relation.

2.1 Why a single P matters

The structural theorems determine an exact contract-conditional finite representation packet, but not *what physical particle world* realizes it or what absolute scale in GeV it occupies. The overlap, modular, gauge, anomaly, and admissibility arguments give the declared Standard Model packet, the exact hypercharge pattern, and a three-color carrier. The generation window is conditional; separate band premises select rank three. Tensoring the selected response band with the declared generation table gives a conditional rank-45 candidate. A distinct local-domain receipt checks a declared tensor-identity operator and conditional gap inheritance without source-selecting the action or transporting the Spin packet. Physical matter-pole and continuum interpretation is open. The carrier-mode theorem is an additional action-level result and does not emit zero-GeV quantum particles. These structural results do not by themselves tell us the numerical values of the W boson mass, the Higgs boson mass, or the up-quark mass.

The role of the pixel closure is to supply one common quantitative scale variable for the entire downstream spectrum. In the synthesis paper, the fine-structure branch asks for the nonzero detuning of a holographic screen cell such that the cell's outer geometric displacement from perfect self-similar equilibrium equals the electromagnetic observation scale emitted by the encoded branch under the common physical carrier identification. That identification remains a separate receipt. The outer side of the closure is

$$P = \varphi + \alpha_{\text{in}}(P)\sqrt{\pi}.$$

The first-principles computation is a five-step source chain. First, the golden-ratio entropy balance of the local screen cell supplies $\varphi = (1+\sqrt{5})/2$. Second, boundary maximum-entropy normalization fixes the width of the leading electromagnetic detuning to $\sqrt{\pi}$. Third, a trial P is sent through the source map

$$M_U(P) = E_P e^{-2\pi} P^{1/6}, \qquad E_{\text{cell}}(P) = \frac{E_P}{\sqrt{P}},$$

followed by the heat-kernel closure

$$\bar{\ell}_{\text{SU}(2)}(t_2(P)) + \bar{\ell}_{\text{SU}(3)}(t_3(P)) = \frac{P}{4},$$

which selects $\alpha_U(P)$ and the running family $\alpha_i(m_Z; P)$. Fourth, electroweak mixing gives the source anchor

$$a_0(P) = \alpha_2^{-1}(m_Z; P) + \frac{5}{3}\alpha_1^{-1}(m_Z; P).$$

Fifth, Ward-projected $\text{U}(1)_Q$ transport gives the Thomson endpoint

$$A_T(P) = T_Q(a_0(P), F_{\text{src}}(P)) = \alpha_{\text{em}}^{-1}(0; P),$$

and the cell closes when

$$P = \varphi + \frac{\sqrt{\pi}}{A_T(P)}.$$

The numerical solve is therefore a root problem for

$$H(P) := P - \varphi - \frac{\sqrt{\pi}}{A_T(P)}.$$

The branch interval is localized from the observer-facing data, the solver evaluates $A_T(P)$ from the declared source map at each candidate point, and the accepted value is the unique zero of H on that interval. The measured endpoint can locate the interval. It cannot replace the root condition. This proves uniqueness only for the declared numerical map. The derivation connecting its root to the observer-supporting physical electromagnetic endpoint is missing. The CODATA-conditioned comparison readout $\alpha^{-1}(0) = 137.035999177(21)$ gives

$$P \simeq 1.6309682094.$$

The same equation is the local ruler for the downstream particle rows. The five-layer OPH diagnostic trunk emits the certified source-side diagnostic point

$$\begin{aligned} P_{\text{fwd}} &= 1.630972095858897\ldots, \\ \alpha_{\text{root}}^{-1} &= 136.994835177413\ldots. \end{aligned}$$

The root is an incomplete declared-map output, not a physical fine-structure prediction. The detailed table records how the diagnostic trunk, endpoint residual, and source-spectral payload fit together. A separate hardware note reports an optical-cavity check of the same declared geometry.

It is a non-discriminating engineering check and supplies no physical endpoint. Once the diagnostic map coordinate is set, the particle question becomes whether all downstream readouts can be written as

$$X_j = G_j(P)$$

with no new sector-specific constants inserted by hand. This paper studies exactly that downstream map.

2.2 How the electroweak branch works

In the implementation used here, the quantitative electroweak branch is the first major readout from P. Its algebraic output family is

$$\{M_W^{\text{chart}},\ M_Z^{\text{chart}},\ \alpha_{\text{em}}^{-1}(q^2),\ \sin^2\theta_W(q^2),\ v\},$$

evaluated from one declared input P on the printed running/matching/threshold/scheme package. The superscript chart marks a mass-chart coordinate, not a certified complex pole. The construction is straightforward in spirit: start from P, build the electroweak running family, select the physical carrier point on that family, and read off the W boson and Z boson pair from that selected point. The Higgs/top critical stage inherits that same electroweak core and does not introduce a new free-input sector. On the declared Ward-projected transport branch, the low-energy electromagnetic row is represented by the Thomson endpoint

$$\alpha_{\text{Th}}^{-1}(P) = \lim_{q^2\to 0} \alpha_{\text{em}}^{-1}(q^2; P)$$

of that same electromagnetic transport family. The forward logical order on this branch is:

$$\begin{aligned} P &\longmapsto (M_U(P), E_{\text{cell}}(P)) \longmapsto \alpha_U(P) \longmapsto (t_U(P), t_{\text{tr}}(P)) \\ &\longmapsto (t_2(P), t_3(P), v_{\text{chart}}(P)) \longmapsto \alpha_i(\mu_*; P). \end{aligned}$$

Here $v_{\text{chart}}(P)$ is the mass-chart coordinate. It is not a renormalized vacuum expectation value in any fixed scheme, and no receipt identifies it with one. The forward transmutation certificate records that the same runtime basis reconstructs the unified diffusion parameter $t_U(P) = 4\pi^2\alpha_U(P)$ and transmutation exponent $t_{\text{tr}}(P) = 2\pi/((N_c+1)\alpha_U(P))$ as the pixel-closure solve itself. The runtime subgraph reads measured electroweak data only for validation. No frozen provenance receipt excludes target dependence or selects one repair law.

The electroweak branch is the place where the named pixel coordinate organizes the weak-sector rows. The reference-fitted inverse adapter displays

$$M_W = 80.3625\ \text{GeV}, \qquad M_Z = 91.1879\ \text{GeV}.$$

The selected-carrier chart, the runtime-target-separated formula candidate, and the reference-fitted coherent repair diagnostic are recorded separately in the support table below.

2.3 What the electroweak branch fixes

Under the declared electroweak running, matching, threshold, and scheme premises, the pixel variable P supplies the source basis $(\alpha_U, \alpha_{2,m_Z}, \alpha_{Y,m_Z}, \eta_{\text{source}}, v)$ used by the candidate chart. The displayed value law is an exact implication of the five-part quotient-path certificate stated below, but the finite carrier does not emit that certificate. The candidate law evaluates the mass-side pair (W, Z) together with the running-family anchor (a_0, s_0, v). The electromagnetic row is physically read only after Ward projection to the unbroken $\mathrm{U}(1)_Q$ channel; its low-energy value is the $q^2 \to 0$ endpoint of the same transport family. The derivation also includes two explicit comparison calculations with distinct provenance: the exact selected-carrier chart and the reference-fitted coherent repair diagnostic. The same support logic applies to the flavor branches, whose premises differ. The electroweak quantitative branch states its source payload, same-scheme remainder, and interval certificate explicitly.

For empirical comparison, a documented piecewise $e^+e^- \to$ hadrons compilation, built from resonance parameters and perturbative QCD, evaluates the subtracted dispersion integral to $\Delta\alpha^{(5)}_{\mathrm{had}}(m_Z) = 0.027609 \pm 0.000112$. Inserting it into the endpoint map with the frozen source anchor and lepton transport packet gives $\alpha^{-1} = 136.3827548175$ on $[136.3670480603, 136.3984651934]$, at $P = 1.6310415204$, on the empirical-closure row class. The comparison to the measured endpoint sits in an explicitly compare-only block: the measured value lies outside the interval, and the certified same-scheme anchor gap is $[0.6198609041, 0.6505569679]$ inverse-alpha units. The anchor $a_0(P) = \alpha^{-1}_{\mathrm{em}}(m_Z^2; P)$ is the one-loop renormalization-group value run from the OPH unification scale to m_Z; the physical five-flavor on-shell value is 128.939, and the difference from the OPH anchor 128.308 is 0.631, at the lower edge of the certified gap. The gap records transport missing from the one-loop anchor; it is not evidence by itself that a source chain closes. A source-only bridge that closes this gap reduces to the OPH hadronic spectral measure, which needs a working hadron construction. The empirical endpoint is a comparison calculation and does not supply that source-only bridge. The exact exterior weak multiplicity is four. Its use as the transmutation factor $\beta_{\mathrm{EW}} = 4$ is conditional on the common screen/electroweak load-carrier identification; overloaded β-ratios appear only on benchmark readouts and are not part of the theorem contract.

3 Results at a Glance

The particle derivation carries the local pixel scale into the directly comparable rows below. Detailed scope statements are collected in the support sections that follow.

The theorem rows and target-anchored rows describe different objects. The theorem table carries a two-modulus quark non-identifiability result and a source-domain charged no-go boundary. The target-anchored rows contain same-family witnesses and diagnostic calculations. The electromagnetic, color, and metric carrier roles have conditional classical-mode receipts but no derived quantum mass rows; the W/Z numbers are convention-dependent chart and adapter coordinates; and the Higgs boson sits on the declared Higgs/top critical surface.

The following sections use the premises and caveats stated in the table.

Non-hadron results. The non-hadron results split by branch as follows:

The values and benchmark checks are

$$\alpha^{-1}(0) = 137.035999177(21), \qquad P \simeq 1.6309682094,$$

$$N_{\mathrm{EW}} = 3.5323546226929906511\ldots \times 10^{122}, \qquad \mathcal{B}_{\mathrm{EW}} = 0, \qquad \epsilon_H = 0.$$

This bridge value is about 6.6 percent above the Planck-Λ central capacity $N_\Lambda \simeq 3.313 \times 10^{122}$, and the physical equality requires an executed public-record fixed point plus the common screen/electroweak load-carrier map. No source-only nonzero particle mass follows. The selected-carrier, value-law, inverse-adapter, and conditional Higgs coordinates appear only in the technical comparison sections below. Target-anchored charged-lepton and quark witness values, including the mixed-convention quark target packet, appear only in the exact-fit comparison calculations. Compare-only absolute neutrino attachments are withheld from public prediction tables. The neutrino comparison surface keeps representative splitting checks such as

$$\Delta m^2_{21} = 7.488059465106851 \times 10^{-5}\,\mathrm{eV}^2,$$
$$\Delta m^2_{31} = 2.5123118727618473 \times 10^{-3}\,\mathrm{eV}^2,$$
$$\Delta m^2_{32} = 2.4374312781107786 \times 10^{-3}\,\mathrm{eV}^2.$$

The table states each derivation chain and the caveat that prevents a stronger theorem claim.

How to read mismatches and exact hits. Exact numerical agreement is not by itself a proof of a blind prediction. The provenance record classifies the W/Z row as target-used frozen-reference reproduction, the charged-lepton triple as an empirically anchored current-family witness, and the mixed-convention quark packet as a target-anchored selected-frame comparison rather than a public source-only prediction. The same record classifies the hierarchy/naturality formula as a zero-residual identity on its declared selected branch, with $\epsilon_H = 0$; the strict source-root and physical-scale certificates are separate premises. Where a row does not match, or where it matches only under constrained premises, the explanations are as follows. The source/root certificate gives $\alpha^{-1}_{\mathrm{root}} = 136.994835177413\ldots$ and $P_{\mathrm{fwd}} = 1.630972095858897\ldots$, but the declared map is incomplete. The certified gauge-width map gives $\alpha^{-1} = 137.035660136946577\ldots$, with gauge-width residual 2.5×10^{-6} relative to the measured 137.035999177(21). The no-hadron packet that combines different pixel coordinates is no fixed point and is excluded from physical output. Source-only closure requires Ward-projected hadronic spectral transport. The separate endpoint-accounting diagnostic is

$$\Delta^{\mathrm{fp}}_{\mathrm{H,cal}} = \alpha_U(P_C)\, C_{24,Q}, \qquad C_{24,Q} = 1.000964785973232625384951114070 2475\ldots.$$

The $C_{24,Q}$ factor is endpoint accounting rather than a source-emitted theorem. At the public endpoint pixel value

$$P \simeq 1.6309682094$$

the endpoint residual package requires

$$\Delta_{\mathrm{source}}(P) = 0.041465861005223389053448715357314044\ldots.$$

That scalar belongs to the source-side hadronic spectral transport and scheme-remainder map. The declared empirical route would use a separately labeled $e^+e^- \to$ hadrons spectral input. No same-scheme integrated endpoint payload is emitted here. The first-principles transport row excludes an inserted comparison endpoint. The electroweak support record names the residual map and the RG/matching/threshold/scheme packet. An observer inside the branch measures the dressed Thomson coupling, not the undressed source diagnostic $1/136.994835\ldots$, because a zero-momentum electromagnetic measurement sees the Ward-projected $\mathrm{U}(1)_Q$ current after charged-lepton vacuum polarization, confined-quark/hadron spectral transport, and same-scheme finite endpoint matching have been included. The electroweak W/Z row is therefore a chart and prescription comparison, not a physical mass benchmark. Charged leptons carry a source-domain no-go because the determinant trace-lift attachment from the electroweak descendants of P to physical charged data is absent. The auxiliary direct-top PDG row differs from the theorem coordinate by 0.20673301656674425 GeV, or 0.28458848947515303 combined standard deviations, because Q007TP and Q007TP4 are distinct extraction codomains; the direct-top response kernel has a source-domain no-go boundary. Neutrino absolute masses are not directly measured, and PMNS-angle residuals are visible comparison tension outside the theorem branch. First-principles hadron masses have no emitted prediction because the required Ward-projected hadronic spectral measure must come from a working OPH hadron construction. Empirical hadron closure rows carry a separate support class.

Local unification surface. The bosonic rows carry one cross-branch statement. The same pixel input P, fixed on the synthesis-paper outer/inner closure relation, fixes the electroweak and criticality trunk

$$P \longmapsto \alpha_U(P) \longmapsto \big(t_U(P), t_{\rm tr}(P)\big) \longmapsto v_{\rm chart}(P) \longmapsto (M_W, M_Z)\,.$$

$$\sigma_{\rm crit} = \alpha_U(P)\cos(2\theta_{W0})/\sqrt{\pi} \longmapsto (m_H, m_t)\,.$$

Here $\alpha_U(P)$ is the branch value selected by the same forward pixel-closure solve, so the bosonic trunk contains no inverse electroweak readback of the internal transmutation data. On the companion gravity side, the same pixel law packages

$$\bar{\ell}_{\rm SU(2)}(t_{2,\rm run}) \;+\; \bar{\ell}_{\rm SU(3)}(t_{3,\rm run}) = P/4.$$

The gravity normalization itself is emitted by the selected scale certificate $\gamma_\star = \ell_\star \nu_{\rm Cs}/c$, equivalently $B_\star = 3\pi/\ell_\star^2$. With $a_{\rm cell} = P\ell_\star^2$, the Newton area law gives $G_{\rm geom} = \ell_\star^2$, so the pixel cancels. The local unification surface separates one explicit familiar-unit readout package from three mathematical surfaces: the W/Z comparison, the conditional declared-surface Higgs/top coordinate, and the gravity-side readout with its strict classical-regime clause. On the gravity side, the stated local extension surface uses the lifted product presentation of the realized quotient branch and identifies

$$\bar{\ell}_{\rm shared} = \bar{\ell}_{\rm SU(2)}(t_{2,\rm run}) \;+\; \bar{\ell}_{\rm SU(3)}(t_{3,\rm run}).$$

On that same surface the pixel law fixes $\bar{\ell}_{\rm shared} = P/4$, and the local SI readout is

$$G_{\rm SI} = \frac{c^3\ell_\star^2}{\hbar}$$

from the selected scale certificate. The χ_ν protected-reserve branch uses the same public endpoint convention as

$$P_\chi = P_{\rm C} = 1.630968209403959\ldots, \qquad \epsilon_{\rm res} = \frac{P_\chi}{24},$$

after the same-collar shared-edge budget and one-class $\mathbb{Z}_6$ trace receipts pass. This does not make $P/4$ a primitive Hilbert-space dimension: $P/4$ is the local screen-cell entropy budget in the particle and hierarchy branches, while $P_\chi/24$ is the shared scalar-reserve density used by the χ_ν collar branch. On that declared extension surface the same familiar-unit package reads

$$L_{\rm loc} = \sqrt{a_{\rm cell}}\,\widehat{L}(P), \qquad t_{\rm loc} = \frac{\sqrt{a_{\rm cell}}}{c}\,\widehat{T}(P),$$

$$E_{\rm loc} = \frac{\hbar c}{\sqrt{a_{\rm cell}}}\,\widehat{E}(P), \qquad \Theta_{\rm loc} = \frac{\hbar c}{k_B\sqrt{a_{\rm cell}}}\,\widehat{\Theta}(P),$$

with dimensionless $\widehat{L}, \widehat{T}, \widehat{E}, \widehat{\Theta}$. Thus, at fixed P, the local ruler is $\sqrt{a_{\rm cell}}$; seconds are that ruler divided by the structural Lorentz output c; and GeV and Kelvin are downstream familiar-unit displays of the inverse local ruler through $\hbar$ and k_B. On that declared extension surface the local scale-readout bundle is

$$\begin{aligned} c &= 299792458\,\mathrm{m/s},\\ G &= 6.674299995910528 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}, \end{aligned}$$

No nonzero particle-mass coordinate is a source-only prediction. The W/Z inverse adapter and the conditional Higgs/top coordinates are compare-only quantities discussed in their technical sections.

The structural Lorentz output is the common invariant null cone and speed $c_\star$. The decimal 299792458 m/s is exact by the SI definition of the metre, not a predicted magnitude. The paper keeps the G row as an exact scale-readback value, with no literal zero-difference identity against the rounded benchmark 6.6743×10^{-11}.

The detailed inventory below records the source-derived theorems, conditional continuations, and target-anchored comparisons sector by sector.

3.1 Detailed particle inventory

Family	Particle	Theorem scope	OPH value or strongest derived benchmark	Open condition
Conditional carrier modes	electromagnetic	action/phase receipt	two transverse classical $k^2 = 0$ modes; $\mu^2_{Q,\mathrm{quad}} = 0$ only in the displayed Maxwell action	physical Hilbert space, positive-residue pole, and stable asymptotic photon receipt absent
Conditional carrier modes	color	perturbative/pre-confinement action receipt	$2 \dim G$ transverse classical $k^2 = 0$ modes; $\mu^2_{\mathrm{YM,quad}} = 0$ only in the displayed pure-Yang–Mills expansion	positive-residue pole and deconfined asymptotic colored sector absent; no free gluon is claimed on the confining branch
Conditional carrier modes	tensor	action/background receipt	two Einstein transverse-traceless classical $k^2 = 0$ modes; $\mu^2_{\mathrm{EH,TT}} = 0$ only on the pure-Einstein branch	metric quantization, positive physical Hilbert space, positive-residue pole, and asymptotic/EFT particle interpretation absent
Electroweak branch	W boson	no source-only physical mass	no OPH-native pole	selected-carrier, conditional value-law, and inverse-calibration coordinates are comparisons; no pole theorem
Electroweak branch	Z boson	no source-only physical mass	no OPH-native pole	same boundary as the W row
Higgs/top critical stage	Higgs boson	no source-only physical mass	no OPH-native pole	requires an independent source root and physical scale, finite quotient-path selection, same-scheme running and threshold transport, target-independent criticality rigidity, a complex pole, and an uncertainty enclosure
Quark family	top quark	separate target-anchored comparison coordinate	withheld from public prediction table	cross-section pole-mass extraction coordinate; it is not a sixth entry in one common running-mass chart and is not selected by the source spread laws
Quark family	up quark	reciprocal-ray candidate rejected	withheld from public prediction table	common-scale dimensionless Yukawa test fails; a source-derived flavor-orbit selector is absent
Quark family	down quark	reciprocal-ray candidate rejected	withheld from public prediction table	same common-scale rejection and six-scalar interface boundary
Quark family	strange quark	held-out reciprocal-ray failure	withheld from public prediction table	part of the 19.9% to 21.6% held-out failure across tested scales
Quark family	charm quark	held-out reciprocal-ray failure	withheld from public prediction table	stored GeV-valued matrices are mixed-convention mass textures, not physical Yukawas
Quark family	bottom quark	reciprocal-ray candidate rejected	withheld from public prediction table	common-scale source transport, Higgs normalization, and flavor-orbit selection absent
Charged leptons	electron	continuation gap	n/a; exact same-carrier centered readback exists once a charged source pair is emitted, but the absolute scale is blocked by the closed common-shift no-go; benchmark target $g_e^\star = 0.0457789$, equivalently $\Delta_e^{\mathrm{abs},\star} = 3.00398633$	close branch-generator splitting, then emit the lift whose descended scalar is $\mu_{\mathrm{phys}}(Y_e)$; within that lift the exact smaller forcing object is the physical identity-mode equalizer, after which $\widetilde{C}_e(Y_e) = \widehat{C}_e(Y_e) + \mu_{\mathrm{phys}}(Y_e)\,\mathbf{1}$, $A_{\mathrm{ch}}(Y_e) = \mu_{\mathrm{phys}}(Y_e)$, and the readouts g_e, Δ_e^{abs} follow
Charged leptons	muon	continuation gap	n/a; same exact centered-readback / common-shift-no-go frontier as the electron row	same $\widehat{C}_e^{\mathrm{cand}} \to$ branch-generator splitting $\to$ lift $\to \mu_{\mathrm{phys}}(Y_e) \to A_{\mathrm{ch}} \to g_e$ closure chain, with the physical identity-mode equalizer beneath the descended scalar
Charged leptons	tau lepton	continuation gap	n/a; same exact centered-readback / common-shift-no-go frontier as the electron row	same $\widehat{C}_e^{\mathrm{cand}} \to$ branch-generator splitting $\to \mu_{\mathrm{phys}}(Y_e) \to A_{\mathrm{ch}} \to g_e$ closure chain, with the physical identity-mode equalizer beneath the descended scalar

Family	Particle	Theorem scope	OPH value or strongest derived benchmark	Open condition
Neutrino comparison	declared f-basis weighted-cycle matrix	rejected target-informed candidate	no flavor-particle mass row; the frozen declared-basis coordinate gives $\theta_{12} = 34.2259^\circ$, $\theta_{23} = 49.7228^\circ$, $\theta_{13} = 8.68636^\circ$, $\delta = 305.581^\circ$, $J = -0.02753$, and, under its declared normal-ordering labels, $\Delta m^2_{21}/\Delta m^2_{32} = 0.03072111$	the upstream flavor kernel is a hand-written template; the physical charged basis and mass-label rule are open; the target-informed selector precludes a blind-prediction claim; the NuFIT 6.1 correlated profile rejects the candidate; Majorana and absolute-attachment values are compare-only
Hadrons	proton	strong-binding construction absent; empirical comparison contract emitted	no first-principles emitted prediction	first-principles prediction requires a working source-derived hadronic backend, such as GLORB/Echosahedron, plus Ward-projected two-current, higher-point, and transition spectral exports, same-scheme remainder, and production systematics; empirical closure rows use separate $e^+e^- \to$ hadrons input
Hadrons	neutron	strong-binding construction absent; empirical comparison contract emitted	no first-principles emitted prediction	same OPH construction gate, with isospin-resolved hadron systematics in the construction support class
Hadrons	neutral pion proxy	strong-binding construction absent; empirical comparison contract emitted	no first-principles emitted prediction	same OPH construction gate; local stable-channel surrogate output is not a paper prediction
Hadrons	$\rho(770)^0$ proxy	strong-binding construction absent; empirical comparison contract emitted	no first-principles emitted prediction	same OPH construction gate, plus finite-volume resonance extraction in the construction support class

4 Source Values, Screen Architecture, and Theorem Premises

The particle derivation uses the three axioms, the screen-capacity closure, the local pixel closure, the regulated screen realization, and the imported OPH theorems stated below.

4.1 Canonical OPH basis

The particle-spectrum derivation uses the three OPH axioms shared with the spacetime and gauge derivation papers [6, 7]. They are restated here for local readability because every particle-family chapter depends on them either directly or through the structural OPH chain imported from those papers.

Axiom 1 (oriented twelve-port observer screen). There exists an observer patch net on an oriented spherical screen: at every finite resolution each local carrier has twelve primitive boundary ports forming the vertices of an oriented triangular boundary with 30 edges and 20 faces, combinatorially the boundary of an icosahedron, and carriers join through typed seams and coherent triple overlaps, refine to an oriented spherical support, and expose local state, readback, records, repair moves, and checkpoints. Formally, for every regulator r in a directed system there is a typed object $\mathfrak{N}_r = (\mathcal{P}_r, \mathcal{A}_r, \mathcal{R}_r, \mathcal{I}_r, \mathcal{U}_r, \mathcal{C}_r, N_r, S_r, b_r)$: a finite patch/overlap category with an isotone local algebra net and central record algebras; a federation of finite carriers, each with twelve primitive pairwise-orthogonal central port projections summing to one and a boundary packet $K = (P, E, F, o)$ with $|P| = 12$, $|E| = 30$, $|F| = 20$, degree-five ports, five-cycle links, and coherently oriented two-face edges, isomorphic to the icosahedral boundary complex with no preferred labels; seam algebras with unital restrictions and coherent triple-overlap cocycles forming the nerve N_r; a finite edge-midpoint refinement support S_r realized orientation-preservingly in S^2 with mesh tending to zero; and a source-bound degree-one bridge $b_r : N_r \to S_r$, all commuting with refinement. The axiom constrains architecture only. It does not assert semantic agreement, any state selection, repair termination, equal state weights, a response law, a rotation group, a current algebra, or any gauge, particle, or metric content; those are theorems, named-realization results, or open interfaces.

Axiom 2 (observer agreement). Observers operating on the screen agree on the meaning of the data they jointly interpret. Formally, the interpretation map $\mathcal{J}_r : \mathsf{Data}_r \to \mathsf{Meaning}_r$ from observer-accessible data to operational meanings is natural with respect to every visible overlap restriction, recharting, seam translation, higher-overlap map, federation map, and refinement map: every declared data-access diagram for accepted public data commutes after interpretation, and $\mathcal{J}_r \circ c_{s\to r} = C_{s\to r} \circ \mathcal{J}_s$ across resolutions. The axiom constrains accepted shared data only, and the domain it quantifies over is supplied by the Axiom 1 interfaces rather than by the axiom itself. It does not imply global state extension, repair termination, confluence, unique normal forms, record durability, Byzantine safety, dissemination, or commutation of state optimization with refinement; raw mismatch and repair precede acceptance.

Axiom 3 (conditional maximum randomness). Everything that observer agreement leaves unconstrained is maximally random. Formally, at finite regulator r a state is a compatible family of local normalized states on the accessible algebra net; $\mathcal{K}_r$ is the nonempty convex set of such families satisfying the finite observer-visible constraints supplied by Axioms 1 and 2 through an A1-generated constraint grammar with a factorization theorem; and, relative to an exact reference family τ_r, an A1-generated observer cover $\mathcal{G}_r$, and strictly positive exact weights $w_{r,P}$ constructed from quotient-visible A1 data by a rule natural under admissible presentation equivalence, the realized state is the information projection. The cover restriction map is injective on $\mathcal{K}_r$; equivalently, at smooth points no nonzero feasible tangent is invisible to the whole cover. The realized state is

$$\rho_r = \arg\min_{\rho\in\mathcal{K}_r} \sum_{P\in\mathcal{G}_r} w_{r,P}\, D(\rho_{r,P}\|\tau_{r,P}).$$

The same projection rule applies separately to each declared optimizer object type (ontic state, inference state, or transition distribution), and statements about one type do not transfer to another. This is equivalent to weighted local entropy maximization when every local reference density is identity-proportional in its declared trace. "Random" means least informative relative to the declared reference and the complete agreement constraint set. The axiom selects one state inside one fixed feasible space. It does not compare unrelated state spaces or field contents, does not imply optimizer compatibility across refinement, and does not imply recovery, alignment, or mixing; each such statement carries its own theorem, interface, or countermodel.

The complete formal basis, including its constraint map and non-implications, is given in the canonical three-axiom reference.

The first axiom's spherical support is The Standard Model gauge paper's screen-first global-support branch. On the producer branch, S^2 becomes available only after spherical-incidence, mesh, cross-ratio, normalization, refinement, and carrier-to-support receipts establish it. The resulting support screen is distinct from both the local twelve-port carrier boundary and the federation overlap nerve.

Two families of premise interfaces enter beside the axioms, each named at the results that consume it. The gravity-adjacent chapters consume the edge-center, collar-recovery, generalized-entropy, and modular-normalization interfaces of the spacetime paper; these are classified derivation targets and physical identifications with their own countermodels, not axioms. The particle chapters consume declared sector completions: the generation count inside the conditional window, scalar attachment and multiplicity, and the completeness statement for extra light sectors are open physical attachments, and the charged-lepton and quantitative selector branches state their declared premises where they appear. The transportable-sector/Tannaka reconstruction remains a separate conditional classification route.

4.2 Regulated screen architecture and patch language

The SM/GR derivation in Ref. [6, 7] states the axioms in algebraic language. Ref. [4] supplies one concrete regulated realization of that language: a federation of finite patch carriers with echosahedral multi-port interfaces, recurrent toroidal subchannels, exposed overlap packets, record algebras, and local repair instruments. The formal observer patch is the bounded access-and-record structure. One echosahedral carrier is a candidate primitive physical realization on the homogeneous branch; the carrier alone does not satisfy the observer definition. This regulated architecture is important for the particle derivation for three reasons.

First, it makes the screen picture operational. A patch is a finite local algebra in a bounded carrier, and an overlap is a boundary-visible algebra with declared shared observables. Second, it makes the edge-sector language concrete. The particle derivation uses edge sectors, transport, refinement, fusion, and overlap data to build the gauge and flavor branches; Ref. [4] shows how those objects can be realized at fixed cutoff by finite gauge-aware patch carriers. Third, it clarifies the measurement interface. The observer-facing measurement package is a theorem-bearing fixed-cutoff statement about a central record algebra, Born probabilities for its event projectors, and Lüders conditioning on that same commuting algebra.

Three geometric objects govern this reference implementation. The local carrier boundary has the twelve-port icosahedral incidence certified on the echosahedral lineage. The federation screen is the finite federation together with its overlap nerve. The support screen is the observer-visible S^2 chart obtained only when global incidence, mesh, cross-ratio, normalization, and refinement receipts hold. A federation of local icosahedral carriers can have a nonspherical nerve, so local A_5 symmetry does not force a global S^2 support.

The framework is presentation-invariant and carrier-sensitive. A different hidden implementation is physically equivalent when its observer-visible quotient agrees. A different visible port graph, topology, record process, or response law may select a different branch. Hardware evidence enters only through a public OPH evidence bundle with stable hashes and verifier receipts.

Physical phase locking is a candidate producer for coherent overlap comparison. To support that interpretation it must produce the accepted repair relation, transactional confluence, public records, and controlled noise bounds. No current theorem identifies phase locking with consensus confluence, modular flow, or an observer clock. Separate support-visible continuum theorems supply the geometric and gauge limits.

4.3 Quantitative inputs and where they enter the particle derivation

The quantitative derivation developed here uses one incomplete declared-map coordinate and one conditional capacity coordinate together with the selected no-G scale certificate:

$$P \equiv a_{\mathrm{cell}}/\ell_\star^2, \tag{1}$$

$$D \equiv \dim \mathcal{H}_{\mathrm{cap},r,D}, \qquad N_{\mathrm{CRC}} \equiv \log D_{\mathrm{CRC}}, \tag{2}$$

$$\gamma_\star \equiv \frac{\ell_\star \nu_{\mathrm{Cs}}}{c}, \qquad B_\star \equiv \frac{3\pi}{\ell_\star^2}. \tag{3}$$

These closure values have distinct downstream dependencies.

Pixel area P. The pixel area is the declared local UV-area ratio. In this derivation it feeds the heat-kernel / edge-law input stage and, through that route, the forward electroweak surface. It is the common upstream numerical variable for the electroweak, Higgs/top, flavor, quark, and hadron-facing branches. The particle paper therefore treats $P = P_\star$ as the inherited root of the incomplete outer/inner declared map

$$P_\star = \varphi + \frac{\sqrt{\pi}}{A_T(P_\star)}$$

from the synthesis paper [2], not as a quantity derived again inside the particle-spectrum argument itself.

Conditional screen capacity N_{CRC}. The screen capacity is not part of the local recovered-core gravity/gauge derivation. It enters the separate screen-capacity branch through the stable cosmic record-closure target

$$\mathfrak{F}_{r,0}(D_\star) = \{D_\star\}, \qquad N_{\mathrm{CRC}} = \log D_\star, \qquad \Lambda_{\mathrm{CRC}}\ell_\star^2 = \frac{3\pi}{N_{\mathrm{CRC}}}, \qquad \Lambda_{\mathrm{CRC}} = \frac{3\pi}{GN_{\mathrm{CRC}}}.$$

The synthesis paper specifies the physical active-readback target directly. At finite cutoff, compatible reachable public record atoms and the globally coupled checkpoint family define a confusability graph G_q and

$$M_0(q) = \alpha(G_q), \qquad \mathfrak{F}_{r,\varepsilon}(D) = \{M_\varepsilon(q) : q \in \widetilde{\Omega}_{r,D}\}.$$

Only a whole-fiber singleton is scalar. A faithful capacity-carrier representation makes that scalar exact map deflationary; a confusability-reflecting capacity embedding makes it monotone, and fixed-D refinement eventually stabilizes. The public checkpoint packet, carrier representation, whole-fiber scalarization, and exact finite-size slack law with one physical zero remain open. Identifying that capacity with de Sitter entropy requires the horizon–record identification; identifying it with the electroweak bridge requires the common screen/electroweak load-carrier identification. The bridge value near 3.532×10^{122} is about 6.6 percent above the Planck-Λ central capacity near 3.313×10^{122}. The last equality is therefore a conditional scale-certified display; it does not determine an SI curvature scale from $P_\star$ and N_{CRC} alone. An independently frozen discrimination scale ρ_{op} is a test of the direct code capacity through $\log M_0 - \pi/\rho_{\mathrm{op}}^2$, not its definition. A diagonal terminal-state count or its argmax does not construct the off-diagonal map. In the particle context, this matters only for the cosmological-capacity discussion that frames why local null data do not determine the cosmological constant. The implemented branch uses the static-patch normalization

$$N_{\mathrm{patch}} = \left(\frac{r_{\mathrm{dS}}}{\ell_P}\right)^2 \simeq 1.05 \times 10^{122}, \qquad N_{\mathrm{scr}} = \pi N_{\mathrm{patch}} \simeq 3.313 \times 10^{122}.$$

This capacity bookkeeping supplies no neutrino result; the weighted-cycle neutrino derivation is a separate branch.

Selected scale certificate. The gravity scale is a separate observation-located certificate. On the declared branch,

$$\gamma_\star = \frac{\ell_\star \nu_{\mathrm{Cs}}}{c}, \qquad B_\star = \frac{3\pi}{\ell_\star^2}, \qquad G_{\mathrm{SI}} = \frac{c^3\ell_\star^2}{\hbar}.$$

If the declared-map root acquires its missing physical identification, the local pixel $P_\star$ identifies $a_{\mathrm{cell}} = P_\star\ell_\star^2$ and $\bar{\ell}_{\mathrm{shared}} = P_\star/4$. In the Newton area law $P_\star$ cancels, so this paper does not treat G as a particle-sector consequence of $P_\star$. $\bar{\ell}_{\mathrm{shared}}$ is a shared-cut density, not the logarithm of an autonomous cell Hilbert factor. The particle branch uses it for cell/edge consistency; it does not select a fixed primitive observer capacity.

Why these are the declared closure/readback values. One of the discipline constraints of the OPH derivation is that particle-sector freedom should not be hidden in a large uncontrolled parameter set. This paper exposes the incomplete declared-map root $P_\star$, the proposed N_{CRC} correctable-public-record closure point, and the selected scale certificate. The same shared-coordinate discipline applies to every conditional branch. For the particle-spectrum branch, the nontrivial common quantitative burden sits on $P_\star$: N_{scr} enters the separate capacity/cosmology side and $\gamma_\star$, equivalently $B_\star$, enters the separate gravity-scale side, whereas the reverse-engineering claim for masses and couplings is that one shared declared-map coordinate should drive the spectrum instead of a long sector-by-sector input list.

4.4 Technical theorem data carried into the particle-spectrum derivation

This paper uses the same theorem premises as *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency and Deriving Standard Model Gauge Structure from Observer Overlap Consistency* [6, 7], together with the regulator-language clarification integrated into the consensus and microphysics appendices. This separation matters only insofar as it separates branch-internal statements from declared inputs.

On the finite identity-reference branch, the third axiom and the declared local constraint grammar give the regulator-scale local Gibbs form, the quasi-local propagation bound, and the endpoint-control estimate for bounded intervals. Refinement stability is supplied by the separate optimizer-pushforward and refinement-tail interface. In regulator language, one works with finite local Hilbert spaces and a boundary-fixed overlap action on cut data. Beyond those regulator premises, the Standard Model gauge paper supplies the support-visible Bisognano–Wichmann (BW) scaling theorem for the Lorentz/null-modular/Einstein branch. Transportability and the fixed-cutoff bosonic sector category are theorem-produced in The Standard Model gauge paper; the refinement/fiber ladder and cofinal admissible compact-gauge witness additionally require its explicit compact-gauge refinement receipt. On the central branch the matching gluing obstruction is the combined zero-obstruction transport criterion: vanishing triangle defect plus trivial represented holonomy of the strictified edge 1-cocycle. It introduces no second theorem-side input. Any local-Gibbs or collar-mixing language on that branch is fixed-cutoff recoverability/support control; it is not the source of the compact-gauge witness. For the BW/geometric side, the target is the support-visible extracted geometric subnet. The Standard Model gauge paper proves the needed scaling theorem by combining regularized support-visible modular transport, weak-$*$ extraction of the cap pair in the Gelfand–Naimark–Segal representation, support-readable modular covariance, BW framing, held-out oriented cross-ratio rigidity, and independently normalized geometric 2π-Kubo–Martin–Schwinger (KMS) convergence with wrong-scale controls. The unregularized full-algebra common-floor route is deliberately not claimed, because off-support directions may collapse without affecting observer-facing matrix elements. Bare finite consensus does not by itself produce the finite cap-normal BW certificate used by that theorem. The theorem also consumes an independent mixed Gelfand–Naimark–Segal common-comparison premise on the same tower. Local maximum entropy and refinement do not produce that premise.

For the particle-spectrum derivation, this has a practical consequence. Structural statements such as compact-gauge reconstruction, the contract-conditional Standard Model quotient, the exact hypercharge lattice, the color count $N_c = 3$, and the declared generation completion are not floating independently of the declared input and theorem premises. Nor do they imply a physical family attachment. The additional classical carrier modes live on explicit quadratic action/background/phase receipts, and particle poles require the stronger quantum receipt. These statements live on a controlled chain whose scope conditions need to be visible in this paper, especially for quantitative closure, continuation, and nonperturbative sectors.

4.5 Imported core theorem packages

This paper uses the following OPH theorem packages without re-proving them in full.

1. **Patch-net and overlap language.** Ref. [3] organizes overlap repair, fixed-point language, cycle holonomy, and gauge-as-gluing in a form that is especially useful for flavor transport and observer-centric interpretation.

2. **Screen-side regulated architecture.** Ref. [4] gives the federated echosahedral patch-carrier model, the regulated patch-net embedding theorem, and the fixed-cutoff edge heat-kernel / Casimir theorem.

3. **Measurement interface.** Ref. [4], together with the integrated measurement appendices and the synthesis paper *Observers Are All You Need* [2], supplies the fixed-cutoff central-record, Born-rule, and Lüders-conditioning package together with the fixed-cutoff Bell/CHSH combined theorems used when the measurement sections explain how observations pick out definite observer-accessible outcomes and how two-wing comparison laws stay on the quantum side of the classical Bell bound.

4. **Contract-conditional Standard Model gauge chain and conditional compact reconstruction.** *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency and Deriving Standard Model Gauge Structure from Observer Overlap Consistency* [6, 7] and the dedicated gauge-group fragment together provide the explicit inverse-port response and rank-15 matter contracts, anomaly balance, and tensor descent that imply

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \qquad 3 \le N_g \le 5, \qquad N_c = 3.$$

   The gauge group, charge lattice, and $N_c = 3$ come from the stated premises alone within this conditional chain. The same Standard Model gauge paper gives the transportable-sector/Tannaka construction as a separate conditional compact-group route. Physical source binding of the finite current and identification with the Tannaka current require the open producer, intertwiner, and laboratory attachment.

5. **Conditional topic notes.** The integrated branch appendices carry useful material on supersymmetry, the finite-quotient baryogenesis source theorem and its open record-generator branch, proton stability, generation structure, and Yukawa hierarchy, but those sections have to be read with their stated claim boundaries intact. They are sources for conditional analyses, not recovered-core theorems.

This reading rule governs the whole paper. Structural gauge and carrier results can be used as structural results. The electroweak branch is a declared quantitative calibration branch. The Higgs/top critical stage is a conditional downstream calculation on its declared running, matching, and threshold surface. The quark, charged-lepton, neutrino, and hadron chapters state conditional or comparison results because their physical premises are incomplete.

5 Gauge Architecture, Standard Model Structure, and Structural Carriers

The particle-spectrum derivation does not start from a list of particles. It starts from the contract-conditional finite gauge branch. The explicit response and rank-15 matter contracts, determinant balance, Spin lift, and tensor descent imply the Standard Model gauge group, charge lattice, and carrier roles before any mass readout. The overlap/transport obstruction calculus and transportable-sector/Tannaka reconstruction form a separate conditional classification route. The target-free matter reduct leaves the generation count unselected. The strengthened finite screen interface selects rank three under its single-object, faithful-action, and operational-cost premises. The physical family interpretation and the extra-sector completeness statement are open. This logical split matters for this paper. The gauge branch does not fix the carriers' quantum-particle masses; the action-level and quantum spectral gates below decide whether a classical mode or particle row exists.

On the declared packet, the gauge/content result used throughout this paper is

$$\frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}, \qquad 3 \leq N_g \leq 5, \qquad N_c = 3.$$

Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency and *Deriving Standard Model Gauge Structure from Observer Overlap Consistency* [6, 7], the longer main derivation surface, and the dedicated gauge-group proof fragment agree on the split. The source-bound finite receipts fix the quotient, hypercharge lattice, and $N_c = 3$ from their stated premises. Intrinsic CP capability together with weak-sector UV completeness gives $3 \leq N_g \leq 5$. The exact screen-band theorem selects rank three on the strengthened finite interface, and the declared Laplacian simulator realizes the corresponding lowest-positive-frequency residue. Witten parity is a consistency check. The finite local-domain receipt checks the separately declared operator $D_\sigma \otimes I_{45}$ and conditional gap inheritance. It does not construct a source-selected local matter carrier or bridge the twelve-port Spin packet to that domain. These steps do not identify physical matter poles or exclude extra light sectors.

The finite response/matter-contract A_5 route and the transportable-sector/Tannaka route are separate chains. Their equality as one physical gauge-current object and physical source binding of the first route are open. The rank-three response band therefore remains a candidate family fiber. Its abstract rank-45 tensor product with the generation table does not attach that band to three physical families.

For the particle zoo this structural branch fixes three carrier roles, not three particle masses. The Standard Model gauge paper's carrier-mode theorem then supplies the following conditional action-level statements. The Maxwell action on the ordinary unbroken electromagnetic vacuum has two transverse classical modes. The pure Yang–Mills action about a flat connection has $2 \dim G$ perturbative transverse modes, but the confined color phase has no asserted gauge-invariant asymptotic gluon. The pure two-derivative Einstein–Hilbert action about Minkowski space has two classical transverse-traceless modes. In every case a quantum particle additionally requires a positive- energy physical Hilbert space, a positive-residue physical two-point pole, and the appropriate asymptotic or deconfinement receipt.

These conditional classical modes should not be confused with quantum-particle mass rows. The group and gravity branches identify the carrier roles; the displayed actions and phases supply their classical propagation; and only a separate quantum spectral receipt can establish them as photon, gluon, or graviton particles. The boson sections address the W, Z, and Higgs rows on their own electroweak and Higgs/top quantitative stages.

6 Electroweak Calibration and the Higgs/Top Critical Stage

The bosonic calculation separates source outputs from comparison coordinates. The electroweak calibration supplies the W boson and Z boson chart, and the criticality calculation supplies conditional Higgs and top coordinates. Neither calculation emits an OPH-native particle pole.

6.1 Electroweak chart on the quantitative branch

The electroweak derivation consists of a single-P running family followed by a reduced two-scalar carrier, a selector on that carrier, an exact carrier mass chart, and a conditional quotient-transport theorem beyond the selected carrier. In practical terms, the construction starts from the declared pixel input P, builds the running electroweak family, reduces it to the selected carrier, reads the W boson/Z boson pair from the selected point on that carrier, and then evaluates the mass pair together with the Ward-projected electromagnetic transport family. On this branch, a source-only prediction must respect that ordering: first certify the shared pixel input P on the declared running/matching/threshold/scheme surface, thereby fixing the source basis

$$(\alpha_U, \alpha_{2,m_Z}, \alpha_{Y,m_Z}, \eta_{\text{source}}, v),$$

with $\alpha_U(P)$, equivalently $t_U(P)$ and $t_{\text{tr}}(P)$, fixed by the same forward pixel-closure solve, and only then emit the electroweak transport family from that source basis. The stated theorem does not contain this full chain. Its W/Z rows are data-comparison adapter coordinates rather than OPH-native poles.

The calibration couplings are read at the source scale 91.5883371732 GeV, a coordinate of the source solve, rather than at the measured Z target. Evaluation of the conditional pole map in the raw pre-carrier basis omits the compact-carrier hypercharge step. The selector-consistent conditional evaluation lands the neutral coordinate near 90.66 GeV, about half a GeV below the converted reference target. The raw-basis and selector-consistent numbers are conditional diagnostics of the imported prescription. Neither is an OPH-native pole.

The same quantitative branch has one exact golden-ratio benchmark. If one writes

$$x(C) := \frac{S_{\text{gen}}(C)}{S_{\text{bulk}}(C)} = 1 + \frac{\langle L_C \rangle}{S_{\text{bulk}}(C)}$$

for the total/bulk/edge hierarchy and imposes exact self-similar balance

$$\frac{S_{\text{gen}}(C)}{S_{\text{bulk}}(C)} = \frac{S_{\text{bulk}}(C)}{\langle L_C \rangle},$$

then x obeys $x^2 - x - 1 = 0$, so the unique positive equilibrium point is

$$x = \varphi := \frac{1 + \sqrt{5}}{2}.$$

Equivalently the order parameter $A_\varphi(x) = x - 1 - \frac{1}{x}$ vanishes there. The synthesis paper records the unique numerical root of the incomplete outer/inner declared map. The point of the equilibrium theorem here is that the proximity to φ has a structural reason. Exact balance is too symmetric to support durable records, structure, and dynamics, so the declared map assigns a small equilibrium-breaking detuning away from that balance point.

The compare-only W/Z adapter rows in the final bundle are

$$m_W = 80.3625 \text{ GeV}, \qquad m_Z = 91.1879 \text{ GeV}.$$

These numbers sit on a target-conditioned validation surface and do not form a target-free prediction theorem. The exact selected-carrier chart gives the local pair

$$m_W^{\text{carrier}} = 80.38629169244275 \text{ GeV}, \qquad m_Z^{\text{carrier}} = 91.18290444674243 \text{ GeV},$$

so the mathematics distinguishes the selected-carrier chart, the candidate value-law pair $(80.3770000154, 91.1879780779)$ GeV, the conditional quotient-transport implication, and the measured-reference inverse repair surface. The rounded pair $(80.377, 91.1879780919)$ GeV is a boundary alias of the candidate, not the inverse adapter. None is a physical pole theorem.

Theorem 6.1 (Measured-pair reparametrization (calibration): coherent electroweak inverse fit)**.** *Let the emitted running/core electroweak basis be*

$$Q_{\text{run}}(P) = (\alpha_{Y,m_Z}(P), \alpha_{2,m_Z}(P), v_{\text{chart}}(P), \eta_{\text{source}}(P)),$$

and write

$$\tau_Y(\tau_2) = -\frac{\tau_2 + 2\eta_{\text{source}}}{1 + 4\tau_2^2}, \qquad n_{\text{fiber}}(\tau_2) = 1 + \frac{\alpha_{Y,m_Z}\tau_Y(\tau_2) + \alpha_{2,m_Z}\tau_2}{\alpha_{Y,m_Z} + \alpha_{2,m_Z}}.$$

Freeze one authoritative target pair

$$\mathcal{T}^\dagger = (M_W^\dagger, M_Z^\dagger), \qquad 0 < M_W^\dagger < M_Z^\dagger,$$

and define the charged anchor

$$\tau_{2,W}^\dagger = \frac{M_W^{\dagger\,2}}{\pi v^2 \alpha_{2,m_Z}} - 1, \qquad \delta\alpha_2^\dagger = \frac{M_W^{\dagger\,2}}{\pi v^2} - \alpha_{2,m_Z}.$$

Then the fiber-parallel hypercharge leg is

$$\tau_Y^\dagger = \tau_Y(\tau_{2,W}^\dagger), \qquad n_{\text{fiber}}^\dagger = n_{\text{fiber}}(\tau_{2,W}^\dagger),$$

$$M_{Z,\text{fiber}}^\dagger = v\sqrt{\pi(\alpha_{Y,m_Z} + \alpha_{2,m_Z})\, n_{\text{fiber}}^\dagger},$$

and the orthogonal neutral-shear scalar is

$$\delta M_Z^\perp = M_Z^\dagger - M_{Z,\text{fiber}}^\dagger,$$

equivalently

$$\delta n^\dagger = \frac{(M_Z^\dagger + M_{Z,\text{fiber}}^\dagger)\,\delta M_Z^\perp}{\pi v^2(\alpha_{Y,m_Z} + \alpha_{2,m_Z})}, \qquad \delta\alpha_Y^\perp = \frac{(M_Z^\dagger + M_{Z,\text{fiber}}^\dagger)\,\delta M_Z^\perp}{\pi v^2}.$$

The fiber-parallel hypercharge motion is

$$\delta\alpha_Y^\parallel = \alpha_{Y,m_Z}\,\frac{8\eta_{\text{source}}(\tau_{2,W}^\dagger)^2 - \tau_{2,W}^\dagger}{1 + 4(\tau_{2,W}^\dagger)^2}.$$

Therefore the unique coherent repair package is

$$\Sigma_{EW}^\dagger = (\delta\alpha_2^\dagger, \delta\alpha_Y^\parallel, \delta\alpha_Y^\perp),$$

equivalently $(\tau_{2,W}^\dagger, \delta n^\dagger)$*. If the selected carrier anchor is the* $\tau_2 = 0$ *fiber point, so that*

$$\alpha_{2,*} = \alpha_{2,m_Z}, \qquad \alpha_{Y,*} = \alpha_{Y,m_Z}(1 - 2\eta_{\text{source}}),$$

then the coherent validation couplings are

$$\alpha_{2,\dagger} = \alpha_{2,*} + \delta\alpha_2^{\dagger}, \qquad \alpha_{Y,\dagger} = \alpha_{Y,*} + \delta\alpha_Y^{\parallel} + \delta\alpha_Y^{\perp},$$

and they emit one coherent quintet

$$M_W^{\dagger} = v\sqrt{\pi\alpha_{2,\dagger}}, \qquad M_Z^{\dagger} = v\sqrt{\pi(\alpha_{Y,\dagger} + \alpha_{2,\dagger})},$$

$$\alpha_{\mathrm{em},\dagger}^{-1} = \frac{\alpha_{Y,\dagger} + \alpha_{2,\dagger}}{\alpha_{Y,\dagger}\alpha_{2,\dagger}}, \qquad \sin^2\theta_{W,\dagger} = \frac{\alpha_{Y,\dagger}}{\alpha_{Y,\dagger} + \alpha_{2,\dagger}}.$$

The target-conditioned electroweak validation law is one unique coherent value-emission law on top of the closed fiber map.

For the reference-fitted inverse-adapter comparison,

$$\begin{aligned}
\alpha_{2,\star} &= 0.03377843630219015,\\
\alpha_{Y,\star} &= 0.009682831911900495,\\
\eta_{\mathrm{source}} &= 0.022147000871961295, v \quad = 246.76711732749683\ \mathrm{GeV},\\
M_W^{\dagger} &= 80.3625\ \mathrm{GeV},\\
M_Z^{\dagger} &= 91.1879\ \mathrm{GeV},
\end{aligned}$$

and the law emits

$$\begin{aligned}
\tau_{2,W}^{\dagger} &= -0.0005918464744071317,\\
\delta\alpha_2^{\dagger} &= -1.9991648436440412\times 10^{-5},\\
M_{Z,\mathrm{fiber}}^{\dagger} &= 91.16822266259587\ \mathrm{GeV},\\
\delta M_Z^{\perp} &= 19.6773374041328\ \mathrm{MeV},\\
\delta\alpha_Y^{\parallel} &= 5.9969727526906094\times 10^{-6},\\
\delta\alpha_Y^{\perp} &= 1.8756950557734103\times 10^{-5},\\
\delta n^{\dagger} &= 4.2716772021678714\times 10^{-4}.
\end{aligned}$$

So the coherent validation couplings are

$$\begin{aligned}
\alpha_{2,\dagger} &= 0.03375844465375371,\\
\alpha_{Y,\dagger} &= 0.00970758583521092,
\end{aligned}$$

and the same coherent branch emits

$$M_W^{\dagger} = 80.3625\ \mathrm{GeV}, \qquad M_Z^{\dagger} = 91.1879\ \mathrm{GeV},$$

$$\begin{aligned}
\alpha_{\mathrm{em},\dagger}^{-1} &= 132.6344411210187,\\
\sin^2\theta_{W,\dagger} &= 0.22333729871365943.
\end{aligned}$$

These two carrier-side scalars belong to the target-conditioned validation surface. They are not the physical electromagnetic readout on the Ward-projected transport branch. That inverse-fit law provides an exact coherent calibration surface: it is a measured-pair reparametrization. On that target-conditioned surface the reference W/Z pair is hit exactly because the basis is solved from that pair; the hit is the defining condition, and the branch is calibration. It carries no predictive force. The forward running/chart solve gives (80.330, 91.119) GeV. The Particle Data Group (PDG) comparison values are mass-dependent-width Breit–Wigner parameters, while complex-pole masses define another convention. As a stated convention diagnostic under the complex-pole conversion of the PDG 2026 reference values, the energy-pole coordinates are $(M_W, M_Z) = (80.3411410, 91.1623040)$ GeV, whereas the older $\sqrt{\operatorname{Re} s}$ coordinates are $(80.3340218, 91.1537725)$ GeV. These two definitions must not be mixed. The running/chart coordinates carry no OPH theory covariance, the physical readout contract is open, and no physical pull or near-hit is assigned. The candidate forward value law has a stronger conditional theorem: it follows uniquely from a finite quotient-path certificate. The certificate is not emitted by the three axioms or by the existing carrier artifacts.

Definition 6.2 (Finite quotient-path certificate)**.** *After quotienting gauge representatives, port labels, scheduler data, and other hidden presentation coordinates, require:*

1. *the real, CP-even, color-singlet, charge-preserving response through two returns is exactly $\mathbb{R}q_2 \oplus \mathbb{R}q_n$, with no third scalar or charged-neutral off-block term;*
2. *the primitive response is bilinear in $(\eta_{\text{source}}, \alpha_U)$, one primitive event has fixed unit response normalization, and the quotient probability measure gives each of the four transmutation slots weight $1/4$, yielding $\lambda_{\text{EW}} = \eta_{\text{source}}\alpha_U/4$;*
3. *explicit carrier path lists, carrying one factor of η_{source} per return and uniform color-orbit measure $1/3$, have primitive, one-return, and two-return incidences $(1, 2/3, 1)$ for q_2 and $(1, 4/3, 2)$ for q_n, while the diagonal $\mathbb{Z}_6$ projector in the declared response representation has normalized trace $1/6$, source amplitude ρ_{EW}, and subtracts $(\rho_{\text{EW}}/6)\eta_{\text{source}}^2$ from both degree-two characters;*
4. *mismatch descent fixes the charged sign as contracting and the neutral sign as uplifting, and the parallel hypercharge fibre is the least-norm minimizer with Gram factor $1+4\tau_2^2$ and residual pairing $\tau_2 + 2\eta_{\text{source}}$;*
5. *the listed paths exhaust the admissible class: deeper paths, alternative central weights, and mixed terms are absent, quotient-trivial, or separated by a positive target-independent selector gap.*

Theorem 6.3 (Conditional quotient-transport value law)**.** *Let the source-only electroweak basis be*

$$Q_{\text{src}}(P) = \big(\alpha_U(P), \alpha_{2,m_Z}(P), \alpha_{Y,m_Z}(P), \eta_{\text{source}}(P), v_{\text{chart}}(P)\big),$$

and define

$$\rho_{\text{EW}} := \frac{\alpha_{2,m_Z} - \alpha_{Y,m_Z}}{\alpha_{2,m_Z} + \alpha_{Y,m_Z}}, \lambda_{\text{EW}} := \frac{\eta_{\text{source}}\alpha_U}{4}.$$

Assume that every entry is emitted on the same source branch, that $\eta_{\text{source}} = \rho_{\text{EW}}\alpha_U$, and that no measured W/Z, measured v, fitted electroweak parameter, or calibrated proxy is an ancestor of the tuple. Hence $\lambda_{\text{EW}} = \eta_{\text{source}}^2/(4\rho_{\text{EW}})$. Assume also the finite certificate of Definition 6.2 and the physical-domain inequalities $\alpha_2' > 0$, $\alpha_Y' > 0$. Then the beyond-selected-carrier transport map is uniquely

$$\tau_2^{\text{exact}} = -\lambda_{\text{EW}}\left(1 + \frac{2}{3}\eta_{\text{source}} + \left(1 - \frac{\rho_{\text{EW}}}{6}\right)\eta_{\text{source}}^2\right),$$

$$\delta n^{\text{exact}} = \lambda_{\text{EW}}\left(1 + \frac{4}{3}\eta_{\text{source}} + \left(2 - \frac{\rho_{\text{EW}}}{6}\right)\eta_{\text{source}}^2\right),$$

with fiber hypercharge transport

$$\tau_Y^{\text{fiber}} = -\frac{\tau_2^{\text{exact}} + 2\eta_{\text{source}}}{1 + 4(\tau_2^{\text{exact}})^2}.$$

The coherent transport shifts are

$$\delta\alpha_2 = \alpha_{2,m_Z}\tau_2^{\mathrm{exact}}, \qquad \delta\alpha_Y^{\parallel} = \alpha_{Y,m_Z}\frac{8\eta_{\mathrm{source}}(\tau_2^{\mathrm{exact}})^2 - \tau_2^{\mathrm{exact}}}{1+4(\tau_2^{\mathrm{exact}})^2},$$

$$\delta\alpha_Y^{\perp} = (\alpha_{2,m_Z} + \alpha_{Y,m_Z})\,\delta n^{\mathrm{exact}}.$$

With

$$\alpha_{2,*} = \alpha_{2,m_Z}, \qquad \alpha_{Y,*} = \alpha_{Y,m_Z}(1-2\eta_{\mathrm{source}}),$$

the coherent transport couplings are

$$\alpha_2' = \alpha_{2,*} + \delta\alpha_2, \qquad \alpha_Y' = \alpha_{Y,*} + \delta\alpha_Y^{\parallel} + \delta\alpha_Y^{\perp},$$

and they emit one coherent mass chart

$$M_W^{\mathrm{chart}} = v\sqrt{\pi\alpha_2'}, \qquad M_Z^{\mathrm{chart}} = v\sqrt{\pi(\alpha_2'+\alpha_Y')}, \qquad v = v_{\mathrm{chart}}(P).$$

$$a_0(P) := \frac{\alpha_{2,m_Z}(P) + \alpha_{Y,m_Z}(P)}{\alpha_{2,m_Z}(P)\alpha_{Y,m_Z}(P)}, \qquad s_0(P) := \frac{\alpha_{Y,m_Z}(P)}{\alpha_{2,m_Z}(P)+\alpha_{Y,m_Z}(P)}.$$

Thus the certificate fixes the candidate mass pair together with the running-family anchor (a_0, s_0, v) *from the source basis. The physical electromagnetic row is read from the Ward-projected unbroken* $\mathrm{U}(1)_Q$ *channel in Theorem 6.5.*

Proof. The first condition reduces every admissible response to the two displayed coordinates. The second fixes their common primitive activity. The third gives the charged and neutral path characters, including the common $\mathbb{Z}_6$ subtraction, and the fourth fixes their signs. The fibre functional

$$\mathcal{J}_Y(t) = \frac{1}{2}(1+4(\tau_2^{\mathrm{exact}})^2)t^2 + (\tau_2^{\mathrm{exact}} + 2\eta_{\mathrm{source}})t$$

is strictly convex, so its unique minimizer is $-(\tau_2^{\mathrm{exact}} + 2\eta_{\mathrm{source}})/(1+4(\tau_2^{\mathrm{exact}})^2)$. Substitution gives the repaired couplings and the one-Higgs mass chart. The exhaustiveness condition excludes every output-changing admissible deformation, which proves uniqueness on the certified class. □

Using the same declared electroweak validation basis

$$\begin{aligned}
\alpha_{2,m_Z} &= 0.03377843630219015,\\
\alpha_{Y,m_Z} &= 0.010131601067241624,\\
\alpha_U &= 0.04112498041477454,\\
\eta_{\mathrm{source}} &= 0.022147000871961295,\\
v &= 246.76711732749683\ \mathrm{GeV},
\end{aligned}$$

the conditional quotient-transport law evaluates to

$$M_W^{\mathrm{chart}} = 80.37700001539531\ \mathrm{GeV}, \qquad M_Z^{\mathrm{chart}} = 91.18797807794321\ \mathrm{GeV},$$

$$\begin{aligned}
a_0 &= 128.30576920234813,\\
s_0 &= 0.23073542347506173.
\end{aligned}$$

This specialization verifies the candidate value-law algebra. It does not derive the finite quotient-path certificate, establish that this calibration tuple belongs to the source-only pixel branch, or identify W, Z as complex propagator poles. A proof requires the finite path lists, quotient canonicalizer, exact incidence and central-trace checks, fibre Gram calculation, deformation enumeration or positive selector gap, and a same-branch target-independence proof.

Proposition 6.4 (Local nondegeneracy of the two-output mass chart)**.** *Define*

$$n_{\text{fib}}(\tau_2) := 1 + \frac{\alpha_2\tau_2 + \alpha_Y\tau_Y(\tau_2)}{\alpha_2 + \alpha_Y}.$$

On the physical domain $1+\tau_2>0$ *and* $n_{\text{fib}}(\tau_2)+\delta n>0$*,*

$$\det \frac{\partial(M_W^{\text{chart}}, M_Z^{\text{chart}})}{\partial(\tau_2,\delta n)} = \frac{M_W^{\text{chart}} M_Z^{\text{chart}}}{4(1+\tau_2)\,[n_{\text{fib}}(\tau_2)+\delta n]} > 0.$$

Holding $(\alpha_2, \alpha_Y, v, \eta)$ *fixed, the two-output chart is locally nondegenerate. This determinant neither proves the two-coordinate response condition nor excludes an additional physical response channel. Within the displayed two-coordinate chart, the missing content is a source selector rather than another fit coordinate.*

Proof. M_W^{chart} is independent of δn, while

$$\frac{\partial M_W^{\text{chart}}}{\partial \tau_2} = \frac{M_W^{\text{chart}}}{2(1+\tau_2)}, \qquad \frac{\partial M_Z^{\text{chart}}}{\partial \delta n} = \frac{M_Z^{\text{chart}}}{2[n_{\text{fib}}(\tau_2)+\delta n]}.$$

The determinant is the product of these diagonal entries. □

Theorem 6.5 (Ward-projected $\mathrm{U}(1)_Q$ transport law and Thomson-limit electromagnetic readout)**.** *Let the source-only running family be fixed by the forward pixel solve on the declared running/matching/threshold/scheme surface, with source basis*

$$Q_{\text{src}}(P) = \big(\alpha_U(P), \alpha_{2,m_Z}(P), \alpha_{Y,m_Z}(P), \eta_{\text{source}}(P), v_{\text{chart}}(P)\big).$$

Assume the realized electric-charge branch, so that

$$Q = T_3 + Y$$

on the realized low-energy branch and color-singlet physical states carry integer Q-charge. Assume further that the post-selector electroweak transport object is the populated quotient-local kernel

$$K_{\text{src}}^{\text{EW}}(q^2;P) = \{\Pi_{AA}(q^2;P), \Pi_{AZ}(q^2;P), \Pi_{ZZ}(q^2;P), \Pi_{WW}(q^2;P)\}$$

on the declared physical observable algebra, together with a Ward projector

$$\mathcal{W}_Q : K_{\text{src}}^{\text{EW}}(q^2;P) \longrightarrow \Pi_Q(q^2;P)$$

onto the unbroken electromagnetic channel satisfying at $q^2=0$

$$\mathcal{W}_Q[\Pi_{AZ}(0;P)] = 0, \qquad \mathcal{W}_Q[\Pi_{AA}(0;P)] \neq 0.$$

Assume the abelian projected edge-sector probabilities obey the $\mathrm{U}(1)$ *heat-kernel law*

$$p_n(q^2;P) \propto e^{-t_Q(q^2;P)\lambda_n}, \qquad \lambda_n = n^2,$$

equivalently

$$g_Q^2(q^2;P) = \frac{t_Q(q^2;P)}{2\pi},$$

and that the scalar readout uses the same source family, declared renormalization scheme, and originating electroweak kernel throughout. Then the unique electromagnetic coupling readout on the Ward-projected branch is

$$\alpha_{\rm em}^{-1}(q^2; P) = \frac{8\pi^2}{t_Q(q^2; P)}.$$

If

$$a_0(P) := \alpha_{\rm em}^{-1}(m_Z^2; P),$$

then the unique zero-normalized affine scalar on the same source-locked family is

$$\delta_\alpha(q^2; m_Z^2; P) := \frac{t_Q(m_Z^2; P)}{t_Q(q^2; P)} - 1,$$

so that

$$\alpha_{\rm em}^{-1}(q^2; P) = a_0(P)\big(1 + \delta_\alpha(q^2; m_Z^2; P)\big).$$

Hence the Thomson-limit electromagnetic readout on that same Ward-projected electromagnetic kernel:

$$\alpha_{\rm Th}^{-1}(P) := \lim_{q^2 \to 0} \alpha_{\rm em}^{-1}(q^2; P) = a_0(P)\, \frac{t_Q(m_Z^2; P)}{t_Q(0; P)}.$$

Consequently, the low-energy electromagnetic row on this branch is read as the Thomson endpoint of the electroweak transport family instead of as a mass-chart byproduct.

Proof. The realized electric-charge branch fixes the charge operator by $Q = T_3 + Y$, and the quotient-protected charge theorem fixes the physical color-singlet Q-lattice to be integral. The abelian edge-sector theorem supplies the heat-kernel extraction rule

$$p_n \propto e^{-t_Q \lambda_n}, \qquad g_Q^2 = \frac{t_Q}{2\pi},$$

so the Ward-projected Q-channel determines a unique transport time $t_Q(q^2; P)$. Because the Ward projector kills A - Z mixing at $q^2 = 0$, the selected channel is the physical unbroken electromagnetic channel. Then

$$\alpha_{\rm em} = \frac{g_Q^2}{4\pi} = \frac{t_Q}{8\pi^2},$$

hence

$$\alpha_{\rm em}^{-1} = \frac{8\pi^2}{t_Q}.$$

Evaluating at $q^2 = m_Z^2$ defines the source-locked anchor $a_0(P)$, and dividing by the same expression at general q^2 yields

$$\delta_\alpha = \frac{t_Q(m_Z^2)}{t_Q(q^2)} - 1.$$

The provenance-equality clause forces one running-family source, one frozen scheme, and one origin kernel for all scalar readouts, so no mixed-family or Z-only surrogate is promotable. The Thomson formula is the $q^2 \to 0$ limit of the same identity. □

Corollary 6.6 (Conditional Ward-projected Maxwell normalization)**.** *On the same Ward-projected* $\mathrm{U}(1)_Q$ *branch, assume in addition the low-energy Maxwell action with the displayed nonzero kinetic normalization. Then the electromagnetic field strength and current obey*

$$F_Q = dA_Q, \qquad dF_Q = 0, \qquad d{*}F_Q = g_Q^2(q^2; P){*}J_Q,$$

with

$$g_Q^2(q^2;P) = \frac{t_Q(q^2;P)}{2\pi}, \qquad \alpha_{\mathrm{em}}^{-1}(q^2;P) = \frac{8\pi^2}{t_Q(q^2;P)}.$$

At the Thomson endpoint, this is the Maxwell normalization used by the public fine-structure row.

Proof. The compact reconstruction supplies the unbroken abelian electromagnetic connection direction $\mathrm{U}(1)_Q$ with charge generator $Q = T_3 + Y$, but does not supply the Maxwell kinetic action. On an abelian factor the compact-gauge curvature restricts to $F_Q = dA_Q$, so $d^2 = 0$ gives $dF_Q = 0$. Varying the quadratic electromagnetic action in the OPH coupling convention gives $d{*}F_Q = g_Q^2 {*}J_Q$. The theorem above identifies the same branch coupling as $g_Q^2 = t_Q/(2\pi)$, hence $\alpha_{\mathrm{em}}^{-1} = 8\pi^2/t_Q$. □

Corollary 6.7 (Source-only admissibility criterion on the Ward-projected electroweak branch)**.** *The Ward-projected electroweak branch supports a source-only fine-structure row only when the populated Ward-projected transport kernel satisfies*

$$\lim_{q^2\to 0} \alpha_{\mathrm{em}}^{-1}(q^2;P) = \alpha_{\mathrm{Th}}^{-1}(P),$$

on the same source-locked family and without any separate endpoint value feeding that branch. Concretely, the evidence packet must contain the source-derived Ward-current spectral measure, source Jacobi kernel, electromagnetic normalization, same-scheme finite remainder, no-target-leak dependency graph, full pixel-map contraction interval, and a prediction interval narrower than the declared direct-measurement interval. Without those objects, the public endpoint is an empirical hadron-closure row.

This criterion is the two-current marginal of the larger source-derived hadronic spectral backend. It is adequate for the Thomson endpoint and HVP branch only if the same source law also records the QCD quotient ensemble, source parameter map, Ward current normalization, systematics accounting, and no-target-leak DAG. Full hadronic precision claims require the higher-point and transition spectral exports from that same backend rather than reusing the scalar fine-structure residual.

The endpoint table records the source-locked anchor from the runtime candidate,

$$a_0(P) = 128.307965473286248209961108741756 7\ldots,$$

and the OPH-plus-empirical hadron-closure value

$$\alpha_{\mathrm{emp}}^{-1}(0) = 136.3827548174946, \qquad \alpha_{\mathrm{emp}}^{-1}(0) \in [136.3670480603, 136.3984651934].$$

The source derivation contains no built-in reference inverse-α value. The measured endpoint 137.035999177(21) is excluded as a source-only closure-solve input and is a separate comparison coordinate. The source-side calculation emits the certified value $\alpha_{\mathrm{root}}^{-1} = 136.994835177413\ldots$, so its difference from the endpoint readout is an endpoint-and-matching comparison packet. This certificate establishes a unique root of the incomplete declared numerical map. It does not derive a relation between that root and the physical Thomson endpoint.

Remark 6.8. *The empirical closure row is a Thomson-limit inverse fine-structure evaluation using external hadron data. The source anchor* $a_0(P) = 128.3079654732862482099611087417567\ldots$ *is the electroweak-scale running-family value at* m_Z^2*, while the measured row is separate. The 2022 CODATA/NIST reference value is* $\alpha^{-1} = 137.035999177(21)$ *[8].*

The measured-reference inverse repair surface provides diagnostic validation and carries no predictive force. The companion carrier-side pair

$$\alpha^{-1}_{\mathrm{em},\dagger} = 132.6344411210187,$$
$$\sin^2\theta_{W,\dagger} = 0.22333729871365943$$

is mass-chart bookkeeping on that target-conditioned validation surface. It is not the physical electromagnetic readout on the Ward-projected $\mathrm{U}(1)_Q$ branch.

The electroweak quantitative branch contains an underdetermination theorem for the quadratic repair family, a color-balanced descent, a candidate mass emitter, and the Ward-projected electromagnetic transport theorem above the source-locked running-family anchor. All of them use one source family, one declared renormalization scheme, and one originating electroweak kernel. The forward transmutation calculation reconstructs the same $\alpha_U(P)$, $t_U(P)$, and $t_{\mathrm{tr}}(P)$ as the pixel-closure solve without reading them back from measured couplings. This separation does not establish target independence. This construction supplies neither the finite quotient-path certificate nor a physical pole.

6.2 Hierarchy interpretation on the local transmutation branch

The selected branch certifies a dimensionless electroweak hierarchy/naturality relation. The large cosmic horizon ratio and the electroweak hierarchy belong to different OPH branches. The proposed global capacity branch N_{CRC} comes from the direct correctable-public-record map. Its de Sitter interpretation and electroweak identification require the horizon–record identification and the common screen/electroweak load-carrier identification, respectively. The local pixel/transmutation readout is the ratio

$$\frac{v}{E_\star} = P_\star^{-1/2}\exp\!\left[-\frac{2\pi}{4\alpha_U(P_\star)}\right].$$

The integer 4 has an exact representation-theoretic witness on the selected exterior package: three color copies of the weak doublet Q plus one lepton doublet L. Identifying this multiplicity with the transmutation coefficient is conditional on an intertwiner between the port and weak carriers and on equality of their normalized load traces. It is not the $SU(2)$ beta-function coefficient, and its physical load interpretation cannot be inferred from the dimension count alone. The hierarchy certificate records the interval

$$I_U = [0.041123336195630494,\ 0.041125336195630496],$$

the Krawczyk inclusion

$$K(I_U) \subset [0.0411243357185544983,\ 0.0411243366727064662] \subset \mathrm{int}(I_U),$$

and a strictly negative derivative enclosure $[-10.995768, -10.985284]$. Thus the local source equation has a unique zero in that enclosure. On the public endpoint branch, at the CODATA-located comparison pixel P_C,

$$\alpha_U(P_C) = 0.041124336195630495,$$
$$\frac{v}{E_\star} = 2.0199803239725553 \times 10^{-17}.$$

The source-only branch excludes the public Thomson endpoint upstream and has certified forward pixel $P_{\mathrm{fwd}} = 1.630972095858897\ldots$. These two named branches must not be merged. The dimensionless ratio is generated by the ordinary exponential transmutation factor controlled by the unified diffusion coupling; a weak scale in GeV additionally requires an independently source-closed $E_\star$.

This is also the OPH reading of Higgs naturalness on the declared electroweak/criticality surface. The observer-visible scalar mass is the normal-form readout

$$m_H = H_{\mathrm{OPH}}(P_\star)$$

on the selected branch. The split into a bare Higgs mass plus a cutoff correction is a regulator coordinate split outside the observer-facing physical variable. A scalar relevant deformation absent from the retained branch constraints is off-branch; a scalar deformation present on the selected branch has its value fixed by the branch equations. The claim inherits the electroweak/criticality quantitative surface and the $\mathcal{R}_U$ precision policy. The selected exact source-to-Higgs coarse-graining square has $\epsilon_H = 0$, with $\epsilon_H \in [0,0]$, as stated below.

6.3 Local/global resonance continuation for the hierarchy

The local/global hierarchy construction compares the electroweak transmutation exponent with a proposed global capacity branch at the same joint pair $(P_\star, N_{\mathrm{CRC}})$. Conditional on an executed direct public-record fixed point and the common screen/electroweak load-carrier identification, the target relation is

$$t_{\mathrm{tr}}(P_\star) = \frac{P_\star}{12}\log\left(\frac{N_{\mathrm{CRC}}}{\pi}\right),$$

or equivalently

$$\frac{v}{E_{\mathrm{cell}}} = \left(\frac{N_{\mathrm{CRC}}}{\pi}\right)^{-P_\star/12}.$$

This packages the mathematical electroweak hierarchy bridge as a resonance between the local declared-map root and a conditional global screen-capacity coordinate. That screen branch supplies twelve curvature ports. The exterior package supplies four weak-doublet copies, and the common screen/electroweak load-carrier map is the physical identification that makes their additive load the screen load. The oriented 24-slot register and product-adjoint count are independent bookkeeping facts and carry no load or clock implication.

For comparison, the observer-visible product adjoint has the independent count. On the realized product branch,

$$m_{\mathrm{rep}} = 2\dim(\mathfrak{su}(3)\oplus\mathfrak{su}(2)\oplus\mathfrak{u}(1)) = 2(8+3+1) = 24.$$

The factor 2 orients each product-adjoint channel and does not follow from the screen register. This count does not enter the direct capacity or common-load proof. The $\mathfrak{su}(5)$ adjoint has the same single-orientation integer for a different support: its X/Y mixed gauge channels are absent from the OPH product branch.

After whole-fiber scalarization and faithful capacity-carrier representation, the direct active-capacity map is deflationary. It therefore has no positive two-sided Banach attractor on a domain containing smaller capacities. Under confusability-reflecting capacity extension, however, iteration from the declared maximum boundary dimension stabilizes finitely at the greatest fixed point. This order-theoretic selection, imported from the synthesis paper, replaces the smooth tick-map alternative. A finite-size slack law with one physical zero is a stronger open receipt. An independently produced ρ_{op} remains available as a commuting-square test against $\log M_0$; no derivative is intrinsic at finite integer dimension.

Theorem 6.9 (Direct public-capacity/common-load electroweak bridge)**.** *Assume $N_\star = \log M_0(\mathfrak{U}_{N_\star})$ is supplied by the independent correctable-public-record closure. Assume the screen-sieve branch independently supplies*

$$\Gamma_{\rm scr} = \frac{P}{12}\log\left(\frac{N_\star}{\pi}\right),$$

the common screen/electroweak load-carrier map identifies $\Gamma_{\rm scr} = \log(E_{\rm cell}/v)$, and the independent electroweak source relation gives $\log(E_{\rm cell}/v) = \pi/[2\alpha_U(P)]$. Then the bridge residual

$$\mathcal{B}_{\rm EW}(P, N_\star) := \alpha_U(P)\log(N_\star/\pi) - \frac{6\pi}{P} = 0.$$

Equivalently, the bridge coordinate is

$$N_{\rm EW}(P) = \pi\exp\left[\frac{6\pi}{P\alpha_U(P)}\right].$$

For the public endpoint branch,

$$N_{\rm EW}(P_\star) = 3.532354622692990651118751296233054760046\,2 \times 10^{122}.$$

For the bridge-map coordinate, conventionally labeled

$$N^{\rm EW}_{\rm CRC} = \pi\exp\left[\frac{6\pi}{P_\star\alpha_U(P_\star)}\right],$$

one has $\mathcal{B}_{\rm EW}(P_\star, N^{\rm EW}_{\rm CRC}) = 0$, and hence

$$\frac{v}{E_{\rm cell}} = \left(\frac{N^{\rm EW}_{\rm CRC}}{\pi}\right)^{-P_\star/12}.$$

The exterior representation witness gives the exact weak-doublet multiplicity $3 + 1 = 4$, and every additive isomorphism-invariant load normalized by $L_P(\mathbb{C}) = P$ gives $L_P(\mathbb{C}^4) = 4P$. Identifying this abstract weak load with the public screen load is exactly the common-carrier receipt. The notation $N^{\rm EW}_{\rm CRC}$ does not identify this mathematical coordinate with a cosmic readback fixed point. That identification requires the source-derived direct public-record producer and the common screen/electroweak load-carrier map. The Planck-Λ central capacity $N_\Lambda \simeq 3.313\times 10^{122}$ is about 6.6 percent lower than the bridge coordinate.

Proof. Equating the screen and electroweak load expressions gives $P\log(N_\star/\pi)/12 = \pi/[2\alpha_U(P)]$. Rearrangement gives $\mathcal{B}_{\rm EW}(P, N_\star) = 0$. Substitution gives the displayed $N_{\rm EW}(P)$ and the hierarchy identity. ☐

Theorem 6.10 (Icosahedral screen sieve and gated capacity composition)**.** *On the certified echosahedral carrier lineage of Ref. [4], assume twelve primitive equal-trace port-center atoms, a declared integer atom-counting grammar on the total-twelve fiber, the normalized central-readback Hilbert–Schmidt cost, oriented $(12, 30, 20)$ incidence, and refinement-cocycle data. Then the counting theorem gives twelve unit ports. The incidence theorem gives their fixed-point-free inverse pairing, the proper A_5 action, and the regular rank-three icosahedral frame, naturally under refinement and relabeling. An equal-weight invariant load X is therefore read locally as $X/12$. For $X = \log(N/\pi)$, local cell entropy $P/4$, and electroweak multiplicity $\beta_{\rm EW} = 4$, define the screen composition coordinate*

$$\Gamma_{\rm scr} = \beta_{\rm EW}\frac{P}{4}\frac{1}{12}\log(N/\pi) = \frac{P}{12}\log(N/\pi).$$

If, in addition, the common screen/electroweak load-carrier map identifies $\Gamma_{\rm scr} = \log(E_{\rm cell}/v)$ *and matches it to the source transmutation exponent* $\pi/(2\alpha_U(P))$*, then*

$$\log(E_{\rm cell}/v) = \frac{P}{12}\log(N/\pi), \qquad \alpha_U(P)^{-1} = \frac{P}{6\pi}\log(N/\pi),$$

which is equivalent to the electroweak bridge residual $\mathcal{B}_{\rm EW}(P,N) = 0$*.*

Proof. The declared counting-realization identity

$$H(q) = 12 + \sum_p (q_p - 1)^2$$

gives the unique all-one allocation with exact gap two. The oriented incidence has distance profile $(1,5,5,1)$, positive automorphism group A_5, and exact Gram matrix $G^2 = 4G$ of rank three, as proved and checked in Ref. [4]. Edge-center collars expose the twelve primitive atoms as central ports. Equal-weight additivity gives $X/12$. Multiplying by $P/4$ and $\beta_{\rm EW} = 4$ gives $\Gamma_{\rm scr} = (P/12)\log(N/\pi)$. The additional identification receipt and the source transmutation equation give the remaining displays. □

The all-one split and gap are exact within the declared integer counting and normalized readback-cost realization. The integer loads also have an operational finite realization. Append-only signed atomic events generate the register values, and an admissible repair transfers one whole unit across a seam with load difference $d \geq 2$. For

$$V(q) = \sum_p q_p^2$$

each such move gives the exact decrement $\Delta V = -2(d-1) < 0$. Hence V proves termination. The minimum number of admissible moves is a derived settling cost, natural under every declared carrier rotation and refinement. A half-unit display rescales the readback units of the same event graph; it does not implement half of an atomic event.

The load-square Lyapunov function V, the seam quadratic, the normalized readback cost H, and the Hessian associated with the third axiom are distinct objects. The positive readback scale remains a unit convention. The proper-A_5 result follows independently from oriented incidence on the declared echosahedral lineage. The port/weak-carrier intertwiner, normalized port-load trace equality, and common screen/electroweak load-carrier map remain physical branch premises. The exact exterior multiplicity alone does not discharge the physical load identity.

This theorem is local to one certified carrier lineage. It does not identify that carrier with a primitive observer, construct the federation nerve, or derive a global S^2 support screen. Those claims require the independent access-and-record, overlap, carrier-to-screen, and refinement receipts stated upstream.

Theorem 6.11 (EW bridge-map fixed point)**.** *Define the log-capacity map*

$$\mathcal{C}_{\rm EW}(P,x) = (1-\lambda)x + \lambda\frac{6\pi}{P\alpha_U(P)}, \qquad 0 < \lambda \leq 1.$$

For fixed $P = P_\star$ *and* $\lambda = 1/2$*, this is a contraction with Lipschitz constant* $1/2$*. Its unique fixed point is*

$$x_\star = \frac{6\pi}{P_\star\alpha_U(P_\star)}, \qquad N_{\rm CRC}^{\rm EW} = \pi e^{x_\star}.$$

Therefore

$$\mathcal{B}_{\mathrm{EW}}(P_\star, N^{\mathrm{EW}}_{\mathrm{CRC}}) = 0.$$

This theorem concerns the deliberately defined map $\mathcal{C}_{\mathrm{EW}}$, not the physical direct public-record producer. Identifying the output with cosmic capacity requires construction of that producer and the common screen/electroweak load-carrier map. The Planck-Λ central value 3.313×10^{122} differs by about 6.6 percent.

Proof. The map has derivative $1-\lambda$ in x, hence is a contraction for $0 < \lambda \leq 1$. Solving $x = \mathcal{C}_{\mathrm{EW}}(P_\star, x)$ gives the displayed $x_\star$. Substitution into $\mathcal{B}_{\mathrm{EW}}$ gives zero. □

Theorem 6.12 (Conditional RG/Higgs naturality identity)**.** *On a declared hierarchy branch satisfying the selected-surface premises, let Q_s be the source hierarchy quotient and Q_H the Higgs/electroweak quotient. Define*

$$\rho_{sH}([x]_s) = [P(x), N(x), \Theta(P(x), N(x)), \Pi_{HT} F_{\mathrm{crit}} F_{\mathrm{EW}}(P(x), N(x)), 0]_H\,,$$

with

$$\Theta(P, N) = \left(\frac{N}{\pi}\right)^{-P/12}.$$

Here F_{EW} denotes the declared electroweak map and F_{crit} denotes the criticality map. For the corresponding normal-form maps and obstruction maps,

$$\rho_{sH} n_s = n_H \rho_{sH}, \qquad \chi_{sH} h_s = h_H \rho_{sH}.$$

Consequently,

$$\varepsilon_H = \max\{\varepsilon^n_{sH}, \varepsilon^h_{sH}\} = 0, \qquad \varepsilon_H \in [0, 0].$$

Proof. The two routes through the source and Higgs quotients preserve the same tuple

$$(P, N, \Theta(P, N), \Pi_{HT} F_{\mathrm{crit}} F_{\mathrm{EW}}(P, N)).$$

A hidden scalar relevant deformation is not an independent physical input on the selected branch: if absent from the retained finite constraint family, it is off-branch by refinement stability; if retained, its value is fixed by the branch equations and belongs to the displayed Higgs/top coordinate. Both controlled squares commute exactly, so the product defect is zero. □

The conditional RG/Higgs naturality map does not itself use the measured weak scale. Measured W/Z, Higgs/top, G, Planck-area, and Λ data are excluded by its declared input contract. The local/global package records the direct-capacity/common-load electroweak projection bridge, mathematical EW bridge-map fixed point, conditional operational-readback implications, and RG naturality square. Conditional on the declared branch, this certifies the dimensionless electroweak hierarchy/naturality identities; it does not construct the physical public-record producer or establish the common screen/electroweak load-carrier identification. The strict source-root certificate is open, and the result does not independently attach $E_\star$ to a physical GeV scale. The finite producer imported from the synthesis paper is

$$\mathfrak{F}_{r,\varepsilon}(D) = \{M_\varepsilon(q) : q \in \widetilde{\Omega}_{r,D}\}, \qquad M_0(q) = \alpha(G_q),$$

with the public checkpoint packet, capacity-carrier representation, whole-fiber scalarization, confusability-reflecting extension/refinement, finite-size slack law, and fixed-point receipts kept explicit. The operational resolution is an independent test. The conditional local carrier has twelve ports and an oriented 24-slot register; the latter supplies no four-load conclusion. The mathematical local/global hierarchy-resonance package is a conditional consistency surface. It does not construct the physical cosmic readback producer. The hierarchy result does not use the local/global resonance as an input.

6.4 Higgs/top critical stage

The Higgs/top critical stage is conditional on the electroweak gauge core. It is described by a downstream split law together with a Jacobian readout map on the declared running, matching, and threshold surface. Write the coupling asymmetry as ρ_{EW}, distinct from the transmutation coefficient $b_{\mathrm{tr}} = 4$. The candidate electroweak tuple

$$(\eta_{\mathrm{source}}, \rho_{\mathrm{EW}}, \lambda_{EW}, \tau_{2,\mathrm{tree}}^{\mathrm{exact}}, \delta n_{\mathrm{tree}}^{\mathrm{exact}})$$

emits the shared scalar

$$\rho_{HT} = \log(1 + \tau_{2,\mathrm{tree}}^{\mathrm{exact}})$$

and the declared residual formulas

$$R_T = -\tau_{2,\mathrm{tree}}^{\mathrm{exact}} \eta_{\mathrm{source}}^2 + \left(1 + \frac{\rho_{\mathrm{EW}}}{28}\right) \eta_{\mathrm{source}}^6 + \frac{\eta_{\mathrm{source}}^8}{14} + \frac{\eta_{\mathrm{source}}^9}{27},$$

$$R_H = \eta_{\mathrm{source}}^5 - \frac{3}{25} \eta_{\mathrm{source}}^6 + \frac{\lambda_{EW} \eta_{\mathrm{source}}^6}{18} + \frac{\eta_{\mathrm{source}}^8}{2\rho_{\mathrm{EW}}}.$$

The split coordinates are

$$\pi_y = \frac{\eta_{\mathrm{source}} + \left(\frac{3}{2} + \frac{\rho_{\mathrm{EW}}}{4}\right) \rho_{HT} + R_T}{\sqrt{\pi}}, \qquad \pi_\lambda = \frac{\eta_{\mathrm{source}} - \left(\frac{4}{3} - \frac{\rho_{\mathrm{EW}}}{54}\right) \rho_{HT} + R_H}{\sqrt{\pi}},$$

so the declared criticality Jacobian reads out

$$\delta y_t(\mu_t) = \pi_y \, y_t^{\mathrm{core}}(\mu_t), \qquad \delta\lambda(\mu_t) = -\frac{16}{9} \pi_\lambda \, \lambda^{\mathrm{core}}(\mu_t).$$

These equations define a physical readout only if a finite split-character certificate supplies one strict source branch and a fixed pre-split carrier, an exact two-coordinate quotient response, explicit characters deriving $\rho_{HT}, R_T, R_H, \pi_y, \pi_\lambda$ and $-16/9$, oriented mismatch descent with canonical normalization and positivity, and an exhaustive deformation quotient proving rigidity or a target-free positive-gap selector. The displayed formulas do not supply that certificate. This gives

$$m_H = 125.1995304097179 \text{ GeV}, \qquad m_t^{\mathrm{crit}} = 172.3523553288312 \text{ GeV}.$$

This is a coordinate on the declared electroweak/criticality Jacobian surface, obtained by back-solving from the measured pair through the synchronization-scale scan, so it is an exact interpolation of the PDG pair, a target-anchored fit; it carries no certified Higgs or top complex pole. The target-free headline is the double-criticality family evaluated below. The same surface emits a companion top coordinate. An exact negative result bounds this branch from below: at fixed P, with $u = 1 - P/24$, the exposed target-free reduct admits both the linear ($y_t = u$) and Born ($y_t = \sqrt{u}$) probability-to-amplitude completions, which agree on every reduct field and differ in the leading Higgs and top pole ratios by $2u(1-u)$ and $u(1-u)/2$. The reduct is therefore not P-complete for the Higgs/top readout. A physical conclusion requires an explicit source object that selects the amplitude lift. The Standard Model gauge paper states and proves this two-completion theorem. The selected-class quark support witness carries a target-anchored running-top comparison row using the PDG 2025 cross-section entry; it is not a separate public source-only top prediction. The bridge to the auxiliary direct-top PDG row is closed as a source-domain codomain no-go; the auxiliary row is compare-only. The one-scalar companion seed

$$\sigma_{\mathrm{crit}} = \frac{\alpha_U \cos(2\theta_{W0})}{\sqrt{\pi}}$$

defines the fixed-ray companion branch with $\pi_y = \pi_\lambda = \sigma_{\mathrm{crit}}$. Separately, the same criticality Jacobian admits a compare-only inverse slice that hits the canonical Higgs/top reference pair exactly: it is an exact interpolation of the PDG pair, and the landing is the defining condition of the slice rather than a derived result. That exact comparison is useful for bookkeeping and does not define the forward branch used for the public rows. The Higgs boson belongs naturally in the boson discussion. The quark-family chapter treats the top as the third-generation up-type quark. The top coordinate reported here is the conditional criticality coordinate on this surface, not a separate public top-mass prediction row. The selected-class quark support wrapper carries only a target-anchored comparison witness under the declared source-only rule.

The local implication from the candidate electroweak tuple through the displayed criticality formulas is executable and single-valued, and this arithmetic grade is certified from raw interval data: the input box contains the eleven declared branch inputs (the electroweak tuple and the criticality core/Jacobian constants, each with units or dimensionless normalization and provenance), the outward-rounded interval extension of every displayed node, the Jacobian interval enclosure over the full box, and a non-singular diagonal readout block with determinant bounded away from zero; the certified statement is scoped to the declared surface and makes no criticality-system existence or uniqueness claim. A physical Higgs mass requires a source root, independent scale, finite quotient-path selection, same-scheme running, split rigidity, top and threshold transport, a complex pole, an uncertainty enclosure, and target-independent provenance. The exact inverse slice is a compare-only validation surface.

6.5 Source-closure and physical-pole envelope

The preceding formulas end at running or declared-surface coordinates. A physical $W/Z/H$ theorem requires the following additional composition.

Proposition 6.13 (Conditional physical clock attachment)**.** *If the same strict source branch emits the dimensionless energy $\varepsilon_{\mathrm{clk}} > 0$ of a declared clock transition whose physical frequency ν_{clk} defines the operational unit, then, with $h_{\mathrm{P}} = 2\pi\hbar$ Planck's constant,*

$$E_\star = \frac{h_{\mathrm{P}}\nu_{\mathrm{clk}}}{\varepsilon_{\mathrm{clk}}}.$$

Equivalently, $\gamma_\star = \ell_\star \nu_{\mathrm{clk}}/c$ gives $E_\star = \hbar\nu_{\mathrm{clk}}/\gamma_\star$. An operator-norm perturbation of at most ϵ changes an isolated selected clock gap by at most 2ϵ. This is a conversion and stability theorem; the required physical reference transition and SI binding are absent.

Proposition 6.14 (Exact finite source-Hamiltonian gap)**.** *Let D_σ be the sign-twisted seam derivative on the connected observer-visible finite local domain, and let $H_\sigma = D_\sigma^* D_\sigma$ use the declared unit-counting measure. The signed-incidence formula makes H_σ integer symmetric and positive semidefinite. A kernel vector would supply a sign-consistent coloring on each connected component. The declared reversing seam transport frustrates every triangle, so the connected complex has no such coloring and the signed incidence rank is* 8,662*. Therefore* $\ker H_\sigma = 0$*. With maximum degree* 12*, the determinant bound gives*

$$\lambda_{\min}(H_\sigma) \geq 24^{-8661} > 0.$$

A sparse numerical refinement gives $\lambda_{\min} = 0.11753670336118684$ with relative residual 1.49×10^{-14}. The decimal is measured rather than certified. The exact statement is strict positivity and the displayed floor.

This source-derived dimensionless gap does not select a physical field fiber or laboratory transition. It therefore supplies no ν_{clk}, SI energy, particle mass, or determinant-line attachment. A sign-consistent four-cycle control has an exact zero mode, so positivity is not hardwired into the instrument [5].

The atomic component of that packet has an exact scalar reduction. Let $P = P_3 + P_4$ project onto the isolated zero-field $6S_{1/2}$, $I = 7/2$ cesium ground manifold, $\dim V_3 = 7$, $\dim V_4 = 9$, and let $Q = 1 - P$. On every real interval where $Q(\widehat{H} - z)Q$ is invertible, define the Feshbach–Schur operator $\mathcal{F}_P(z) = P(\widehat{H} - z)P - P\widehat{H}Q\,[Q(\widehat{H} - z)Q]^{-1}Q\widehat{H}P$.

Proposition 6.15 (Cesium channel scalarization and monotone clock roots)**.** *$z \in \operatorname{spec}\widehat{H}$ exactly when $0 \in \operatorname{spec}\mathcal{F}_P(z)$, with matching multiplicities. Rotational invariance and the multiplicity-free decomposition $V_3 \oplus V_4$ force $\mathcal{F}_P(z) = f_3(z)P_3 + f_4(z)P_4$, and $\mathrm{d}\mathcal{F}_P/\mathrm{d}z \preceq -P$ gives $f_F'(z) \leq -1$. One sign bracket per channel therefore isolates exactly one level, and the clock normal form is $H^{\mathrm{eff}}_{\mathrm{clock}} = E_0 P + \widehat{a}_{\mathrm{Cs}}\,\mathbf{I}\cdot\mathbf{J}$ with $E_0 = (9E_4 + 7E_3)/16$ and $\varepsilon_{\mathrm{Cs}} = E_4 - E_3 = 4\widehat{a}_{\mathrm{Cs}}$.*

The public atomic packet may therefore export two monotone scalar interval evaluators with resolvent certificates and sign brackets in place of a correlated 55-electron matrix. The open clock parents are the atomic-scheme electromagnetic convention, the absolute electron ratio $m_e c^2/E_\star$, the cesium nuclear mass/spin/current/Compton packet, the two scalar evaluators, summable refinement tails, and the no-target provenance freeze. A non-entailment theorem fixes the direction of effort: two source extensions that agree on every named object and differ in one unfixed parent produce different unique gaps, so no unique $\varepsilon_{\mathrm{Cs}}$ is a theorem emitted by the source, and further evaluation of the existing formulas cannot close the clock branch.

Proposition 6.16 (Operational scale metrology)**.** *In SI units $E/\mathrm{GeV} = [E/(h\nu_{\mathrm{Cs}})]\,[h\nu_{\mathrm{Cs}}/\mathrm{GeV}]$, and the second factor is an exact unit conversion. A source-only numerical GeV mass is therefore equivalent to a source-only dimensionless operational clock ratio. Replacing cesium by another clock species adds the burden of predicting that clock's frequency ratio to the defining cesium transition.*

The stored clock-gap candidate reproduces the displayed gravitational coupling to a relative agreement of order 10^{-49} through the clock–gravity identity, so the decimal is a calibration checksum rather than a prediction. The selection mechanism for source laws themselves is closed separately by a source-action rigidity theorem: on a finite quotient with a faithful reference measure, an affinely independent feature basis, and a source-emitted moment vector, the maximum-entropy law is unique, every feasible competitor pays a positive relative-entropy gap with the Pinsker lower bound, and the surviving action freedom consists of additive constants, exact feature redundancies, and BRST-exact terms. The physical Standard-Model operator basis and its moment vector $c_r(P_\star, N_\star)$ are the open inputs of that mechanism.

Proposition 6.17 (Unique frozen RG and threshold transport)**.** *Let the canonically normalized coefficient vector $X(\mu)$ obey a locally Lipschitz beta system on each interval of a frozen finite threshold list. Assume deterministic matching maps, fixed loop order, field content, renormalization scheme, threshold locations, and decoupling order. Then one source initial condition determines one low-energy coefficient vector. If L_j bounds the beta-system Lipschitz constant and K_j bounds threshold map j, perturbations obey the corresponding product of $e^{L_j \Delta t_j}$ and K_j, plus the declared truncation and matching remainders.*

Proof. Picard–Lindelöf gives existence and uniqueness between thresholds. Grönwall bounds perturbations on each interval; applying each deterministic Lipschitz threshold map and inducting through the ordered list gives the stated composition. $\square$

The printed running/matching packet is a declared-convention contract. It does not satisfy this source theorem because its beta provenance, threshold origins, matching interval composition, and truncation enclosure are open.

Theorem 6.18 (Conditional complex-pole theorem)**.** *For $B = W, Z, H$, let $\Gamma_B^{\mathrm{phys}}(s, \xi)$ be the BRST-complete physical inverse two-point block, including every declared mixing field after the Ward/Slavnov–Taylor projection, analytically continued to a declared Riemann sheet, and set*

$$D_B(s, \xi) = \det \Gamma_B^{\mathrm{phys}}(s, \xi).$$

Assume a reference determinant has exactly one simple physical zero inside a certified contour, no zero lies on the contour, and the full determinant obeys $|D_B - D_{B,0}| < |D_{B,0}|$ there. In the neutral sector also assume the Ward-protected photon line has been separated from the massive eigenvalue. Then the enclosed pole is unique and stable. With the exact convention

$$s_B = \left(M_B - \frac{i}{2}\Gamma_B\right)^2, \qquad M_B > 0, \quad \Gamma_B \geq 0,$$

it determines one mass and width. If a Nielsen identity has the form $\partial_\xi D_B = C_B^{(\xi)} D_B$, the simple pole is gauge-parameter independent [19, 28]. For W and Z, interpretation as a physical resonance additionally requires a dressed BRST-invariant current amplitude whose Laurent coefficient at that pole is nonzero; for H, the analogous scalar-amplitude coupling is a separate condition. A simple zero of the inverse-propagator determinant alone does not establish either coupling.

Proof. Rouché's theorem preserves the number of enclosed zeros under the certified perturbation. Simplicity gives the implicit-function stability bound and a locally simple determinant zero. Differentiating $D_B(s_B(\xi), \xi) = 0$ and using the Nielsen identity gives $ds_B/d\xi = 0$. An invertible analytic field redefinition multiplies D_B by a nonzero analytic factor and therefore preserves the pole set. □

No source-emitted $W/Z/H$ self-energy kernels, analytic-sheet receipt, contour enclosures, widths, residues, or pole-convention uncertainty map are present on the displayed mass-chart surface. Schema fields named for the W, Z, and top pole masses are labels only and do not supply this theorem.

Physical scope of the icosahedral Standard Model and W/Z results

The finite icosahedral package is an exact conditional recognition result. Incidence determines the antipode J. Under the explicit contract that an admissible response is a signed central involutive graph automorphism implementing inverse-port readback, the responses are exactly $\pm J$; their common sign is conventional. Conditional also on the rank-15 matter contract with its unique charge-conjugate projector pair, the port-current, anomaly, Spin, and tensor-descent certificates fix the $3+2$ block structure, hypercharge lattice, and $\mathbb{Z}_6$ quotient from the stated premises alone. The finite source packet therefore selects the odd-Weyl Spin typing and the maximal faithful $\mathbb{Z}_6$ quotient on its declared carrier. The kernel is computed on every realized tensor and is insensitive to the projector representative. This finite selection supplies no laboratory identification of its current or flux sectors and no physical seam action. Equality with the independently reconstructed Tannaka current, attachment of the band to three physical chiral families, exclusion of extra light sectors, four-dimensional topological attachment, scalar attachment and dynamics, and construction of a chiral quantum field theory remain work in progress.

The conditional field-theory implications are explicit. A finite local action gives an exact finite gauge-invariance and locality theorem, and the familiar electroweak tree kernel is conditional on a separate canonical continuum and action-normalization bridge. An exact finite measure criterion and an exact finite Hamiltonian criterion are two parallel branches over that action. A separate formal perturbative branch carries the strict finite-order W/Z pole theorem. A nonperturbative continuum completion gives an observable-sector reconstruction implication and a distinct continued-sheet resonance-stability implication. The measure and perturbative branches are parallel descendants of the finite local action. Neither implies the other, and a perturbative pole does not imply the continuum completion.

These quantum field theory (QFT) theorems state what follows from typed action and quantization packets. They do not show that the target-free source emits those packets. The source-selected action and normalization, complete measure, target-clean perturbative matching, physical current amplitudes, source law and covariance, uncertainty enclosure, operational clock, and continuum tower with its continued-sheet packet are work in progress. The frozen external Standard Model action and effective-field-theory interval provide bounded validation inputs for the perturbative protocol. The proof-bearing validation also requires two genuinely independent raw loop engines, counterterm and BRST generation from the complete action, Nielsen identities, an artifact-resolving third verifier, nested complex-ball arithmetic, certified contours and Laurent data, dressed-current residues, and complete mutation replay. The external packet is a validation input rather than an OPH source producer. No source-native dimensionless or physical-unit W/Z pole follows from it.

Exact source-side frontier

The finite fifteen-state matter representation fixes the one-copy quadratic indices

$$\sum_{\text{Weyl}} T_{\mathrm{SU}(3)} = 2, \qquad \sum_{\text{Weyl}} T_{\mathrm{SU}(2)} = 2, \qquad \sum_{\text{Weyl}} d_3 d_2 Y^2 = \frac{10}{3}.$$

After importing the standard four-dimensional one-loop functional in the convention

$$\frac{\mathrm{d}g_i}{\mathrm{d}\ln\mu} = \frac{b_i g_i^3}{16\pi^2},$$

these indices give

$$b_Y = \frac{20}{9}N_g + \frac{1}{6}N_H, \qquad b_2 = -\frac{22}{3} + \frac{4}{3}N_g + \frac{1}{6}N_H, \qquad b_3 = -11 + \frac{4}{3}N_g.$$

Here N_g counts matter families and N_H counts scalar doublets. The declared choice $(N_g, N_H) = (3, 1)$ gives $(b_Y, b_2, b_3) = (41/6, -19/6, -7)$. The finite source model does not select that physical multiplicity pair.

On the declared one-doublet, rank-three family branch, four-dimensional power counting classifies the renormalizable scalar and Yukawa coefficient space as

$$\mathbb{R}_{>0} \times \mathbb{R}^2 \times \mathrm{Mat}_3(\mathbb{C})^3.$$

The three Yukawa matrices have complex dimension 27. This is an exact operator-basis theorem, with no selected coefficient, vacuum coordinate, or map from the source chart to a renormalized vacuum expectation value. A bounded check of the selected charged-response and pole-residue simulator pair finds no explicit scalar-completion input in those two artifacts. That check is not an exhaustive producer classification and does not prove physical scalar non-identifiability.

The invariant screen order-unit line and the central order-unit line of the four-copy weak multiplicity algebra are each one-dimensional. There is a unique positive unital normalized-trace isomorphism between them, natural under the declared refinements. This abstract line isomorphism does not identify a physical common load, choose the scalar sector, or normalize a laboratory current.

A source clock would convert a dimensionless gap $\epsilon_{\mathrm{clk}} = \Delta E_{\mathrm{clk}}/E_\star$ through

$$\Delta E_{\mathrm{clk}} = h\nu_{\mathrm{clk}}, \qquad E_\star = \frac{h\nu_{\mathrm{clk}}}{\epsilon_{\mathrm{clk}}}.$$

For $0 < \epsilon_{\mathrm{lo}} \le \epsilon_{\mathrm{clk}} \le \epsilon_{\mathrm{hi}}$, the exact interval is

$$\frac{h\nu_{\mathrm{clk}}}{\epsilon_{\mathrm{hi}}} \le E_\star \le \frac{h\nu_{\mathrm{clk}}}{\epsilon_{\mathrm{lo}}}.$$

The interval inversion is exact. Proposition 6.14 supplies a source-derived dimensionless spectral gap on the finite local domain. It supplies neither a physical reference transition nor its frequency, so the SI interval has no source value.

Quantization steps and the W/Z landing

The Standard Model field-theory implications split into a perturbative branch and an observable-reconstruction branch. Their dependency structure is:

$$\begin{aligned}
\textsf{source matter and declared action} &\longrightarrow \textsf{finite local classical action},\\
\textsf{finite local classical action} &\longrightarrow \textsf{finite chiral quantum object},\\
\textsf{finite local classical action} &\longrightarrow \textsf{formal BV/ST restoration} \longrightarrow \textsf{strict finite}-\textsf{order } W/Z,\\
\textsf{nonperturbative observable tower} &\longrightarrow \textsf{observable reconstruction} \longrightarrow \textsf{continued}-\textsf{sheet resonance}.
\end{aligned}$$

Source provenance for the classical action additionally requires the finite matter result and a source/action-identity receipt. A finite chiral measure or Hamiltonian does not follow from perturbative restoration, and perturbative restoration does not follow from either finite quantum construction. Strict finite-order poles do not supply a nonperturbative observable tower or a continued-sheet resonance theorem.

Construction	Mathematical implication	Required input	State
Finite matter packet	finite representation, charge, anomaly, lattice, and selector consequences on the declared packet	physical source selection and attachment	exact finite result; physical attachment is work in progress
Local classical action	finite local classical G_6 action is gauge invariant and local; the standard tree kernel also needs canonical continuum normalization	source-selected action, coefficients, regulator, normalization, and ancestry	conditional implication; source production is work in progress
Finite quantum object	an equivariant determinant-line section or a noncollapsing constrained Hamiltonian supplies an exact finite quantum object	full operator, measure/current or Hamiltonian/nonvacuum packet, and refinement controls	criteria proved; constructions are work in progress
Perturbative renormalization and poles	stable anomaly-free BV/ST theory is formally restorable order by order; strict W/Z poles follow from the complete finite-order packet	counterterm basis, matching and Fleischer–Jegerlehner (FJ) engines, identities, currents, and numerical freeze	implications proved; external validation and OPH source production are work in progress
Observable reconstruction	Osterwalder–Schrader reconstruction and a separate continued-sheet resonance theorem	reflection-positive tower and analytic-continuation packet	implications proved; constructions are work in progress

Theorem 6.19 (Finite local classical G_6 action)**.** *Let K_r be a finite oriented spin four-complex with bounded incidence, positive cell weights, declared boundary conditions, paired edge orientations, and declared spin transports. Put $U_e \in G_6 = S(U(3) \times U(2))$ on oriented edges and the declared Higgs and left-handed matter variables on vertices. Require every matter action to descend to a well-defined representation of G_6, every plaquette term to be a declared class function of its G_6 holonomy, and every finite difference to be gauge covariant with defined endpoint data. Then a finite sum of these plaquette, Higgs, fermion, and invariant Yukawa terms is local and exactly gauge invariant.*

Proof. Plaquette holonomies transform by conjugation, covariant edge differences transform at their endpoints, and class functions and invariant contractions remove those transformations. In integer hypercharge normalization the three Yukawa sums are

$$1 + 3 - 4 = 0, \qquad 1 - 3 + 2 = 0, \qquad -3 - 3 + 6 = 0.$$

Every term has bounded cell support. □

Corollary 6.20 (Canonically normalized electroweak tree kernel)**.** *If, in addition, the long-wavelength map sends the finite Higgs kinetic form to $(D_\mu H)^\dagger D^\mu H$ with canonical generator normalization and broken background $H_0 = 2^{-1/2}(0, v)^T$, then*

$$w = \frac{g^2 v^2}{4}, \qquad \mathcal{M}_N^2 = \frac{v^2}{4} \begin{pmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{pmatrix}, \qquad z = \frac{(g^2 + g'^2) v^2}{4},$$

and the other neutral eigenvalue is zero.

Remark 6.21 (Classical-action boundary)**.** *This is a classical existence template after the complex, fields, coefficients, boundary data, and normalization bridge are supplied. It does not show that OPH selects them, construct a chiral measure, remove mirrors or doublers, or produce a complex pole.*

Theorem 6.22 (Equivariant determinant-line criterion)**.** *At a fixed finite stage, let $D_r(U)$ be a local gauge-covariant, γ_5-Hermitian Ginsparg–Wilson operator on a connected admissible gauge-field component on which the chiral-projector rank is constant, with the declared spectral gap and locality bounds. A basis-independent, gauge-invariant finite chiral measure exists precisely when the determinant line admits a nowhere-zero gauge-equivariant section in the required locality and smoothness class. For a nonfree gauge action this is an equivariant statement over the action groupoid, including stabilizer actions. In local connection form the measure current must reproduce the projector curvature and obey global loop integrability. Flatness with trivial holonomy is only a sufficient special case.*

Proof. Changes of Weyl basis are transition functions of the determinant line. A nowhere-zero equivariant section cancels them and descends to the gauge quotient; conversely a basis-independent gauge-invariant phase supplies compatible nonzero fiber vectors. The curvature equation and loop condition are the local and global integrability conditions for that section. □

Theorem 6.23 (Finite Hamiltonian soundness)**.** *Let $\mathcal{H}_{\mathrm{kin}}$ be finite dimensional and let local Gauss generators exponentiate to the complete local G_6 action. Define*

$$C_G = \sum_{x,a} (G^a_x)^\dagger G^a_x, \qquad P_{\mathrm{phys}} = \mathbf{1}_{\{0\}}(C_G), \qquad 1 < \operatorname{rank} P_{\mathrm{phys}} < \dim \mathcal{H}_{\mathrm{kin}}.$$

Suppose a bounded-range self-adjoint H_r commutes with this action and has a unique certified physical ground state Ω_r with a positive gap. If a bounded self-adjoint gauge-invariant observable O_r has strictly positive variance in Ω_r, and the packet also supplies its claimed chiral index, positive mirror gap, primitive completeness, and complement-complete refinement controls, then $P_{\mathrm{phys}}\mathcal{H}_{\mathrm{kin}}$ is a nonvacuous finite unitary gauge theory with a positive-energy nonvacuum physical excitation and the declared chiral/mirror properties.

Proof. The vector

$$(O_r - \langle \Omega_r, O_r \Omega_r \rangle)\Omega_r$$

is physical, orthogonal to the ground state, and nonzero by the variance condition. The remaining conclusions are exactly the separately supplied index, gap, completeness, and refinement clauses. Thus neither an identity projector nor a vacuum-only physical sector passes. □

Remark 6.24 (Finite-quantum boundary)**.** *The determinant-line and Hamiltonian results are criteria and soundness theorems. Neither complete G_6 construction is supplied. Anomaly arithmetic alone instantiates neither theorem.*

Theorem 6.25 (Formal Slavnov–Taylor restoration)**.** *Let S_0 be the complete canonically normalized Standard-Model BV action, including the gauge-fixing and antifield sectors, and suppose $(S_0, S_0) = 0$. Fix a regulator/scheme satisfying the quantum action principle, locality, and the declared power counting. Assume stability under renormalization in a complete permitted counterterm basis with fixed normalization conditions, classification of the applicable local ghost-number-one BRST cohomology, vanishing of the declared perturbative anomaly class, and a separate check of all applicable global anomalies. Then finite local counterterms may be chosen recursively so that the formal series*

$$\Gamma = S_0 + \sum_{n \geq 1} \kappa^n \Gamma_n, \qquad \kappa = (16\pi^2)^{-1},$$

satisfies the renormalized Slavnov–Taylor identity order by order.

Proof. At order n, the quantum action principle makes the breaking a local ghost-number-one functional Δ_n. Wess–Zumino consistency makes it closed under the linearized Slavnov–Taylor operator. The cohomology classification splits it into an anomaly representative plus an exact term. Anomaly clearance removes the first, and an allowed finite counterterm cancels the second. Stability and the normalization conditions fix the remaining invariant freedom, so induction proves the formal statement. □

Remark 6.26 (Perturbative boundary)**.** *This is a formal power-series theorem. It supplies no convergence theorem, finite chiral measure, finite constrained Hamiltonian, or Wightman construction. A numerical implementation must produce its regulator-specific restoration transcript and verify the Ward, Slavnov–Taylor, and Nielsen identities.*

Lemma 6.27 (Conditional first-order FJ coordinate change)**.** *Freeze the potential normalization, bare VEV shift, tadpole prescription, field and parameter counterterms, mass arguments, mixing coordinates, and gauge-fixing convention. If the complete finite change is*

$$p_L = p_F + \kappa\, \delta p^{(1)} + O(\kappa^2)$$

and $s(p) = s_0(p) + \kappa s_1(p) + O(\kappa^2)$*, equality of the exact pole in the two coordinates implies*

$$s_{1,F} = s_{1,L} + \delta p^{a(1)} \partial_a s_0.$$

Proof. Substitute $p_L(p_F)$ and Taylor expand through first order. The sum must include every transformed parameter, normalization, mass argument, counterterm, and mixing coordinate; a VEV-only substitution is insufficient. □

Theorem 6.28 (Strict charged and neutral pole coefficients)**.** *On one scheme, contribution mask, and resonance sheet, write*

$$\Gamma^T_W = s - w + \kappa \Pi^{(1)}_{WW} + \kappa^2 \Pi^{(2)}_{WW} + O(\kappa^3).$$

For $s_W = w + \kappa s_{W,1} + \kappa^2 s_{W,2} + O(\kappa^3)$*,*

$$s_{W,1} = -\Pi^{(1)}_{WW}(w), \qquad s_{W,2} = \Pi^{(1)}_{WW}(w)\Pi^{(1)\prime}_{WW}(w) - \Pi^{(2)}_{WW}(w).$$

For the massive root of the full photon–Z *matrix with tree value* $z \neq 0$*,*

$$s_{Z,1} = -\Pi^{(1)}_{ZZ}(z),$$

$$s_{Z,2} = \Pi^{(1)}_{ZZ}(z)\Pi^{(1)\prime}_{ZZ}(z) - \Pi^{(2)}_{ZZ}(z) + \frac{\Pi^{(1)}_{ZA}(z)\Pi^{(1)}_{AZ}(z)}{z}.$$

Thus the one-loop-squared neutral mixing product is excluded at strict one loop and is one mandatory term of the complete strict-two-loop mask.

Proof. Insert the root series and compare powers of κ. In the neutral sector use the Schur complement of the photon block; both off-diagonal entries begin at order κ. □

Theorem 6.29 (Nielsen control and physical current pole)**.** *Suppose the inverse matrix and Nielsen insertions are holomorphic near a simple massive root on the frozen sheet and, through retained order N,*

$$\partial_\eta \Gamma^T = \Lambda_\eta \Gamma^T + \Gamma^T \widetilde{\Lambda}_\eta + O(\kappa^{N+1}).$$

Then

$$\partial_\eta \det \Gamma^T = \operatorname{tr}(\Lambda_\eta + \widetilde{\Lambda}_\eta) \det \Gamma^T + O(\kappa^{N+1}), \qquad \partial_\eta s_p = O(\kappa^{N+1}).$$

If the simple left/right kernel vectors ℓ, r have $\ell^\dagger \Gamma^{T\prime}(s_p) r \neq 0$, and the dressed renormalized BRST-invariant current vertices obey $J_L^\dagger r \neq 0$ and $\ell^\dagger J_R \neq 0$, then the gauge-invariant current amplitude has Laurent residue

$$\frac{(J_L^\dagger r)(\ell^\dagger J_R)}{\ell^\dagger \Gamma^{T\prime}(s_p) r}.$$

Proof. Use $\partial_\eta \det \Gamma = \operatorname{tr}(\operatorname{adj} \Gamma\, \partial_\eta \Gamma)$; the adjugate identity remains valid at a singular matrix and avoids dividing by the determinant. The simple-root implicit equation gives the omitted-order gauge variation. The rank-one Laurent expansion of Γ^{-1}, contracted with the dressed current vertices, gives the amplitude pole. This is not a positivity claim for an unstable elementary field. ☐

Theorem 6.30 (Gauge-invariant observable reconstruction)**.** *A compatible cofinal Schwinger family for a complete declared gauge-invariant observable algebra that satisfies distributional convergence, Euclidean covariance, graded symmetry/locality, reflection positivity, clustering, growth/regularity, noncollapse, and refinement Cauchy control reconstructs a positive Hilbert space, cyclic vacuum, positive-energy translations, and the corresponding observable-sector Wightman distributions and local graded net.*

Remark 6.31. *This observable-sector implication does not by itself construct colored local fields, charged infrared sectors, confinement, asymptotic completeness, an S-matrix, or a second-sheet resonance.*

Theorem 6.32 (Resonance and residue stability)**.** *On one common continued sheet, write*

$$G_r(s) = \frac{N_r(s)}{D_r(s)} + G_{r,\mathrm{reg}}(s).$$

Let a Jordan contour and its interior lie in a common domain on which $D_r, N_r, G_{r,\mathrm{reg}}$ and their limits are holomorphic. Assume locally uniform convergence of these data and D_r', one simple enclosed zero s_r, uniform nonzero contour and derivative bounds, and the Rouché inequality

$$\sup_C |D - D_r| < \inf_C |D_r|.$$

Then the limiting denominator has one simple zero s_*, $s_r \to s_*$, and, if $N(s_*) \neq 0$,

$$\operatorname*{Res}_{s=s_r} G_r(s) = \frac{N_r(s_r)}{D_r'(s_r)} \longrightarrow \frac{N(s_*)}{D'(s_*)} \neq 0.$$

Proof. Rouché preserves the zero count for the holomorphic denominators. Compactness and uniqueness give root convergence. Uniform numerator and derivative convergence with the lower derivative bound gives residue convergence. Ordinary uniform convergence of the meromorphic quotient through its own poles is neither assumed nor valid. ☐

Remark 6.33 (Producer boundary)**.** *These results specify conditional implications. A target-free shared-ξ external Standard Model harness supplies a declared action and active census, a frozen effective-field-theory interval, dual rule tables, direct and converted FJ blocks, and payload-replayed Ward and Slavnov–Taylor controls. It supplies no artifact-resolved complex-ball W/Z contour and Laurent packet, dressed-current residue, independent-ξ cross-check, or complete identity-control transcript. Those validation objects are work in progress. The source-selected classical action, either complete finite quantum object, the OPH matching and current packet, and the observable tower with continued-sheet data are also work in progress.*

Strict-one-loop W/Z pole-map kernel

The finite-order theory map is explicit and machine checked. Let the renormalized one-doublet electroweak input at scale Q be $\theta(Q) = (g, g', v_F, \ldots)$, with canonical Higgs kinetic term and $v_F > 0$, and set

$$w = \frac{g^2 v_F^2}{4}, \qquad z = \frac{(g^2 + g'^2) v_F^2}{4}.$$

Use the inverse-propagator convention

$$\Gamma^T(s) = s - m_0^2 - \Delta^T(s) = s - m_0^2 + \Pi^T(s), \qquad \Delta^T = -\Pi^T,$$

where each $\Delta^{(1)}$ includes its one-loop factor, counterterms, tadpoles, and the complete declared strict-one-loop mask. Relative to the coefficient convention of Theorem 6.28,

$$\Delta_{ij}^{(1)}(s) = -\kappa \Pi_{ij}^{(1)}(s), \qquad \kappa = (16\pi^2)^{-1}.$$

The two notations are translations of one pole equation, not independent results.

Proposition 6.34 (Strict-one-loop charged and neutral pole map)**.** *Assume the tree roots $w, z > 0$ are simple and the declared one-loop entries are holomorphic near them on the frozen analytic sheet. Then*

$$s_W^{[1]} = w + \Delta_{WW}^{(1)}(w), \qquad s_Z^{[1]} = z + \Delta_{ZZ}^{(1)}(z).$$

In the neutral tree-level photon–Z basis,

$$\Gamma_N^T(s) = \begin{pmatrix} s - \Delta_{AA}^{(1)}(s) & -\Delta_{AZ}^{(1)}(s) \\ -\Delta_{ZA}^{(1)}(s) & s - z - \Delta_{ZZ}^{(1)}(s) \end{pmatrix} + O(\epsilon^2).$$

The product $\Delta_{ZA}^{(1)}\Delta_{AZ}^{(1)}$ has loop power two and is excluded from a strict-one-loop root. Its leading Schur-complement contribution is

$$-\frac{\Delta_{ZA}^{(1)}(s)\Delta_{AZ}^{(1)}(s)}{s},$$

which belongs only in a separately complete two-loop map together with the genuine two-loop entries and pole-iteration derivatives.

Proof. Write $s_W = w + \epsilon\sigma_W + O(\epsilon^2)$ in the charged inverse entry. Its order-ϵ coefficient is $\sigma_W - \delta_{WW}^{(1)}(w)$. For the neutral determinant, write $s = z + \epsilon\sigma_Z + O(\epsilon^2)$. The order-$\epsilon$ coefficient is $z[\sigma_Z - \delta_{ZZ}^{(1)}(z)]$; both off-diagonal entries start at order ϵ, so their product starts at order ϵ^2. □

For $s_V = m_{V,0}^2 + \Delta_V^{(1)}$, the strict energy-pole coefficients are

$$\delta M_V^{(1)} = \frac{\operatorname{Re}\Delta_V^{(1)}}{2m_{V,0}}, \qquad \Gamma_V^{(1)} = -\frac{\operatorname{Im}\Delta_V^{(1)}}{m_{V,0}}.$$

They are distinct from the exact coordinate transform of the truncated complex number. On the lower-half-plane branch,

$$M_V = \sqrt{\frac{|s_V| + \operatorname{Re} s_V}{2}}, \qquad \Gamma_V = \sqrt{2(|s_V| - \operatorname{Re} s_V)}, \qquad s_V = (M_V - i\Gamma_V/2)^2.$$

Applying this nonlinear square root exactly resums kinematic powers of the one-loop coefficient. It is a useful display coordinate, not a strict two-loop calculation and not the object to compare in a finite-order Nielsen test.

Proposition 6.35 (Evidence cannot self-attest)**.** *An untrusted input boolean asserting an external Fleischer–Jegerlehner, matching, source-law, gauge/BRST, clock, or ancestry property cannot certify that property. A physical-claim verifier must resolve an independent hash-bound witness, validate it, and bind it to the exact numerical subject, order, mask, scheme, and analytic sheet.*

Proof. Choose a subject for which the external property is false and set the untrusted boolean to true while preserving every relation recomputed by the verifier. If the verifier resolves no independent witness, it follows the same accepting path. Hence acceptance would admit a false instance. □

The released fail-closed receipt implements these rules and rejects self-attested physical flags, unrelated-but-self-consistent poles, substituted empty fixtures, inflated tolerances, corrupted redundant fields, and altered neutral diagnostics. Its archived SMDR order-one fixture at $Q = 160$ GeV evaluates to

$$s_W^{[1]} = (6459.842027569383 - 160.532752773045i)\ \mathrm{GeV}^2,$$
$$s_Z^{[1]} = (8222.835212344102 - 218.292761806439i)\ \mathrm{GeV}^2.$$

The corresponding strict readouts are $(M_W, \Gamma_W) = (80.374161202712, 2.007425074735)$ GeV and $(M_Z, \Gamma_Z) = (90.680036075608, 2.402420059845)$ GeV. These numbers reconstruct an archived backend row. They carry target ancestry and supply no independent self-energy evaluation, no source-selected vacuum normalization, no complete neutral matrix, no source covariance, no independent gauge or Becchi–Rouet–Stora–Tyutin receipt, and no source clock. The strict one-loop pole map is therefore conditional, and it is not source-native physical, so no physical conclusion follows. What is proved is the implication from a complete declared renormalized strict-one-loop packet to the separated pole and mass and width readouts. The construction of that antecedent from the source is work in progress.

A separate target-free shared-ξ Standard Model harness binds a declared action and active census to a frozen effective-field-theory interval. It contains two independent rule tables, direct and converted Fleischer–Jegerlehner blocks, and payload-replayed Ward and Slavnov–Taylor controls. The rule tables are coordinate variants inside one harness rather than two independently implemented raw loop engines. A proof-bearing external validation additionally requires the second engine, counterterms generated from the complete action, BRST-generated ghost vertices, finite symmetry restoration, symbolic Nielsen identities, an artifact-resolving third verifier, rigorously nested complex balls, certified W/Z contours and Laurent data, dressed-current residues, and complete mutation replay. These inputs are work in progress. The harness therefore establishes no physical W/Z pole and supplies no OPH-native source packet.

Theorem 6.36 (Rigidity, dependency, and W/Z/H composition criterion)**.** *Let $\mathcal{F}_{\mathrm{EW}}$ be the fully enumerated class of source maps that preserve the declared locality, symmetry, dimensional, refinement, and provenance constraints, modulo gauge and trivial reparameterization. Suppose either the pole readout is constant on this class or a target-independent selector has one minimizer with a positive winner gap. Suppose further that a hash-bound acyclic dependency graph records the source objects, selectors, conventions, uncertainty model, and outputs, and measured $W/Z/H$ data and calibrated proxies are absent from every source ancestor set.*

If, in addition, a unique source root and independently source-closed physical $E_\star$ emit the canonical electroweak and criticality coefficients, the finite quotient-path and split-character certificates hold, and the declared renormalization-group and complex-pole hypotheses hold, then OPH determines one source-separated pole triple (s_W, s_Z, s_H) with a certified uncertainty enclosure.

Proof. Rigidity or the positive-gap selector removes alternative source maps. A deterministic leaf depends only on its ancestors, so the dependency graph gives formal source separation. The unique root and scale emit one coefficient packet, declared running and matching transport it uniquely, and the pole theorem gives one stable pole per particle. Composition proves the claim. □

This criterion is not satisfied by the displayed numbers. The finite quotient-path and split-character certificates are assumptions, the public and source-only pixel branches differ, and the physical scale, renormalization-group transport, complex pole, and target-independence proof are open.

The missing object is a source-emitted certificate. The structural theory admits target-free analytic counterfamilies $\tau_2 = -c\eta^2$, $\delta n = d(1-\rho_{\mathrm{EW}})\eta^2$ and $(\pi_y, \pi_\lambda) \mapsto (\pi_y + a\eta^N, \pi_\lambda + b\eta^N)$, all on the same open physical domain but with different mass-chart outputs. Likewise, a fixed running mass is compatible with both $s - m_R^2$ and $s - m_R^2 - \epsilon$, which have different poles. These countermodels prove that two-channel exhaustion and analyticity do not emit the electroweak character, criticality split, or physical kernel. Nor is a formal path realization sufficient: any finite polynomial can be encoded by assigning one weighted path to each monomial. A non-vacuous certificate must independently fix its transitions, path measure, signs, quotient action, exhaustive census, rigidity gap, and target-independence proof.

If the source selector is proved deterministic and globally unique, the source law is a delta measure and no Monte Carlo source propagation is required. If the source emits a nondegenerate law, the simulator must propagate that law through the pole map. Scale variation and truncation envelopes are not source covariance, and neither a point estimate nor componentwise intervals determine the covariance of $(\Re s_W, \Im s_W, \Re s_Z, \Im s_Z)$. Summable clock-Hamiltonian refinements give a unique limiting gap; fixed Lipschitz RG segments compose uniquely; Rouché, Nielsen, and analytic conjugacy isolate gauge-independent poles. If a determinant defect is bounded by ϵ, $a = D'(s_0) \neq 0$, and $|D''| \le K$, every r with $\epsilon < |a|r - Kr^2/2$ encloses the unique displaced pole and propagates to mass/width bounds through the selected square root. The four absent source objects are therefore the factorized clock packet, independently weighted electroweak carrier, criticality split-character carrier, and Becchi–Rouet–Stora–Tyutin (BRST)-complete pole-kernel packet. A runtime dependency graph proves computational separation only. A source-only claim also requires a target-independent specification.

The electroweak chapter therefore presents four mathematically distinct surfaces: the selected-carrier chart, the candidate value law, the conditional finite quotient-path theorem, and the measured-reference inverse surface used for diagnostic W/Z values. A physical pole surface requires the additional source-root, scale, RG/scheme, two-point-kernel, rigidity, uncertainty, and provenance receipts.

6.6 Repair-tuple selection and the color amplitude/loop split

The candidate electroweak tuple sits inside a two-parameter freedom that the stated premises do not remove. With $\rho_{\mathrm{EW}} = (\alpha_2 - \alpha_Y)/(\alpha_2 + \alpha_Y)$, the coherent quadratic family is

$$\tau_{2,\mathrm{tree}}^{\mathrm{exact}} = -c\,\eta_{\mathrm{source}}^2, \qquad \delta n_{\mathrm{tree}}^{\mathrm{exact}} = d\,(1-\rho_{\mathrm{EW}})\,\eta_{\mathrm{source}}^2,$$

and an open neighborhood of distinct (c, d) values preserves the positive mass-chart domain and defines consistent electroweak repairs. The charged leg is the $\mathrm{SU}(2)_L$ coupling correction $\delta\alpha_2 = \alpha_2\tau_2$; the neutral leg is the hypercharge screening correction carried by δn.

The compact-gauge reconstruction fixes the color count but does not fix the channel assignment or normalization that would close this freedom. Doplicher–Roberts/Tannaka reconstruction returns the color triplet sector with statistical dimension $d(\rho_3) = N_c = 3$, and its standard conjugate intertwiner satisfies $R^*R = d(\rho_3) = N_c$, so R has norm $\sqrt{N_c}$ while $R/\sqrt{N_c}$ is the isometry; a closed color loop carries the full dimension N_c. The two repair legs sit at different levels. The charged leg is the broken, mass-generating $\mathrm{SU}(2)_L$ channel, driven by the color-singlet condensate amplitude whose large-N_c scaling is the decay-constant scaling $\sqrt{N_c}$ assigned to one standard conjugate intertwiner. The neutral leg would be the unbroken hypercharge screening channel, a vacuum-polarization loop that closes the color line and carries raw trace weight N_c. Under these extra hypotheses the alternative quadratic model has

$$c = \frac{\sqrt{N_c}}{2} = \frac{\sqrt{3}}{2}, \qquad d = \frac{N_c}{2} = \frac{3}{2},$$

where the chart coefficient $d = N_c/2$ represents the raw neutral trace because $(\alpha_2+\alpha_Y)(1-\rho_{\mathrm{EW}}) = 2\alpha_Y$.

The resulting comparison is a target-informed, fixed-slice diagnostic of that alternative model. Profiling the charged coefficient against the running-tree W coordinate is independent of the criticality Jacobian core and inherits the declared α_2, v, running, and scheme surface. It selects $c = 0.8670$, while profiling the neutral coefficient returns $d = 1.461$. Both values are target-conditioned fit outputs. Their proximity to $\sqrt{3}/2$ and $3/2$ is an internal calibration observation, not physical W/Z evidence, and no experimental confidence band is assigned across the noncommensurate chart and mass conventions. That slice is not a test of the complete candidate value law, which changes both coordinates and includes higher path characters; it therefore neither excludes that law nor proves or excludes a separate running-tree companion.

The color-balanced rule does not equal the complete candidate value law in Theorem 6.3; it gives a different W coordinate. A proof of its amplitude and trace hypotheses would establish that alternative quadratic model, not the finite quotient-path law. A source derivation requires the two-channel quotient, exhaustive charged and neutral path lists, the $(1, 2, 1)$ and $(1, 4, 2)$ incidences with their $1/3$ color measure and $1/6$ central trace, the fibre Gram form and residual pairing, an exhaustive deformation class, and a positive selector gap. Measured W/Z agreement cannot replace that certificate. The pre-repair carrier must be specified independently of the desired outputs as an observer-like self-reading patch with bounded local state, typed ports, readback records, and admissible feedback or repair moves. It may contain neither the target masses nor the desired incidence and central-trace coefficients. Independent verification must recompute path legality, exhaustion, exact response rank and nullspace, central action, oriented mismatch first variation, the fibre Gram form, and the deformation census. The order of $\mathbb{Z}_6$ supplies the averaging coefficient $1/6$, not automatically a normalized trace $1/6$: the checker must separate the trivial physical matter-kernel action from the declared center-label/transport trace space and derive the trace, source amplitude, and subtraction sign in the latter.

6.7 Selector rigidity and the discrete repair-law boundary

The candidate electroweak selector closes the continuous (c, d) freedom of the preceding subsection. Writing $x = \tau_{2,\mathrm{tree}}^{\mathrm{exact}}$, $y = \delta n_{\mathrm{tree}}^{\mathrm{exact}}$, and $\kappa = (\alpha_Y + \alpha_2)/\alpha_Y$, the selector

$$J_{\mathrm{EW}}(x, y) = \frac{x^2\left[1 + (2\eta_{\mathrm{source}} + x)^2 + 4x^2\right]}{1 + 4x^2} + \frac{\kappa^2}{4}(1 + 4x^2)y^2$$

admits the exact factorization

$$J_{\mathrm{EW}} - \frac{3}{4}x^2 - \frac{\kappa^2}{4}y^2 = \frac{x^2}{4(1 + 4x^2)}\left[8(x + \eta_{\mathrm{source}})^2 + 8\eta_{\mathrm{source}}^2 + 1 + 4\kappa^2(1 + 4x^2)y^2\right],$$

so $J_{\mathrm{EW}} \geq \frac{3}{4}x^2 + \frac{\kappa^2}{4}y^2$ with equality exactly at the origin, and $x^2 + y^2 \geq r^2$ forces $J_{\mathrm{EW}} \geq \min\{3/4, \kappa^2/4\}r^2$. The selector picks the zero deformation with a quantitative gap. The candidate value-law tuple has $J_{\mathrm{EW}} \approx 3.12 \times 10^{-7} > 0$, so it is incompatible with the selector, and the residual electroweak freedom is one discrete choice between two incompatible laws rather than a continuous chart family.

On the source-only branch the two branch points give the tree/chart coordinates

$$\begin{aligned} &\text{zero-selector law:} \quad M_W/E_\star = 6.579631 \times 10^{-18}, \quad M_Z/E_\star = 7.463335 \times 10^{-18}, \\ &\text{carrier value law:} \quad M_W/E_\star = 6.578870 \times 10^{-18}, \quad M_Z/E_\star = 7.463750 \times 10^{-18}, \end{aligned}$$

with unclosed-clock displays (80.3301, 91.1191) and (80.3208, 91.1242) GeV and discrete ambiguity widths of 9.3 MeV on M_W and 5.1 MeV on M_Z. These are running/chart coordinates. The PDG mass targets use a mass-dependent-width Breit–Wigner convention, and complex-pole masses use another convention. The exact convention map distinguishes $M = \mathrm{Re}\sqrt{s}$ from the legacy $\sqrt{\mathrm{Re}\, s}$ coordinate. The running/chart coordinates therefore have no valid physical pull or near-hit interpretation and carry no theory covariance. Numerical W/Z proximity cannot decide between the laws; the decision object is a source-law selection principle, and the quantitative mechanism of the source-action rigidity theorem applies once the electroweak feature basis and moment vector are emitted.

The two-loop comparison gives $(79.115335, 89.802735)$ GeV. It combines a minimal supersymmetric Standard Model one-loop baseline with a Standard Model two-loop-minus-one-loop increment, an inconsistent hybrid prescription. A partial Feynman-gauge pole calculation gives approximately $(79.53284, 89.71232)$ GeV. Its definition of v, tadpole treatment, field-content and threshold matching, scale choice, and higher-order corrections are open. The outputs establish neither a unique scheme conversion nor a unique 1–2% defect, and they do not exhaust the physical prescription family.

A conditional finite electroweak root/Cartan carrier emits the value-law coefficients exactly: the transitive C_3 color action has invariant measure 1/3, the regular $\mathbb{Z}_6$ register has rank-one trace 1/6, the transitive four-slot register has slot measure 1/4, the fixed independent-product source law gives $\lambda_{\mathrm{EW}} = \eta_{\mathrm{source}}^2/(4\rho_{\mathrm{EW}})$, and the depth-two path census reproduces the candidate quadratic response polynomials. A companion SU(3)-Casimir carrier emits the declared criticality normalizations, $\kappa_\lambda = C_F^2 = 16/9$, the response coefficients $3/2 + \rho_{\mathrm{EW}}/4$ and $4/3 - \rho_{\mathrm{EW}}/54$, and the logarithmic character $\rho_{HT} = \log(1 + \tau_2)$ from multiplicativity with unit derivative. Both carriers are target-exposed candidates without a source-independent quotient-path selection or census certificate.

The declared criticality Jacobian entries follow exactly from the core, $\partial m_t/\partial y_t = m_{t,0}/y_0$ and $\partial m_H/\partial\lambda = m_{H,0}/(2\lambda_0)$, and the exact readout is $m_t = m_{t,0}(1 + r_y)$ and $m_H = m_{H,0}\sqrt{1 - r_\lambda}$ with the exact linearization error $m_{H,0}r^2/(2(1 + \sqrt{1 - r})^2)$. The synchronized criticality core is target-conditioned: its synchronization scale minimizes an objective containing the Higgs and top comparison values. Without that switch, the literal candidate one-loop source equations give the target-free tree coordinates $m_t^{\overline{\rm MS}}/E_\star = 1.267758 \times 10^{-17}$ and $m_H^{\rm tree}/E_\star = 9.427653 \times 10^{-18}$, with unclosed-clock displays 154.78 GeV and 115.10 GeV and a QCD-converted top display of 164.13 GeV. These are one-loop tree coordinates on the declared surface; they are not physical poles and do not satisfy the pole conditions of §6.5.

The candidate one-loop coordinates form a zero-continuous-parameter family. The literal core derives its top boundary from the gauge sector through the double-criticality condition $\lambda(\mu_b) = 0$, $\beta_\lambda(\mu_b) = 0$, which fixes $y_t(\mu_b) = [(2g_2^4 + (g_2^2 + g_Y^2)^2)/16]^{1/4}$. The candidate branch imposes this at the gauge-unification scale μ_U, while the electroweak transmutation in the same model anchors v at the pixel energy $E_{\rm cell} = E_\star/\sqrt{P}$, so the candidate model carries two different high-scale anchors. Evaluating the same law at the named source scales gives, at one loop, $(m_t, m_H) =$ (164.1, 115.1) GeV at μ_U, (170.4, 127.8) at $E_{\rm cell}$, and (170.7, 128.3) at $E_\star$; a benchmark-validated two-loop upgrade shifts these to (169.4, 119.4), (175.7, 131.8), and (175.9, 132.3). The family brackets the measured pair in both channels at both loop orders, so the candidate deficit decomposes into the boundary-scale choice plus loop truncation, with no continuous freedom anywhere. Along the fit-free curve the Higgs coordinate at the measured top mass is 125.7 GeV at two loops, within 0.5 percent of measurement and inside the declared tree-to-pole matching band; the implied boundary scale is 4.8×10^{17} GeV, between μ_U and $E_{\rm cell}$. No boundary-scale selection theorem is supplied; numerical agreement cannot select the scale, and the declared-surface values obtained by back-solving from the measured pair stay classified as target-anchored fits.

The triple-criticality equation $\mathrm{d}\beta_\lambda/\mathrm{d}\ln\mu = 0$ has no solution along the flow: the family derivative is dominated by $-24y_t^3\beta_{y_t} > 0$ and stays bounded away from zero across the whole window, so every family point is a clean λ minimum and the boundary scale requires source structure. A conditional variational selection principle uses three premises. The criticality boundary is a record that reconciles the model's two anchor records, the gauge-unification record at μ_U and the transmutation record at $E_{\rm cell}$. The reconciliation cost is quadratic in renormalization time, and the two anchor records carry equal capacity. These premises give a strictly convex cost with a unique minimizer at the log-midpoint $\sqrt{\mu_U E_{\rm cell}} = E_\star e^{-\pi} P^{-1/6}$, hence $(m_t, m_H) =$ (172.63, 125.77) GeV at two loops; the implication is proved exactly on rational sample points.

The quadratic-cost premise is exact under the declared Gaussian record model: the one-loop chart is inverse-affine, so $1/\alpha(t)$ is exactly linear in renormalization time, and the Kullback–Leibler divergence between fixed-resolution Gaussian records with an affine stored coordinate is exactly $(s^2/2\sigma^2)(t - t_i)^2$, a quadratic cost with no leading-order approximation. Axiom 3 supplies relative-entropy minimization after the reference family, affine observable, trace convention, variance, and resolution have been declared. The Gaussian model and its normalization are separately declared, and their derivation from the complete three-axiom schema is open. The reconciliation placement is exact within a separately declared port-additive mismatch and cost-minimizing repair model. In that model, a record with two scale-parents settles at the capacity-weighted log-mean. Equal capacity follows if the anchor registers have equal slope and readback resolution, which is another declared condition. These record and repair choices are declared inputs, and their derivation from the three axioms is open. The finite criticality carrier must establish that the boundary record carries exactly two parent ports, one to the gauge-unification register and one to the transmutation register, and that those anchor registers belong to the same register class at equal refinement depth. Given the declared Gaussian record model and the stated repair premises, these two census facts select the boundary scale, and with it $(m_t, m_H) =$ (172.63, 125.77) GeV, with no remaining continuous parameter. The W/Z value law requires the same finite carrier census. The equal-capacity assumption is measurable: a capacity asymmetry moves the m_H coordinate by about 2.1 GeV per e-fold. A three-loop implied scale therefore provides a discriminating measurement of that assumption.

A conditional selection theorem follows. If the boundary record reconciles the two anchor records with port-additive quadratic cost in renormalization time and equal anchor capacities, the boundary scale is exactly the log-midpoint. The inverse-affine one-loop chart and the declared Gaussian maximum-entropy record model give the exact quadratic reconciliation cost. The declared port-additive repair model places a two-parent record at the capacity-weighted log-mean. The theorem therefore consumes those record and repair models together with two finite carrier facts: the two-parent port structure of the boundary record and the equal-class equal-depth relation of the anchor registers. A carrier certificate of the census class supplies those two facts.

6.8 Target-conditioned electroweak chart envelope

Evaluating two target-conditioned repair candidates at the endpoint pixel and across the empirical-closure interval gives the following comparison envelope. It is not a source interval: its candidate set is not an exhaustive source-derived deformation class, and it mixes the public endpoint branch with the source-only question.

$$
\begin{aligned}
m_H &\in [125.183,\ 125.232]\ \text{GeV}, &&\text{measured } 125.13 \pm 0.11,\\
m_t &\in [172.278,\ 172.352]\ \text{GeV}, &&\text{measured } 172.1 \pm 0.6,\\
c_W &\in [80.3692,\ 80.3774]\ \text{GeV}, &&\text{PDG BW coordinate } 80.3692 \pm 0.0133,\\
c_Z &\in [91.1880,\ 91.1983]\ \text{GeV}, &&\text{PDG BW coordinate } 91.1880 \pm 0.0020.
\end{aligned}
$$

The Higgs and top entries are target-conditioned comparisons on their declared mass scheme. The W/Z entries are chart coordinates. The PDG targets use mass-dependent-width Breit–Wigner parameters, and complex-pole masses use another convention, so no physical coverage or pull is assigned. At the endpoint pixel the two repair selections give Z-chart coordinates 91.18798 and 91.18801 GeV. The selection spread between the two displayed laws is roughly eight MeV in the W chart, thirty keV in the Z chart, sixteen MeV in m_H, and thirty MeV in m_t. These figures compare two ansätze; they do not bound the full admissible deformation class.

6.9 Residual attribution across the electroweak rows

On this comparison envelope, the Higgs and top residuals are target-conditioned mass-scheme diagnostics. No W/Z standard-deviation assignment is valid because the chart coordinates, PDG mass-dependent-width parameters, and complex poles use different conventions. The W-chart spread is repair-selection freedom: the two candidate repair laws differ by about eight MeV. The m_H and m_t offsets track the same repair-selection width folded through the declared Higgs and top Jacobian cores.

The Z-chart row is the most sensitive adapter case. The empirical-closure pixel inherits the frozen electromagnetic anchor deficit through the fine-structure endpoint, and c_Z tracks the pixel with $\mathrm{d}c_Z/\mathrm{d}P \approx 123$ GeV per unit P. Two propagation branches separate the mechanism. The direct branch injects the certified anchor shift into the running couplings and re-evaluates the candidate W/Z chart. That branch shifts c_Z by -66 to -87 MeV on the hypercharge line and by roughly -230 to -302 MeV on the proportional line, exceeding the adapter displacement by an order of magnitude, so it is excluded. The pixel branch moves the endpoint to the measured fine-structure value and propagates the induced pixel shift through $\mathrm{d}P/\mathrm{d}A_{\mathrm{Th}}$. That branch moves the empirical pixel onto the calibration pixel to within 4×10^{-7} in P, equivalently -0.05 MeV in the Z chart. This is an internal adapter closure check; it supplies no physical mass or pole.

For this comparison surface, a closed fine-structure anchor bridge would collapse the endpoint pixel interval onto the calibration pixel and remove this particular chart displacement. It would not supply the finite quotient-path certificate, the independent scale, the RG/scheme certificate, or the complex-pole gate. The anchor bridge is the object developed in the fine-structure companion, and its source branch is the subject of Section 6.10.

6.10 Source branch of the fine-structure anchor bridge

The certified anchor gap [0.620, 0.651] in inverse fine-structure units names the shift the source side supplies at the electroweak anchor for the empirical-closure endpoint to reach the measured fine-structure value. The gap is defined by that measured requirement, so inserting it back is circular. The non-circular source route continues the declared running convention to the next order and tests whether the induced anchor shift lands inside the certified gap.

That test is executed. The one-loop anchor is reproduced from the declared coefficients and the transmutation-certificate boundary data to a residual of 10^{-11}. The standard two-loop running system, integrated down the same 33.2 e-folds with a top-Yukawa convention scan, shifts the anchor by [+1.62, +2.14] in inverse fine-structure units for every convention. The shift has the correct sign and overshoots the certified gap by a factor of two and a half to three and a half. Restoring the balance requires the threshold and scheme-conversion data, which the matching contract classifies as hidden fit parameters when they are undeclared. The balance requirement is quantitative: the threshold map removes about 2.14 inverse-alpha, equivalent under naive coefficient accounting to a single effective threshold near 458 GeV. The anchor bridge has no source scheme-lock or threshold map; its endpoint is empirical.

7 Flavor Transport, Generation Structure, and the Yukawa Dictionary

The gauge branch supplies a canonical rank-three response band under its separate premises. It does not prove physical family attachment or produce the full flavor dictionary. This paper therefore needs a separate flavor chapter. The imported input here is the color carrier $N_c = 3$ together with the declared physical identification of the rank-three candidate. This chapter is the bridge between that conditional input and the matter-family chapters for quarks, charged leptons, and neutrinos.

The flavor derivation is deliberately constructive. It does not claim that the full OPH flavor observable is theorem-level. Instead, it builds the object chain that a closed theorem would have to pass through. The derivation chain has the following main architecture: start from a refinement-indexed family transport kernel, derive a centered generation-bundle branch generator, lift that to same-label transport data, derive the induced overlap-edge transport cocycle, reduce the cocycle to a persistent flavor observable, and then push that observable into common sector-response objects for the downstream quark, charged-lepton, and neutrino derivations.

Mathematically, this derivation is where "why this family exists" becomes a concrete technical question. The finite Standard Model packet gives a rank-three candidate; the flavor derivation is where one tries to turn that candidate and the declared count into a physical attachment, transport, splitting, suppression, phase, and excitation data. The relevant objects are the intermediate transport and spectral structures, not the final fermion masses themselves, that the mass readouts consume.

The active chain, in slightly compressed form, is:

1. normalize a refinement-indexed family transport kernel;
2. derive a centered generation-bundle branch generator on the conditionally declared three-generation charged bundle;
3. lift this to same-label edge-line transport and then to an overlap-edge cocycle with explicit defect and gap bookkeeping;
4. reduce those data to projectors, spectral gaps, pair suppressions, and cycle phases;
5. push the resulting family object into sector-response objects and then into suppression/phase tensors for the downstream matter derivations.

Three branch-specific claim boundaries are attached to this region of the derivation. The shared excitation dictionary is the common proof-facing base. Above that base, the charged branch carries exact centered readback, a closed common-shift no-go, the declared same-label q_e readback, a source-side determinant character for any fixed formal exponent vector

$$S_M = \sum_e M_e^{\mathrm{ch}} \log q_e,$$

a determinant-line lift on theorem-grade physical charged data, and an algebraic mass readout from theorem-grade $A_{\mathrm{ch}}(P)$. The theorem branch does not emit a theorem-grade sector-isolated charged determinant exponent vector, and it does not identify a source-side determinant character with the physical charged determinant line. The charged-lepton theorem gap is the determinant trace-lift attachment $3\mu(r) = S_M(r)$ on the charged determinant channel. The charged determinant channel has a source-domain no-go boundary: the uncentered trace lift is not emitted. The neutrino branch carries a target-informed weighted-cycle candidate above a template family kernel, with compare-only bridge and absolute-attachment diagnostics. The quark branch carries a theorem-grade source-spread obstruction: its ordered profile shapes leave a free $(\mathbb{R}_{>0})^2$ fiber. Selected-class descent proves representative independence but does not choose either positive modulus. Target-anchored mixed-convention mass coordinates and GeV-valued mass textures are comparison data; they are neither a source-only running sextet nor physical dimensionless Yukawa matrices. A separate compare-only microphysical bridge records edge-statistics transport on a diagnostic surface. Together these boundaries mark where the flavor derivation leaves the common transport backbone and passes to branch-specific closure contracts.

8 Quark Family Derivation

The quark branch carries a source-only non-identifiability theorem on the public physical quark frame class f_P chosen by P. The source equations determine the ordered up- and down-sector profile rays but leave their endpoint spans as two independent positive moduli. Selected-class descent does not remove this freedom, so no numeric mass row is emitted. Same-family and restricted common-refinement artifacts reproduce their chosen targets only after those moduli are obtained by target inversion. Their rows also mix renormalization conventions, while their GeV-valued matrices are mass textures rather than physical dimensionless Yukawa matrices. This theorem does not classify all public quark frame classes.

8.1 What quarks are in this derivation

Quarks are the color-charged elementary constituents of hadronic matter. In nature, the up quark and down quark dominate protons and neutrons, while the strange quark, charm quark, bottom quark, and top quark appear in progressively heavier and more unstable sectors. In the OPH particle derivation, the quark branch has a closed downstream algebraic readout conditional on a source-only physical spread datum. The target-free source equations do not emit that datum: they fix two ordered profile rays and leave their positive endpoint spans independent. The compatible fiber is $(\mathbb{R}_{>0})^2$, so the six numeric rows are not source-only predictions.

The quark route starts from the shared flavor excitation dictionary, internalizes the target-free mass bridge on the quark mass-profile ray, and proves selected-bridge-fiber representative independence for the attached physical spread datum on f_P, and applies the affine mean law once both positive moduli are supplied. The stored target-anchored matrices have GeV-valued singular values drawn from several comparison conventions. They are mass textures, not physical dimensionless Yukawa matrices. The same-label left-handed selector surface closes to the singleton σ_{ref}. That lower object is a negative sheet-selector statement on the selected quark sheet. It does not break the independent positive-rescaling action on the two sector profiles.

8.2 Emitted quark rows

Running quark masses are coordinates on a renormalization-group trajectory. Finite renormalizations and scale changes alter those coordinates while preserving physical amplitudes, so OPH can emit an RG-covariant trajectory or invariant but cannot derive the human choice of an $\overline{\mathrm{MS}}$ chart. The chart must be declared after source emission. The top row is a pole extraction coordinate rather than a sixth member of one common running-mass packet. All six rows are withheld because the spread pair is non-identifiable from the source domain. A physical Yukawa construction would additionally transport every running coordinate to one common scale, convert the top coordinate, emit the running Higgs expectation value in the same scheme, and apply $y_q(\mu) = \sqrt{2}\, m_q(\mu)/v(\mu)$.

8.3 Target-free mass bridge and selected-class theorem

The same-label left-handed solver surface closes to the singleton σ_{ref}. This is a negative same-sheet selector statement: same-sheet rephasing preserves CKM moduli, so it does not move that selected sheet to the physical CKM shell. Same-sheet overlap scans, chirality-swapped basis diagnostics, and other non-sector-attached orbit improvements are compare-only and do not change the selected-class theorem.

A separate target-free mass bridge is internalized on the emitted quark mass-profile ray. On the minimal light branch

$$y_u = c_u\, \varepsilon^6, \qquad y_d = c_d\, \varepsilon^6, \qquad \varepsilon = \frac{1}{6},$$

the light-quark overlap-defect theorem emits

$$\Delta_{ud}^{\mathrm{overlap}} = \frac{1}{6} \log \frac{c_d}{c_u}.$$

The emitted same-family mass object is the ray

$$\mathcal{R}_{ud}^{\mathrm{mass}} := \{(c_u, c_d) \in \mathbb{R}^2_{>0} : c_d/c_u \text{ fixed}\},$$

and on that ray the same scalar is equivalently the one-scalar law

$$\Theta_{ud}^{\mathrm{mass}} := \frac{1}{6} \log \frac{c_d}{c_u}.$$

The exact scalar identities are

$$\Delta_{ud}^{\text{overlap}} = \frac{t_1}{5}, \qquad \log\frac{c_d}{c_u} = \frac{6}{5}\,t_1, \qquad t_1 = 5\,\Delta_{ud}^{\text{overlap}} = \frac{5}{6}\log\frac{c_d}{c_u}.$$

The mass-profile package is therefore functorial:

$$\begin{aligned}
\eta_Q^{\text{centered}} &= -\frac{1-x_2^2}{27}\,t_1,\\
\kappa_Q &= -\frac{t_1}{54},\\
x_2 &= -0.5175863354681689.
\end{aligned}$$

The odd source package is also forced:

$$\begin{aligned}
\beta_{u,\text{diag},B}^{\text{source}} &= \frac{t_1}{10},\\
\beta_{d,\text{diag},B}^{\text{source}} &= -\frac{t_1}{10},\\
\mathbf{r}_u &= \left(-\frac{t_1}{10},\,0,\,+\frac{t_1}{10}\right),\\
\mathbf{r}_d &= \left(+\frac{t_1}{10},\,0,\,-\frac{t_1}{10}\right),
\end{aligned}$$

For any supplied positive spread pair one obtains

$$\tau_u = \frac{\sigma_d}{10(\sigma_u+\sigma_d)}\,t_1, \qquad \tau_d = \frac{\sigma_u}{10(\sigma_u+\sigma_d)}\,t_1.$$

These identities do not select σ_u or σ_d. The mass bridge lies outside the selected-class theorem boundary. The pure-B payload pair is a lower realization object beneath the exact theorem, and $\Theta_{ud}^{\text{mass}}$ is its derived scalar wrapper.

Write the same-label left-handed physical carrier as

$$\Sigma_{ud}^{\text{phys}} := \left\{(\sigma_{\text{id}}, \tau, U_{u,L}, U_{d,L}, V_{\text{CKM}}, I_{\text{CKM}}) : V_{\text{CKM}} = U_{u,L}^{\dagger} U_{d,L}\right\}/\!\sim,$$

where

$$(U_{u,L}, U_{d,L}, V) \sim (U_{u,L}D_u,\ U_{d,L}D_d,\ D_u^{\dagger} V D_d)$$

for diagonal $D_u, D_d \in U(1)^3$. On the selected public quark frame class f_P, represented on the realized branch by the explicit common-refinement transport-frame class $[F_0^{\dagger}F_1]$, the exact $\Sigma_{ud}^{\text{phys}}$ element and the attached exact sigma datum are independent of the declared representative inside the selected bridge fiber. Therefore the exact sigma readout descends uniquely to the selected comparison domain on f_P. It is not a public source-only mass prediction because the physical sigma datum is target-derived.

Sigma descent non-selection. Let $R_{\text{decl}}(f_P)$ be the declared selected bridge fiber over the public quark frame class. If a map

$$\Sigma : R_{\text{decl}}(f_P) \to \mathbb{R}^4$$

is constant on that fiber, then it descends to a well-defined public datum $\overline{\Sigma}(f_P)$. This proves representative independence only. It does not select the value of $\overline{\Sigma}(f_P)$, because every constant vector in $\mathbb{R}^4$ would also descend. Therefore a value obtained from the declared running-quark target surface is target-derived after descent.

Common-scale physical rejection of the reciprocal-ray candidate. A physical comparison first converts all six quark eigenvalues to dimensionless Yukawa singular values in one effective theory, scheme, and scale. Using the Standard Model $\overline{\text{MS}}$ values at M_Z tabulated by Antusch, Hinze, and Saad [11], the two ordered log-gap ratios are

$$\rho_u = 1.1108888543, \qquad \rho_d = 0.7519410008, \qquad \rho_u \rho_d = 0.8353228768.$$

The reciprocal-ray law requires the last product to equal one. It is therefore falsified on the properly typed target surface. Even after granting the four endpoint Yukawas (y_u, y_t, y_d, y_b), the minimax reciprocal shape misses the held-out y_c and y_s by 21.556% at M_Z. The result persists under running: the products are 0.836092, 0.837752, 0.845227, and 0.843611 at 10^3, 10^5, 10^{12}, and 10^{16} GeV, with best held-out errors between 19.93% and 21.42%. The stored $\rho = 1.294285$ formula misses the charm coordinate by 56.5% at M_Z, even under the endpoint grant.

The generic physical interface is therefore

$$(\mu_u, \sigma_u, \rho_u, \mu_d, \sigma_d, \rho_d) \in \mathbb{R}^6,$$

with an exact inverse to the six ordered positive Yukawas. The three-scalar theorem is exact only inside its imposed reciprocal-ray, affine-mean, fixed-shared-scale subfamily. The two-spread counterfamily below is a valid restricted non-identifiability lower bound, but it is not a parameterization of the whole physical interface.

A separate selector analysis sharpens the missing source object. Generation-blind flavor-singlet scalars cannot emit a fixed nonzero bifundamental Yukawa matrix. Simultaneous S_3 invariance restricts a fixed matrix to the span of I and J, which has a degenerate eigenspace and aligned up/down frames. A physical construction therefore needs a source-derived flavor-orbit selector, such as a spontaneous invariant functional or genuine bifundamental boundary data, together with a quark–Higgs carrier. The calibration simulator contains no quark, Higgs, or Yukawa coupling and its enumerated observables have zero Fisher information for these coordinates. Its 64k runs are at the calibration null: the two fusion-tower readings are 1.495245 and 1.484301, while the dense first/full readings are 1.498275 and 1.497823. Those values are not quark-mass evidence.

Restricted source-spread non-identifiability theorem. Remove every running-mass target, exact-witness, fitted-spread, and compare-only ancestor. Grant the remaining source packet its strongest ordered three-point shape law. For $\rho_{\text{ord}} > 0$, define

$$v_u = \frac{1}{3(1+\rho_{\text{ord}})}\big(-(2\rho_{\text{ord}}+1), \rho_{\text{ord}}-1, \rho_{\text{ord}}+2\big),$$
$$v_d = \frac{1}{3(1+\rho_{\text{ord}})}\big(-(\rho_{\text{ord}}+2), 1-\rho_{\text{ord}}, 2\rho_{\text{ord}}+1\big).$$

Both vectors have zero trace and unit endpoint span. Their adjacent-gap ratios are ρ_{ord} and ρ_{ord}^{-1}, respectively. Conversely, those three conditions determine each profile up to one positive endpoint span. Hence every compatible pair is

$$E_u = \sigma_u v_u, \qquad E_d = \sigma_d v_d, \qquad (\sigma_u, \sigma_d) \in (\mathbb{R}_{>0})^2.$$

The group $(\mathbb{R}_{>0})^2$ acts freely and transitively by independent rescaling of the two spans while fixing the source shape data. The requested four-tuple

$$\left(\frac{\sigma_u+\sigma_d}{2}, \frac{\sigma_u-\sigma_d}{2}, \sigma_u, \sigma_d\right)$$

is not invariant under this action. No unique source-only spread package follows from the stated source data.

Non-definability of the quark spectrum from the stated data. This ambiguity is a mathematical non-definability result. Fix any generic admissible one-Higgs three-generation package and write

$$Y_q = U_{q,L}\ \mathrm{diag}\Big(e^{\mu_q+\sigma_q v_{q,1}}, e^{\mu_q+\sigma_q v_{q,2}}, e^{\mu_q+\sigma_q v_{q,3}}\Big)\, U_{q,R}^{\dagger}, \qquad q\in\{u,d\},$$

with $\sum_i v_{q,i}=0$. For every $\lambda_u, \lambda_d > 0$, replace σ_q by $\lambda_q\sigma_q$ while holding the frames fixed. Gauge representations, anomaly cancellation, hypercharges, one-Higgs Yukawa completability, the CKM matrix, intrinsic CP capability, and the weak-sector counting clause are unchanged. In addition

$$\det\exp(\lambda_q\sigma_q v_q) = \exp\left(\lambda_q\sigma_q \sum_i v_{q,i}\right) = 1,$$

so determinant normalization does not select either modulus.

Every member of the family carries the same gauge group, matter content, color count, and generation count, so no stated structural datum separates them. which contains no Yukawa eigenvalue. The members are nevertheless physically inequivalent because the declared physical-equivalence relation preserves Yukawa invariants. Axiom 3 also does not remove the family: its gauge-invariant local constraint values and associated multipliers are not numerically specified and no map from P to quark Yukawa multipliers is emitted. Thus the stated axioms and declared completions plus fixed P admit a free $(\mathbb{R}_{>0})^2$ family of equally admissible quark spectra. There is no smallest positive rescaling; its infimum is zero, which would give the massless or degenerate boundary rather than the observed spectrum. Therefore the stated axioms do not define a unique quark mass spectrum. This conclusion uses no additional axiom and can be overturned only by deriving, from the existing axioms, a source functional that is nonconstant on this explicit counterfamily.

The kernel interface: the exact content of the missing derivation. Two companion theorems make that boundary executable. First, the admissibility battery recorded for the family-transport branch (positive-semidefinite hermitian descendants, open three-cluster gaps at every level, a simple centered spectrum, the conjugacy-Riesz margin, persistent projector labeling, and overlap-edge amplitudes above the floor) places no constraint on the spectrum of the centered compressed branch generator: for every pair $(r,s)\in(\mathbb{R}_{>0})^2$ there is a certificate-passing two-level kernel whose generator has raw gap ratio exactly r and spectral span exactly s. The construction places the target spectrum by unitary conjugation, $T = Q\ \mathrm{diag}(\sqrt{\mu})\, Q^{\dagger}$, so the descendant spectrum is exact, and shrinks the refinement drift geometrically until the Riesz margin passes. The persistence certificates are therefore a filter, not a generator: a kernel derivation that merely passes them cannot emit the ordered ratio constant, the mean-law coordinate, or the spans.

Second, the three-scalar interface on record is exact only on its imposed reciprocal-ray subfamily. At fixed shared scale g_{ch}, that subfamily factors through

$$(r,\sigma_u,\sigma_d)\in(\mathbb{R}_{>0})^3, \qquad \rho_{\mathrm{ord}} = \frac{3}{2+r}, \qquad x_2 = \frac{r-1}{r+1},$$

with the rays, the affine mean coefficients A_{ud}, B_{ud}, the sector means, and the six coordinates closed-form in the triple, and with the explicit left inverse $\sigma_u = \frac{1}{2}\ln(m_t/m_u)$, $\sigma_d = \frac{1}{2}\ln(m_b/m_d)$, and $r = 3/\rho - 2$ at $\rho = \ln(m_c/m_u)/\ln(m_t/m_c)$; the forward map is injective and the executed round trip closes at machine precision inside that subfamily. It is not the general interface of two ordered three-point spectra.

Indeed, for any $x \neq \pm 1$, let

$$L(x) = \mathrm{ctr}(-1, x, 1), \qquad Q(x) = \mathrm{ctr}(1, x^2, 1).$$

Their Gram determinant is

$$\det \mathrm{Gram}(L, Q) = \frac{4}{3}(1 - x^2)^2.$$

Thus L, Q form a basis of the centered three-vector plane, and every centered sector spectrum has unique coordinates $E_q = a_q L + b_q Q$. Once both a_q and b_q are free, x, and hence r, is a basis choice rather than an additional invariant of the spectrum. The general common-scale eigenvalue interface has six scalar coordinates: two centered coordinates and one mean for each sector. A proposed r plus four centered coordinates plus two means is a redundant seven-coordinate chart. The reciprocal-ray law removes one centered coordinate per sector and thereby recovers its special three-scalar chart.

By the freedom theorem and the non-definability theorem above, the stated source data select neither the ray triple nor the six-scalar physical interface. The executable acceptance harness is only a fail-closed score of the reciprocal-ray subfamily.

The edge data do not remove the ambiguity. Even after granting source values $S_{13}, S_{23}, \delta_{21} > 0$, every positive pair can be written as

$$\sigma_u = S_{13} + c_u \delta_{21}, \qquad \sigma_d = S_{23} + c_d \delta_{21}$$

for suitable (c_u, c_d). The source equations emit no rule fixing these two coefficients. The specific coefficients used by the numerical candidate are therefore a declared ansatz from a hand-written family-transport template, not an OPH-derived kernel.

Define

$$\sigma^{ud}_{\mathrm{seed}} := \frac{\sigma_u + \sigma_d}{2}, \qquad \eta_{ud} := \frac{\sigma_u - \sigma_d}{2},$$
$$A_{ud} := \frac{1}{2(1 + \rho_{\mathrm{ord}} - x_2^2)},$$
$$B_{ud} := \frac{1}{2\Big(1 - x_2^2 - \frac{x_2^2}{1+\rho_{\mathrm{ord}}}\Big)},$$
$$\rho_{\mathrm{ord}} = 1.294284936377706.$$

Then

$$g_u = g_{\mathrm{ch}} \exp\!\big(-(A_{ud}\sigma^{ud}_{\mathrm{seed}} - B_{ud}\eta_{ud})\big), \qquad g_d = g_{\mathrm{ch}} \exp\!\big(-(A_{ud}\sigma^{ud}_{\mathrm{seed}} + B_{ud}\eta_{ud})\big).$$

The Jacobian from (σ_u, σ_d) to $(\log(g_u/g_{\mathrm{ch}}), \log(g_d/g_{\mathrm{ch}}))$ has determinant $-A_{ud}B_{ud} \neq 0$. The two free moduli therefore change the mass readout; they are not a gauge redundancy. Given a source-only physical spread datum, the affine mean law emits (g_u, g_d) algebraically on f_P. The target-derived packet is retained as a mixed-convention comparison witness. Its dimensionful matrices are mass textures. The conditional selected-class route can be written as

$$f_P + \Sigma^{\mathrm{source}}_{ud}(P) \Longrightarrow (\sigma_u, \sigma_d, \sigma^{ud}_{\mathrm{seed}}, \eta_{ud})_{\mathrm{phys}} \Longrightarrow (g_u, g_d) \Longrightarrow (m_u, m_d, m_s, m_c, m_b, m_t),$$

followed, only after common-scale RG transport and division by the running Higgs expectation value, by dimensionless Yukawa matrices. The first arrow is obstructed by the free $(\mathbb{R}_{>0})^2$ action on the stated source domain.

Target-anchored S_3 two-mode witness. An ansatz built with visible target coordinates reproduces the six mixed-convention comparison coordinates numerically. Its exact mathematical component starts from the transposition Cayley graph of S_3. The adjacency spectrum is $3, 0, -3$ with multiplicities $1, 4, 1$, so the Laplacian spectrum is $0, 3, 6$ and

$$\frac{e^{-3\tau} - e^{-6\tau}}{1 - e^{-3\tau}} = e^{-3\tau}.$$

This identity is a finite-group theorem. No OPH theorem identifies the three distinct isotypic heat values with three generations or supplies the proposed heat time

$$\tau_f = \frac{P}{4} - \frac{\pi\alpha_U}{5}.$$

The ansatz sets $w = \pi\alpha_U$, $r = e^{-3\tau_f}$, $\rho = 3/(2+r)$, and $x = (r-1)/(r+1)$, and then uses

$$\begin{aligned}
e_u &= S_{13} + \frac{\rho\delta_{21}}{1+\rho}, & e_d &= S_{23} + \frac{\delta_{21}}{2(1+\rho-x^2)},\\
\bar\sigma_u &= e_u - w\Delta S_{13}, & \bar\sigma_d &= e_d - w(1-\Delta S_{13}),\\
a_u &= e_u + \frac{w}{2}, & a_d &= e_d - \frac{w}{5},\\
b_u &= b_u^{\mathrm{ray}} - \frac{w\rho}{10}, & b_d &= b_d^{\mathrm{ray}} - \frac{w}{4}.
\end{aligned}$$

The ray coordinates have the symbolic form

$$b_u^{\mathrm{ray}} = a_u\frac{-\rho x + \rho - x - 1}{(1+\rho)(x^2-1)}, \qquad b_d^{\mathrm{ray}} = a_d\frac{-\rho x - \rho - x + 1}{(1+\rho)(x^2-1)}.$$

The ansatz feeds $\bar\sigma_u, \bar\sigma_d$ into the candidate affine mean law above and sets $E_q = a_qL + b_qQ$. With the declared comparison inputs, this gives

$$\begin{gathered}
r = 0.3179975211, \qquad \rho = 1.2942205385, \qquad x = -0.5174535369,\\
(a_u, b_u) = (5.6430129216, -4.9927828217), \qquad (a_d, b_d) = (3.3952170977, -1.8369623926).
\end{gathered}$$

Conditional normalized-trace lemma. Let $\ell : M_d(\mathbb{C}) \to \mathbb{C}$ be complex linear, invariant under all unitary conjugations, and normalized by $\ell(I) = 1$. Unitary conjugacy of rank-one projectors and additivity over an orthonormal resolution of I give $\ell(A) = \mathrm{Tr}(A)/d$, so every rank-one channel has weight $1/d$. Likewise, a width operator on $M_d(\mathbb{C})$ invariant under the full $U(d) \times U(d)$ left-right action is scalar by Schur's lemma; total Hilbert–Schmidt width t therefore assigns t/d^2 to each unit slot. Consequently, *if* the physical heat, up-odd, down-odd, up-even, and down-even responses are respectively identified with one isotropic slot in $M_5(\mathbb{C})$ of total width $5w$ and rank-one modules of dimensions $2, 5, 10, 4$, the coefficients $w/5, w/2, w/5, \rho w/10, w/4$ follow. The full invariance and normalization of the response law, those five module assignments, their signs and orientation, and exclusion of competing assignments are hypotheses outside OPH. Thus this lemma closes the denominator arithmetic conditional on the physical channel functor; it does not remove target dependence from the formula.

The per-particle comparison table is not a source-only prediction. It is a target-anchored diagnostic: the evaluator emits dimensionless coordinates, inserts an unproved one-GeV unit, and compares them to a table that mixes light-quark $\overline{\mathrm{MS}}$ values at 2 GeV, heavy-quark self-scale values, and a separate top extraction.

The raw diagonal residual sum against the supplied central values and nominal standard deviations is 1.1653, with a maximum relative residual of 0.2946%. It is not a likelihood or goodness-of-fit statistic: discovery used these targets, there is no source-theory covariance, and the rows are not one common-scale spectrum. In the target-conditioned 219,615-member denominator grammar, the selected tuple $(5, 2, 5, 10, 4)$ is the unique minimum of that raw residual sum, but eight formulas tie its best maximum error. The grammar contains the target-exposed inputs, graph, entries, assignments, signs, and functional forms, so it is not a global look-elsewhere correction.

The evaluator is runtime-target-separated but target-dependent. $S_{13}, S_{23}, \delta_{21}, \Delta S_{13}$, and $g_{\rm ch}$ descend from the explicitly hand-written family-transport template. The last quantity is the dimensionless template-eigenvalue mean plus its minimum gap; treating it as GeV supplies the missing unit by hand. The selected pixel branch depends on an internal quark-spectrum continuation. The edge laws add log-overlap suppression to a linearly scaling $TT^\dagger$ gap, and ΔS_{13} is a selected basis-dependent matrix entry. The ansatz is target-informed and supplies no physical mass law. It does not contradict the non-identifiability theorem.

Sufficient source conditions for flavor. The proof analysis isolates a sufficient conditional completion without instantiating it. A physical theorem would require all of the following on one acyclic, target-free source dependency graph:

1. a unique interval-certified source root for $P, \alpha_U(P)$, with no dependency on an internal quark continuation;
2. a physical S_3 family-carrier attachment, the heat-time law $\tau_f = P/4 - \pi\alpha_U/5$, the ρ, x dictionary, and the two edge-response laws;
3. a source-labeled simple-spectrum family generator and charged seed with exact common-refinement intertwiners, monomial transport of the label rays, positive common normalization, and selected edge magnitudes bounded away from zero;
4. a physical response-channel functor that proves the normalized-trace hypotheses, module assignments, signs, orientation, competing-channel exclusion, and the affine mean law on that carrier, on a domain where the A, B denominators do not vanish;
5. a common-scale physical readout with $m_q(\mu) = y_q(\mu)v(\mu)/\sqrt{2}$, including field normalization, units, and the running Higgs expectation value in the same scheme;
6. a source-only RG packet with existence and non-blowup across the required intervals, a locally Lipschitz beta system, fixed threshold ordering and matching maps, top conversion, and the declared comparison charts.

Given these hypotheses, composition of the single-valued maps makes the final sextet unique. This is a sufficient proof-obligation contract, not a proved-minimal OPH theorem. The numerical table is not its corollary: the evaluator's coordinates are dimensionless and no instance of the final two receipts exists. What the counterfamily proves as a necessary condition is narrower and decisive: any successful source functional must be nonconstant on the independent (λ_u, λ_d) centered-spread rescaling orbit.

Target-conditioned cumulant specification. The representation-slot cumulant ansatz declares

$$F = \mathbb{C}^3_{\rm perm} \oplus V_{\rm std} \cong \mathbf{1} \oplus 2V_{\rm std}, \qquad \dim F = 5, \qquad \dim F_0 = 4,$$

and the composite response dimensions

$$(29, 432, 22, 32, 840, 1008, 432, 1584).$$

These integers are arithmetically correct for the declared direct sums and tensor products. If a continuous Gaussian source space, full unitary isotropy, reversible independent sheets, the listed effect ranks, and the listed signs are additionally assumed, the second-cumulant identity gives

$$\log \mathbb{E}[e^X] = \frac{1}{2}\,\mathrm{Var}\,X = \frac{r}{d}w^2.$$

This conditional identity does not construct the physical effects or select the modules. In particular, $\dim \mathrm{End}_{S_3}(\mathbf{1} \oplus 2V_{\mathrm{std}}) = 5$, so S_3-invariance does not force one scalar covariance on the two-copy multiplicity space. Pooling heterogeneous sums such as $\mathrm{End}(F) \oplus F_0$ into a single $1/29$ response requires an additional symmetry that mixes physically distinct blocks. The proposed F also has heat spectrum $0^{(1)}, 3^{(4)}$, not the regular representation's additional sign-sector value $6^{(1)}$; the physical regular-heat-to-family attachment is open. Finally, a genuine Gaussian covariance term is nonnegative. The negative w^2 entries require a separate signed response or subtraction law, not orientation reversal alone.

With $w = \pi\alpha_U$, the declared cumulant formulas are

$$\bar{\sigma}_u = 3P + \left(5 + \frac{4}{15}\right) w + \frac{w^2}{29}, \quad \bar{\sigma}_d = 2P + \frac{4}{15}w - \frac{w^2}{432},$$
$$a_u = 3P + \left(5 + \frac{4}{5}\right) w + \frac{w^2}{22}, \quad a_d = 2P + \left(1 + \frac{1}{32}\right) w + \frac{w^2}{420},$$

and

$$\frac{g_{\mathrm{ch}}}{v} = 2\exp\left[-2\pi + \frac{P}{1008} + \frac{w^2}{432} - \frac{w^2}{1584}\right].$$

Conditional on these formulas and the inherited heat-time, ray, even-response, affine A, B, and exponentiation laws, the downstream map is deterministic. Using the electroweak v-display gives a maximum residual of 0.2943587% and a raw diagonal nominal-residual sum of 1.1613966 against the same mixed-convention comparison table. Neither number is a likelihood: the calculation uses visible target-derived effective coordinates, there is no common-scale renormalization-group packet or theory covariance, and the supplied cumulant comparison changes the nominal u-row uncertainty relative to the preceding bundle.

Target dependence and ablation are decisive. The target-inferred denominators for four visible effective coordinates are approximately

$$29.19665, \qquad 428.18579, \qquad 21.90604, \qquad 836.47734,$$

close to the selected $29, 432, 22, 840$, while no target-independent grammar of alternative module expressions is specified. The zero-w^2, zero-δ_g ablation has the lower maximum residual 0.2142293% and raw residual sum 0.6694007. This does not make the ablation a physical theory; it shows that the detailed covariance data is not selected by the numerical agreement. The pixel input depends on an internal quark model, and the selector uses mixed sources.

The ansatz is an explicit, falsifiable target-conditioned specification. The claimed rescaling-orbit statement must also be scoped correctly:

$$\|\lambda E_q - E_q\|^2 = (\lambda - 1)^2 \|E_q\|^2$$

has its minimum at one only on the orbit through the declared cumulant vector. For a generic base vector C_q, the minimizer of $\|\lambda C_q - E_q\|^2$ is $\langle C_q, E_q\rangle / \|C_q\|^2$, and no OPH dynamics requires minimizing this postulated residual. The target-conditioned module ansatz does not satisfy the source conditions above or define a physical quark-mass law.

Finite maximum-entropy boundary and conditional register rigidity. The finite-MaxEnt premise is false. On a finite spectrum, maximizing entropy under mean and covariance constraints gives a discrete exponential-quadratic Gibbs law, not a Gaussian density. Explicitly, on support $\{-2, -1, 0, 1, 2\}$,

$$p_A = (1/12, 1/6, 1/2, 1/6, 1/12), \qquad p_B = (1/16, 1/4, 3/8, 1/4, 1/16)$$

both have mean zero and variance one, while $\kappa_4(p_A) = 0$ and $\kappa_4(p_B) = -1/2$. The actual finite-support MaxEnt law at variance one has a nonzero fourth cumulant (approximately -0.46815). Hence neither finite maximum entropy nor the first two moments entails the Gaussian two-cumulant truncation. Gaussianization can be recovered only conditionally from an exported triangular array with a Lindeberg or adequate mixing/bounded-dependency condition, normalization, and covariance convergence; no such array is emitted by the source.

The quark-register proposal instead states an exact primitive-path certificate. Conditional on its nine declared path hypotheses, normalized trace fixes each path weight as rank over the declared register dimension, including

$$-\frac{1}{5}, \frac{4}{15}, \frac{1}{29}, -\frac{1}{432}, \frac{4}{5}, \frac{1}{22}, \frac{1}{32}, \frac{1}{420}, -\frac{1}{10}, -\frac{1}{4}, \frac{1}{1008}, \frac{1}{432}, -\frac{1}{1584}.$$

Exact record projectors also allow only the zero or unit scalar multiplier, and inert tensor refinement preserves normalized trace weights. This is a useful rigidity theorem, but its hypotheses contain the required physical content: the typed register construction, exhaustive primitive-path catalogue, effect ranks, structural multiplicities, signs, winding character, refinement intertwiners, implementation invariance, and positive-gap branch selector. A neutral finite register can be changed while preserving the broad structural signature and changing the normalized response, so the three axioms alone do not select this register. That countermodel is scoped to the broad signature; it is not a realization of every stronger carrier condition one might add.

The selector results sharpen the same boundary. An equivariant section requires a stabilizer-fixed candidate in each orbit fiber; an ambiguous transitive fiber need not have one. Conversely, a complete finite invariant candidate class with a unique positive-gap minimizer has a stable selector when refinement error is below half the gap. The proposal supplies neither a physically complete candidate class nor a source-derived cost and gap. The absolute-scale rescaling no-go, finite-renormalization scheme ambiguity, interval-root schema, and piecewise-RG uniqueness theorem are likewise exact obstruction or conditional well-posedness statements. No actual target-free source map with strict interval inclusion, operational clock, or fixed beta/threshold/matching packet accompanies the proposal. The register composes algebraically with the target-conditioned cumulant ansatz and authorizes no numerical quark-mass claim.

Oriented family-transport carrier test. The primitive oriented family-transport proposal supplies explicit complex 3×3 matrices T_0, T_1, but the simulator test must not manufacture them through a fitted readout. The fixed direct observable is therefore the mean of the natural three-label permutation matrices carried by the saved oriented S_3 edges. Its singular-value ratios are invariant under family permutations, unitary row/column phases, and an overall scale, so they give a necessary comparison before any entrywise match is considered. The proposal requires

$$s(T_0)/s_1 = (1, 0.55147\ldots, 0.25517\ldots), \qquad s(T_1)/s_1 = (1, 0.54625\ldots, 0.27451\ldots).$$

A 4,096-patch BW run, an independent 4,096-patch fusion run, a dense 65,536-patch population run, and a 65,536-patch BW run contain 830,066 edges in total. Their corresponding triples are

$$(1, 0.00522, 0.00134), \quad (1, 0.00552, 0.00361), \quad (1, 0.00179, 0.00002), \quad (1, 0.00071, 0.00028).$$

Source-node block bootstraps retain the same near-rank-one result. Every state lies within the fixed 0.02 Haar-null tolerance and more than 0.5407 from both proposed targets under the fixed 0.05 tolerance. The recorded state contains only edge endpoints, integer S_3 labels, and geometry: it exports no complex oriented family amplitude and no paired coarse/fine edge intertwiner. Hence all direct T_0, refined T_1, and joint physical-emission receipts are false. The conclusion is scoped to the natural direct carrier. A source-derived complex lift could define a different carrier. Choosing its phases or paths to reconstruct the proposed matrices would make the simulator test circular.

Maximal source-derived quark package. Let $\mathcal{P}_q$ denote the three axioms together with the declared finite matter, transport, and quark-source premises used in this section. The quark-side result has four layers:

1. the emitted mass-profile ray
$$\mathcal{R}_{ud}^{\mathrm{mass}} = \left\{ \lambda \left(\frac{1}{5}, -\frac{1-x_2^2}{27} \right) : \lambda \geq 0 \right\}, \qquad \lambda = t_1;$$
2. the same-label left-handed selector value
$$\sigma_{ud} = \sigma_{\mathrm{ref}},$$
which is a negative sheet-selector statement;
3. the separate target-free mass bridge
$$\mathcal{P}_q \vdash \Delta_{ud}^{\mathrm{overlap}} = \frac{1}{6} \log \frac{c_d}{c_u}, \qquad \mathcal{P}_q \vdash \Theta_{ud}^{\mathrm{mass}} = \frac{1}{6} \log \frac{c_d}{c_u};$$
4. the selected-class support wrapper on f_P, which proves representative independence for the attached physical spread datum. It does not select either modulus. Conditional mass readout would also require an RG-covariant trajectory and a declared comparison chart; physical Yukawa matrices require the additional common-scale dimensionless conversion.

Separate same-family and common-refinement artifacts reproduce their chosen target rows exactly. Their sigma datum is obtained by inversion of those rows. The associated GeV-valued matrices are target-anchored mass textures. A separate calculation also backreads a mass-side scalar after identifying coefficient ratios with target mass ratios. None of these surfaces breaks the free source-spread action.

8.4 Quark Theorem Boundary

The conditional downstream closure route is

$$f_P \quad + \quad \Sigma_{ud}^{\mathrm{source}}(P) \Longrightarrow \left(\sigma_u, \sigma_d, \sigma_{\mathrm{seed}}^{ud}, \eta_{ud}\right)_{\mathrm{phys}} \Longrightarrow (g_u, g_d) \Longrightarrow (m_u, m_d, m_s, m_c, m_b, m_t),$$

The first arrow is not identified by the source theory. The last arrow denotes scheme-labelled mass coordinates only after an RG trajectory and comparison chart are specified. Physical dimensionless Yukawa matrices lie one common-scale normalization beyond it. Selected-fiber descent and global frame classification are logically separate from both obstructions.

The conditional matter lift and the conditional port-current algebra do not cut this fiber. Both certificates are coefficient-blind: the lift fixes the gauge representations, the chirality certificate, the anomaly traces, and exactly one Yukawa invariant line per declared channel, the port-current receipt carries no Yukawa-adjacent datum, and independent positive rescaling of the coefficients along the two hadronic invariant lines fixes every certified conclusion of both. The spread fiber survives the certified structure set with its coordinates realized as the free scalar coefficients along those lines. A machine-checked transport certificate records the survival together with the exclusion of the twelve frozen orbit-selector candidates, and flips fail-closed if any certified datum ever moves under the rescaling. A cut of this fiber requires a physical binding of the response representation, the attachment of the screen action to three physical families, or a selector admitted under the frozen single-comparison discipline.

The same certificates permit and type a possible register relation at the unification scale; they supply neither a physical equality between independent Yukawa coefficients nor a physical generation order. The declared one-scalar package has one invariant line on each of the down-conjugate and charged-lepton-conjugate channels through the same scalar, together with a certified zero line on the channel that would use the up scalar for the down-type field. Thus any separately supplied relation of this form uses the down-type and charged-lepton channel pair rather than the forbidden up-scalar channel. Conditional on the declared alphabet and the two transitive-color-orbit constraints of measure balance and register faithfulness, exhaustive enumeration selects the unordered weight multiset $\{1/3,\ 1,\ 3\}$ with no measured mass or angle in its solve path. A separate comparison-only calculation of all six assignments ranks

$$(y_b/y_\tau,\ y_s/y_\mu,\ y_d/y_e) = (1,\ 1/3,\ 3)$$

as uniquely least discrepant. That target-informed empirical ranking does not derive the generation order from the source axioms. Conditional on the adopted assignment and the charged-lepton triple, common down-sector running cancels exactly on the declared one-loop branch, so the sharp RG-protected result is

$$\frac{m_s}{m_d} = \frac{1}{9}\left.\frac{m_\mu}{m_e}\right|_{\mu_U} = 22.9743.$$

FLAG 2024 Tables 11–12 give $m_s/m_{ud} = 27.227(81)$ and $m_u/m_d = 0.465(24)$ for $N_f = 2+1+1$, and 27.42(12) and 0.485(19) for $N_f = 2+1$ [12]. Their derived central values $m_s/m_d = 19.9438$ and 20.3594 make the conditional route value 15.2% and 12.8% high. Across all six assignments, the distinct light-family coefficient-ratio menu is $\{1/9,\ 1/3,\ 3,\ 9\}$, and both FLAG rows reject every assignment under the conservative experimental-only comparison gate. The unavailable input covariance and absent OPH theory uncertainty preclude a covariance-aware significance or a theory-wide falsification conclusion. This result applies only to the declared common-transport assignment family. Different coefficient relations, coefficient alphabets, charged-family attachments, or generation-dependent threshold transport define other hypothesis classes. The retained conclusions are the conditional pairing theorem, the target-free unordered multiset under the declared measure-balance and register-faithfulness constraints, and the exact positive-chamber Koide identity. The stipulated charged-lepton response model is neither blind nor source-derived. The same conditional chart gives $m_b(m_b) = 6.03\,\mathrm{GeV}$, $m_s(2\,\mathrm{GeV}) = 140\,\mathrm{MeV}$, and $m_d(2\,\mathrm{GeV}) = 6.1\,\mathrm{MeV}$, respectively 44.2%, 50.3%, and 30.1% above the recorded comparison coordinates. These absolute values are not source-only results, and the third-generation normalization and threshold packet are open.

Scoped residual-axis no-go. Hermitian circulants on the same C_3 face fiber commute and have the same discrete-Fourier eigenbasis. A direct same-fiber residual construction therefore gives identity mixing. For a direct construction from two distinct real three-dimensional residual axes, the six fivefold, ten threefold, and fifteen twofold unoriented icosahedral axes give the exhaustive acute-angle spectrum

axis families	acute angles in degrees
5×5	63.4349
5×3	$37.3774,\ 79.1877$
5×2	$31.7175,\ 58.2825,\ 90$
3×3	$41.8103,\ 70.5288$
3×2	$20.9052,\ 54.7356,\ 69.0948,\ 90$
2×2	$36,\ 60,\ 72,\ 90$

The smallest nonzero entry is 20.9052°, with sine 0.356822. The compare-only PDG 2024 $K_{\mu 2}$ coordinate $|V_{us}| = 0.2250(4)$ [13] corresponds to $\arcsin |V_{us}| = 13.0029^\circ$ at its central value, which is absent. The enumeration therefore excludes direct equality between the Cabibbo angle and an acute angle between two of these 31 real axes. It does not exclude spinorial representations, higher-order symmetry breaking, additional dynamical corrections, or general overlap models. The $31.7175^\circ = \arctan(1/\varphi)$ entry provides an exact geometry self-check.

The register-Clebsch branch's $\sqrt{m_d/m_s} = 0.2086$ display is the Gatto–Sartori–Tonin texture estimate formed from the same rejected mass ratio. The branch supplies no up/down Yukawa-matrix pair or relative left-handed eigenbasis, so the display is not an independent Cabibbo prediction.

8.5 Strong-CP branch

The selected-class quark wrapper carries target-anchored mass textures on the public quark frame class f_P. Strong CP is a separate phase-side invariant. The stated source domain does not derive the bare QCD angle θ_{QCD}, does not emit the physical anomaly-invariant combination $\bar{\theta}$, and does not prove that the physical strong-CP phase vanishes.

This phase problem is independent of the source-spread and common-scale Yukawa obstructions. A closure would require a source-emitted quark mass matrix at one declared scale, its physical determinant-line phase contribution, and a theorem fixing the topological-angle contribution on the realized branch. The present GeV mass textures do not supply that input.

9 Charged-Lepton Family Derivation

The charged-lepton derivation differs from the quark derivation in a precise way. It does not emit public charged masses from P. It emits an exact same-family witness, a closed common-shift no-go, the declared same-label q_e readback, a determinant-line lift on theorem-grade physical charged data, and a downstream algebraic mass readout from theorem-grade $A_{\mathrm{ch}}(P)$. For any fixed formal source exponent vector $M^{\mathrm{ch}}_{\bullet}$, the same-label readback defines a source-side determinant character

$$S_M = \sum_e M^{\mathrm{ch}}_e \log q_e.$$

The theorem branch does not emit a theorem-grade sector-isolated charged determinant exponent vector, and it does not identify a source-side determinant character with the physical charged determinant line. The source-only theorem therefore supplies no electron, muon, or tau mass.

The mathematical split is equally precise. The charged-lepton derivation starts from the ordered charged package, proves that the realized support is a one-dimensional linear subray, exposes the canonical quadratic support-extension direction, maps that into the charged excitation gaps, closes a two-scalar support-extension law shell, isolates the smaller eta source-readback primitive on that same carrier, and then builds the log-spectrum and forward shape/scale surface. Those centered objects are common-shift invariant. The determinant line fixes the physical affine scalar once theorem-grade physical charged data are present, and the mass readout is then algebraic. The charged theorem boundary has two distinct pieces: physical selection of the latent charged sector-response candidate as $\widehat{C}_e$, and source-to-physical determinant attachment. For a fixed formal source exponent vector $M^{\mathrm{ch}}_{\bullet}$, that attachment is the identity

$$3\mu(r) = \sum_e M^{\mathrm{ch}}_e \log q_e(r),$$

equivalently zero determinant-normalization defect

$$N_{\det}(P) = s_{\det}(P) - \sum_e M^{\mathrm{ch}}_e \log q_e(P),$$

on the charged determinant channel.

Physically, the electron is the light charged lepton that makes atoms and chemistry possible, the muon is its heavier unstable cousin, and the tau lepton is the heaviest charged lepton. A physical charged-lepton derivation would have to explain how those three rows emerge from the shared flavor dictionary without Koide-assisted fitting. The theorem-grade branch must select the latent charged sector-response candidate $\widehat{C}^{\mathrm{cand}}_e$ as $\widehat{C}_e$ through the branch-generator splitting theorem. On theorem-grade physical Y_e, a refinement-stable uncentered lift collapses the determinant-line section and affine anchor to one descended physical affine scalar $\mu_{\mathrm{phys}}(Y_e)$, with

$$\widetilde{C}_e(Y_e) = \widehat{C}_e(Y_e) + \mu_{\mathrm{phys}}(Y_e)\,\mathbf{1}, \qquad s_{\det}(Y_e) = 3\mu_{\mathrm{phys}}(Y_e), \qquad A_{\mathrm{ch}}(Y_e) = \mu_{\mathrm{phys}}(Y_e).$$

Theorem 9.1 (Charged same-carrier source-pair readback)**.** *Fix the ordered charged carrier*

$$(-1, x_2, 1),$$
$$x_2 = -0.5175863354681689.$$

If the charged source pair

$$(\eta_{\mathrm{ext}}, \sigma_{\mathrm{ext}}) = (\eta_{\mathrm{source\_support\_extension\_log\_per\_side}}, \sigma_{\mathrm{source\_support\_extension\_total\_log\_per\_side}})$$

is emitted on that carrier, then the centered charged logs are

$$e_{\mathrm{log,centered}} = -\frac{(3+x_2)\sigma_{\mathrm{ext}} - \eta_{\mathrm{ext}}}{6},$$

$$\mu_{\mathrm{log,centered}} = \frac{x_2\sigma_{\mathrm{ext}} - \eta_{\mathrm{ext}}}{3},$$

$$\tau_{\mathrm{log,centered}} = \frac{(3-x_2)\sigma_{\mathrm{ext}} + \eta_{\mathrm{ext}}}{6},$$

and therefore the charged masses are

$$m_e = g_e\, e^{e_{\mathrm{log,centered}}}, \qquad m_\mu = g_e\, e^{\mu_{\mathrm{log,centered}}}, \qquad m_\tau = g_e\, e^{\tau_{\mathrm{log,centered}}}.$$

This theorem is exact on the same-carrier shell, but the absolute values are not emitted. The visible scalar order is

$$\eta_{\mathrm{ext}} \quad \text{then} \quad \sigma_{\mathrm{ext}},$$

and the charged absolute scale g_e is unresolved. The scalars η_{ext} and σ_{ext} are the first same-carrier residuals. If the latent candidate $\widehat{C}_e^{\text{cand}}$ is selected as the physical operator, then η_{ext} and σ_{ext} become charged spectral invariants instead of separate primitive goals, and the absolute-scale burden is pushed to one affine-covariant absolute charged anchor A_{ch}. In the local chain, $\widehat{C}_e$ itself is undeclared. The available object is only the centered compressed generation-bundle candidate $\widehat{C}_e^{\text{cand}}$. Its missing theorem is the compression-descendant commutator condition after the central split. The centered common-shift quotient is closed negatively, so centered data alone do not emit the affine anchor A_{ch}.

Theorem 9.2 (Charged absolute-scale underdetermination)**.** *Let*

$$E_e^{\text{centered}} = \left(e_{\text{log,centered}}, \mu_{\text{log,centered}}, \tau_{\text{log,centered}}\right)$$

be the centered charged log triple emitted from the charged source pair $(\eta_{\text{ext}}, \sigma_{\text{ext}})$*. Then for every* $c \in \mathbb{R}$*,*

$$Y_e(c) := \exp(c)\,\mathrm{diag}\!\left(e^{E_e^{\text{centered}}}\right)$$

has the same charged spectral invariants

$$\eta_{\text{ext}}, \qquad \sigma_{\text{ext}}, \qquad \gamma_{21}, \qquad \gamma_{32},$$

the same centered log vector, and the same ratio data. Only the absolute masses scale:

$$(m_e, m_\mu, m_\tau) \longmapsto e^c\,(m_e, m_\mu, m_\tau).$$

Hence the charged OPH chain determines only the quotient class

$$E_e^{\text{centered}} \in \mathbb{R}^3/\langle(1,1,1)\rangle,$$

not the absolute scalar $g_e = e^c$*.*

The proof is immediate from the centered sum rule

$$e_{\text{log,centered}} + \mu_{\text{log,centered}} + \tau_{\text{log,centered}} = 0,$$

which implies

$$\det Y_e^{\text{shape}} = 1, \qquad \det Y_e = g_e^3.$$

All emitted charged invariants depend only on differences of log entries, so a common shift in the $(1,1,1)$ direction leaves them unchanged. The stated source domain therefore gives a no-go for the P-driven charged mass: the theorem fixes the centered shape, while the determinant-line landing fixes the absolute normalization.

The charged absolute-scale branch is explicitly typed. The charged scale is a linear quantity

$$g_e = e^{\mu_e^{\text{abs}}},$$

so the log-coordinate μ_e^{abs} must not be mixed directly with centered log gaps. The same-family writeback therefore records the type-consistent shell

$$\begin{aligned}\mu_{e,\text{seed}}^{\text{abs}} &= \log(0.9231656602589082)\\ &= -0.07994658034676537,\end{aligned}$$

$$\mu_{e,\text{cand}}^{\text{abs}} = \mu_{e,\text{seed}}^{\text{abs}} - \gamma_{\min} = -0.38231224060567365,$$

$$g_e^{\mathrm{cand}} = e^{\mu^{\mathrm{abs}}_{e,\mathrm{cand}}} = 0.6822819838027987.$$

This is a representation-consistency shell only, not a charged-mass theorem. It fixes the linear-vs-log coordinate discipline. This surface does not emit a theorem-grade value law for g_e. The checked shortcut

$$\Delta_e^{\mathrm{abs}} = 0.30236566025890826,$$
$$g_e = 0.6822819838027987$$

is not a theorem-grade closure. It merely chooses one representative on the common-shift orbit and does not land on the physical charged masses.

A physical completion requires a theorem-grade affine-covariant section A_{ch} of the quotient map, satisfying

$$A_{\mathrm{ch}}(\ell + c\,\mathbf{1}) = A_{\mathrm{ch}}(\ell) + c.$$

The clean realization of that section is an uncentered charged response lift carrying a determinant line, in which

$$A_{\mathrm{ch}} = \tfrac{1}{3}\log\det(Y_e) = \tfrac{1}{3}\,\mathrm{tr}(\log Y_e).$$

Once such an anchor exists,

$$g_e = e^{A_{\mathrm{ch}}(\ell)}, \qquad \Delta_e^{\mathrm{abs}} = \log g_{\mathrm{ch}}^{\mathrm{shared}} - A_{\mathrm{ch}}(\ell).$$

Proposition 9.3 (Twenty-four-register rate no-go)**.** *An oriented register with twenty-four slots, a twenty-four-stage automaton, or a decomposition into twenty-four formal update steps determines no physical frequency, Hamiltonian gap, or mass scale. Indeed, rescaling every physical generator by $H \mapsto \lambda H$, $\lambda > 0$, preserves the register, transition graph, grading, and settled quotient while multiplying all generator gaps and rates by λ. A repair iteration counter is likewise not an operational clock. A physical rate requires a source-derived clock instrument, calibrated comparison line, and residual bound.*

Theorem 9.4 (Conditional charged determinant-clock attachment)**.** *Let L_{24} be a one-dimensional source-derived operational clock line whose calibrated positive gap has norm Δ_{24}. Suppose there is a refinement-natural, quotient-visible norm-preserving line isomorphism*

$$\Theta_e : L_{24}^{\otimes 3} \longrightarrow \det M_e$$

to the physical charged-lepton determinant line. Then the common charged mass scale is fixed by

$$g_e := |\det M_e|^{1/3} = \Delta_{24}.$$

More generally, a separately derived normalization factor in Θ_e propagates as its cube root. Thus the implication from a calibrated clock-line attachment to the determinant scale is exact.

Proof. The tensor-cube norm is $\|L_{24}^{\otimes 3}\| = \Delta_{24}^3$. Norm preservation gives $|\det M_e| = \Delta_{24}^3$; take the positive cube root. $\square$

The register count does not construct L_{24}, calibrate Δ_{24}, or produce Θ_e. Those are the removable physical clock, determinant-descent, and normalization gates. The determinant scale is work in progress. Koide balance fixes a shape condition and is independent of this affine scale attachment.

Positive-chamber face-circulant identity and conditional finite-GNS balance. The regular C_3 face fiber has the general Hermitian carrier

$$C = aI + \rho\left(e^{i\delta}R + e^{-i\delta}R^2\right), \qquad a, \rho \geq 0,$$

where R is the regular cyclic shift, $R^3 = I$. The commutant of the regular C_3 action is spanned by I, R, R^2, which makes C the general Hermitian C_3-equivariant carrier. Fourier diagonalization gives

$$r_k = a + 2\rho\cos\left(\delta + \frac{2\pi k}{3}\right), \qquad k = 0, 1, 2.$$

In a positive spectral chamber, $m_k = sr_k^2$ and $\sqrt{m_k} = \sqrt{s}\, r_k$. The roots-of-unity identities

$$\sum_{k=0}^{2}\cos\left(\delta + \frac{2\pi k}{3}\right) = 0, \qquad \sum_{k=0}^{2}\cos^2\left(\delta + \frac{2\pi k}{3}\right) = \frac{3}{2}$$

give the exact Koide identity

$$Q = \frac{\sum_k m_k}{(\sum_k \sqrt{m_k})^2} = \frac{1 + 2(\rho/a)^2}{3},$$

so $\rho/a = 1/\sqrt{2}$ is exactly equivalent to $Q = 2/3$ in the positive-eigenvalue chamber. At balance, positivity holds for $|\delta| \leq \pi/12 \pmod{2\pi/3}$, and throughout that chamber Koide is independent of δ. Outside it, physical square roots are absolute eigenvalues and the physical Koide ratio differs from the signed trace expression. Equivalently, if $E_0 = 3a^2$ and $E_c = 6\rho^2$ are the singlet and charged-plane Hilbert–Schmidt powers, then

$$Q = \frac{1 + E_c/E_0}{3}, \qquad Q = \frac{2}{3} \iff E_c = E_0.$$

The positive square-root-mass vector lies on the 45° Koide cone about the democratic direction. Circulant symmetry leaves the relative norm of the singlet and charged plane free.

The finite response packet sets that norm under its declared hypotheses. Let

$$\mathcal{V}_K = \mathbf{1} \oplus \chi \oplus \bar{\chi}, \qquad \mathcal{H}_{\rm or} = \mathbb{C}^2, \qquad \mathcal{A}_K = B(\mathcal{V}_K \otimes \mathcal{H}_{\rm or}) \simeq M_6(\mathbb{C}).$$

A one-state record cannot distinguish orientation reversal, so $\mathbb{C}^2$ is the minimal faithful orientation record. Minimality of this response-local register is an explicit declared hypothesis. With q_+ one oriented outcome, define

$$E_+ = P_0 \otimes I_{\rm or} + P_c \otimes q_+, \qquad Z_0 = P_0 \otimes I_{\rm or}, \qquad Z_c = P_c \otimes q_+.$$

The blocks Z_0, Z_c both have rank two. Born–Lüders conditioning of the unique MaxEnt state $I_6/6$ on E_+ gives

$$p_0 = p_c = \frac{1}{2}.$$

The orientation-blind singlet has event action zero, while the selected charged orientation has action $\ln 2$. Charged multiplicity two times its weight $1/2$ equals the singlet weight. The unique normalized trace on $M_6(\mathbb{C})$ fixes these probabilities without the free trace parameter present on $\mathbb{C} \oplus M_2(\mathbb{C})$.

In $L^2(B(\mathbb{C}^3), \tau_3)$, the operators I, R, R^2 are orthonormal, and

$$\mathcal{J}(e_0) = I, \qquad \mathcal{J}(e_+) = R, \qquad \mathcal{J}(e_-) = R^2$$

is unitary and intertwines the C_3 action, orientation reversal, and the two block projections. The canonical GNS square-root amplitude is

$$\xi = \sqrt{p_0}\, e_0 + \sqrt{\frac{p_c}{2}} \left(e^{i\delta} e_+ + e^{-i\delta} e_- \right),$$

so its response has $a = \sqrt{p_0}$ and $|b| = \sqrt{p_c/2}$. Hence

$$\frac{|b|^2}{a^2} = \frac{p_c}{2p_0} = \frac{1}{2}, \qquad \frac{|b|}{a} = \frac{1}{\sqrt{2}}, \qquad Q = \frac{2}{3}.$$

An orientation action gap ΔS gives the general form

$$\frac{p_c}{p_0} = 2e^{-\Delta S}, \qquad \frac{|b|^2}{a^2} = e^{-\Delta S}, \qquad Q(\Delta S) = \frac{1 + 2e^{-\Delta S}}{3}.$$

For comparison, the target-informed minimal complete public response coordinate has

$$Q_{\mathrm{MCPR}} = 0.6666644634090389, \qquad \frac{|b|}{a} = 0.7071044442750720,$$

whose modulus is 3.30 ppm below $1/\sqrt{2}$. The PDG 2026 central charged-lepton masses give

$$Q_{\mathrm{PDG}} = 0.6666644634026367.$$

The numerical proximity is diagnostic. The response architecture and phase were stipulated with target knowledge, so this comparison is not a source-only prediction [14].

A physical Koide implication is conditional on faithful trace-preserving u.c.p. maps Φ and Ψ from the source process to a physical response process and back, with $\Psi\Phi = \mathrm{id}$. Kadison–Schwarz applied to both maps puts every source element in the multiplicative domain of Φ, so Φ is a $*$-monomorphism and an exact L^2 isometry. If it also intertwines Z_0, Z_c with physical response records and sends the conditioned source state to the normalized positive square-root-mass response $C = (Y_e^\dagger Y_e)^{1/4}$, then it transports $p_0 = p_c$ to $E_0 = E_c$ and proves physical Koide. For an accepted finite chiral checkpoint with three left and three right family modes, positive kinetic metrics, and a committed neutral Higgs direction, the reversible-checkpoint conditions force $\Phi = \mathrm{Ad}_{J_L \oplus J_E}$, preserving chirality and the regular C_3 action. If $M_F = X_F^2$ is the positive source response, then

$$\widehat{Y}_e = \frac{\sqrt{2}}{v} J_L M_F J_E^\dagger, \qquad \mathcal{M}_L = J_L M_F J_L^\dagger.$$

The source and canonical physical square-root responses have equal block powers. This closes the finite balanced shape implication on supplied physical charged data, not the determinant scale. The extended branch $\mathrm{OPH}^+_{\mathrm{ch}}$ adds graded physical completion and quotient source-law selection. Given an exhaustive declared carrier class with a positive winner gap, a source-closed BV/BRST continuum, and an interval or contraction certificate for the charged QFT self-map, it selects one accepted charged sector and one dressed mass readout. The stable electron mass is a dressed spectral lower edge; the muon and tau masses are real parts of specified dressed resonance roots. A balanced C_3 fixed point gives $Q = 2/3$. A C_3-symmetric fixed point with charged attenuation $\chi_\star$ gives

$$Q = \frac{1 + e^{-2\chi_\star}}{3}.$$

An off-plane response requires the full response operator. These extended-branch principles and the QFT certificate are additional conditions beyond the three axioms. The relation $\mathrm{Hom}_{C_3}(\mathbf{1}, \chi \oplus \bar{\chi}) = 0$ prevents the neutral source grammar from producing a labeled charged-family vector through symmetry. A charged source tensor, connection, or selected orbit must supply the phase and individual ratios.

The count $(N_c + 1)/(2N_c N_g) = 2/9$ is arithmetic, and the value $2/9$ is the empirical Brannen–Koide fitted phase [9, 10]; the integer factorization is conditioned on that value. A physical phase theorem requires a regular-C_3 family bundle, an oriented family connection whose declared loop has holonomy $\exp[i\beta_{\mathrm{EW}}/(2N_c N_g)]$, and an attachment identifying that holonomy with the phase of the charged Fourier coefficient. Hypercharge acts as a common scalar on the generation copies, while the shift R permutes them. Pure finite A_5/C_3 holonomy yields cube-root phases. Construction of the required continuous family connection and its source selector is work in progress. Under the stated connection and attachment hypotheses, $\delta = 2/9$. The phase of one equal link differs by a factor of three from the total triangular holonomy. With the scale-free normalization $a = 1$, the ordered roots

$$r_k = a + 2\rho\cos\left(\delta + \frac{2\pi k}{3}\right)$$

become

$$\begin{aligned} r_e &= 0.040349908219207475, \\ r_\mu &= 0.5802119201475368, \\ r_\tau &= 2.3794381716332555. \end{aligned}$$

Under those assumptions the bridge selects the same-carrier pair

$$\begin{aligned} \eta_{\mathrm{ext}} &= -6.729586682888832, \\ \sigma_{\mathrm{ext}} &= 8.154061112725994, \end{aligned}$$

with ordered gap values

$$\begin{aligned} \gamma_{21} &= 5.33160859254774, \\ \gamma_{32} &= 2.822452520178255, \\ \kappa_{\mathrm{ext}} &= -4.59605680397234, \end{aligned}$$

and centered logs

$$E_{\mathrm{log,centered}} = [-4.495223235091244,\ 0.836385357456495,\ 3.6588378776347503].$$

Against the charged references, this continuation-centered shape has residual norm

$$\left\| E^{\mathrm{cont}}_{\mathrm{log,centered}} - E^{\mathrm{ref}}_{\mathrm{log,centered}} \right\| \approx 2.13 \times 10^{-5},$$

so this compare-only branch is a near-exact centered-shape closure up to one common absolute scale. The ppm-scale shape agreement carries a large pull in measurement units: the same $\delta = 2/9$ ansatz misses the measured m_μ/m_e ratio by 9.83 ppm, approximately 440σ of the measurement precision, so the row is excluded as a prediction and stands as an approximate fit. The theorem branch is unpromoted because the public affine absolute anchor is external to this near-exact centered charged-shape branch. The compare-only common scale required for exact absolute masses is

$$g_e^\star = 0.04577885783568762.$$

Equivalently, relative to the stored shared-budget seed

$$g_{\mathrm{ch}}^{\mathrm{shared}} = 0.9231656602589082,$$

the missing affine absolute anchor on the determinant-line route would have to take the target value

$$\Delta_e^{\mathrm{abs},\star} = \log \frac{g_{\mathrm{ch}}^{\mathrm{shared}}}{g_e^\star} = 3.003986333402356.$$

This identifies the charged theorem boundary. The compare-only branch nearly solves the centered charged shape. The derivation does not select the latent candidate $\widehat{C}_e^{\mathrm{cand}}$ as the physical $\widehat{C}_e$, and it lacks one affine-covariant absolute anchor A_{ch} that would turn that centered readback into public charged masses on the theorem branch.

Icosahedral face-incidence carrier. The screen geometry supplies a sharper candidate carrier than a scalar twelve-port moment. The twenty outward-oriented icosahedral faces form A_5/C_3, and the face stabilizer cyclically permutes the three corners. Thus each face has a canonical local regular-C_3 fiber, and an equivariant Hermitian circulant has a face-representative-independent unordered spectrum. This is an exact geometric lemma. It is not a physical generation attachment: the sixty face-corner flags form the regular A_5 torsor, not one canonical three-dimensional matter-family space.

A conditional theorem package declares the diagonal affine repair map

$$T(\kappa, \chi_\rho, \zeta_\delta) = (s_\kappa - q_\kappa \kappa,\ s_\chi + q_\chi \chi_\rho,\ s_\zeta + q_\zeta \zeta_\delta).$$

For that declared map it proves $\|DT\|_\infty = 0.0015356510519\ldots < 1$, so Banach's theorem closes existence, uniqueness, and iterative stability of its fixed point. This is a conditional closure, not a derivation of the repair dynamics. A source-multiplier family $T_{\boldsymbol{\lambda}}$ with the same displayed symmetry, analytic degree, Jacobian, and contraction but different masses exists. It proves scoped selector non-identifiability under those properties, not non-entailment from every OPH axiom. Exact zeroth-order block balance gives $|b|/a = 1/\sqrt{2}$, whereas the endpoint operator uses $|b|/a = e^{-\chi_\rho}/\sqrt{2}$; a source-derived bare-to-endpoint repair bridge is work in progress.

Proposition 9.5 (Finite charged-register realization)**.** *The stipulated charged register-and-path class is nonempty. There is an explicit finite algebraic model with register dimensions*

$$(50, 31, 10, 512, 77, 21, 27, 5),$$

connected transition graphs, normalized tracial states, rank-one event projectors, and the eight declared path weights

$$\left(\frac{1}{50}, -\frac{1}{31}, -\frac{1}{310}, \frac{1}{512}, \frac{1}{77}, \frac{1}{21}, \frac{1}{27}, \frac{1}{135}\right).$$

Each noncentral event in $B(H_r)$ admits an accepted/rejected central record dilation in $B(H_r) \otimes D_2$. The graph family is covariant under the sixty proper icosahedral rotations, and normalized path weights are unchanged under inert ancillary stabilization $X \mapsto X \otimes I_k$.

Proof. For each declared connected graph, the diagonal matrix units and both directed units on every edge generate E_{ij} along graph paths and hence generate the full matrix algebra. The normalized trace of a rank-one event is the reciprocal register dimension, and tensor products multiply the two depth-two weights. For an event P, the pinching map

$$\mathcal{E}_P(X) = PXP + (I-P)X(I-P)$$

is a unital trace-preserving conditional expectation. Its instrument writes $P\rho P$ and $(I-P)\rho(I-P)$ into orthogonal accepted/rejected pointers in the central D_2 factor. Explicit permutation intertwiners preserve graph incidence and ranks for all sixty proper rotations. Finally, $\tau_{dk}(P \otimes I_k) = \tau_d(P)$, and the partial-trace coarse map retracts the inert embedding. These facts construct one model of the declared schema. □

Proposition 9.5 proves formal nonemptiness. It supplies no derivation of the charged source. The eight dimensions, path automaton, coupling degrees, grading, signs, clock, and scalar response are model inputs. The verifier exhausts that automaton, not the physically admissible OPH path space. The construction resolves the abstract event-versus-central-record issue at fixed cutoff. Physical response selection, cofinal screen refinement, and charged-sector attachment are work in progress.

Proposition 9.6 (Conditional charged nature and singularity transport)**.** *Let M_F be the positive face endpoint. Suppose a physical chiral three-family carrier supplies a natural unitary J_L, canonical kinetic metrics, and*

$$\frac{v^2}{2}\widehat{Y}_e\widehat{Y}_e^\dagger = J_L M_F^2 J_L^\dagger.$$

Then its positive left charged response is $J_L M_F J_L^\dagger$. Suppose in addition that the exact renormalized charged kernel is supplied, its singular lines satisfy the declared infrared and regularity conditions, and a finite-register/Dyson map has singularity readout $J_L M_F J_L^\dagger$. Regular invertible field changes preserve the zero set, and a regular Nielsen factorization $\partial_\xi K = A_\xi K + K B_\xi$ makes each simple zero gauge independent.

Proof. The first statement follows from uniqueness of the positive square root. Multiplication of K on the left and right by analytic invertible matrices multiplies its determinant by nonvanishing factors. Jacobi's identity gives

$$\partial_\xi \det K = (\operatorname{tr} A_\xi + \operatorname{tr} B_\xi)\det K,$$

so a simple zero cannot move while the Nielsen factors are regular. These are the standard mixed-fermion and gauge-independence implications used in the pole literature [18, 19]; the stable charged-line infrared qualification follows the infraparticle boundary [20, 21]. □

Proposition 9.6 closes the downstream logic after its physical premises are supplied. It does not supply them. The displayed square identity is the desired operator attachment, and the singularity premise sets the Dyson readout equal to the face response. The explicit kernel $K_0(s) = sI - M_F^2$ has zero self-energy; it is a free algebraic existence witness rather than the interacting charged-lepton one-particle-irreducible kernel. The physical carrier attachment, interacting kernel, and singularity transport are work in progress. The verifier reconstructs the fixed point and response shape but does not check $M_F = g_{\mathrm{end}} S_F$: a coherent change of the ordered spectrum by factors $(2, 1/2, 1)$, together with the matching mass matrix, Yukawa matrix, and pole roots, preserves the determinant and satisfies the tested mass relations. The package is therefore a conditional conditional theorem. It does not identify physical charged matter, construct an interacting pole, or predict a mass.

These results do not yield physical charged-lepton masses. Such a conclusion requires a quotient-visible face-to-charged-family intertwiner, the $\ln 2$ MaxEnt and ensemble-to-Yukawa bridge, a charged connection deriving both the $2/9$ base phase and its correction, a normalized $\mathbb{Z}_6$ determinant character, an OPH dynamical theorem selecting the physical registers, state, exhaustive path category, grading, clock, and response rather than declaring them, cofinal refinement naturality, one target-independent coherent source tuple, and a mass-scheme map. The icosahedral face carrier, face-incidence theorem, and conditional finite-quotient weights supply only the stated finite ingredients.

9.1 Absolute charged-lepton mass intervals from electromagnetic transport

The missing absolute anchor is one real scale. Fixing A_{ch} is equivalent to fixing the single family rescaling $Y_e \mapsto e^{\kappa} Y_e$, under which every charged mass scales as $m_i \mapsto e^{\kappa} m_i$ while the centered shape and every mass ratio are invariant. The declared charged antecedents are invariant under that rescaling, which is the content of the common-shift no-go: the gauge representations, the anomaly structure, the complexity vector, the determinant algebra, and the same-label readback take the same value on the whole family $\{e^{\kappa} Y_e\}$, so no source object built from those antecedents alone selects κ.

The rescaling is not a symmetry of the declared electromagnetic transport. The charged leptons enter the photon vacuum polarization, and the leptonic packet in the fine-structure endpoint map carries

$$P_{\mathrm{lep}}(\kappa) = \frac{1}{3\pi} \sum_i \left[2 \log \frac{M_Z}{m_i} - \frac{5}{3} \right], \qquad \frac{\mathrm{d}P_{\mathrm{lep}}}{\mathrm{d}\kappa} = -\frac{2}{\pi}.$$

The declared endpoint pixel map $P = \varphi + \sqrt{\pi}/A_{\mathrm{Th}}(P)$ consumes P_{lep}, so the emitted endpoint and the pixel move with κ. The common-shift no-go omits this transport from its antecedent list. Source-only, the system is the stiff curve $(P(\kappa), \kappa)$ with $|\mathrm{d}P/\mathrm{d}\kappa| \approx 6 \times 10^{-5}$, and the absolute masses are work in progress on the source branch.

On the empirical-closure branch the transport identifies κ. Inverting the on-shell decomposition

$$a_0 + g = A^{-1}(0)\,(1 - P_{\mathrm{lep}}(\kappa) - \Delta_{\mathrm{had}} - \Delta_{\mathrm{top}})$$

with the frozen anchor $a_0 = 128.308$, the certified anchor-gap interval $g \in [0.620, 0.651]$, the hadronic payload $\Delta_{\mathrm{had}} = 0.027609 \pm 0.000112$ pinned to the published data-driven compilation of Keshavarzi, Nomura and Teubner (Phys. Rev. D 101, 014029), the target-anchored mass ratios, and the measured $A^{-1}(0)$ supplied as a compare-only exclusion inside the interval solve path gives

$$\kappa \in [-0.070, +0.061], \qquad \kappa_{\mathrm{central}} = -0.004.$$

An independent 100-digit decimal calculation brackets π, pads the transcendental evaluation, and rounds every endpoint outward. The resulting target-anchored diagnostic intervals are

$$\begin{aligned}
m_e &= 0.5089\ [0.4765,\ 0.5434]\ \mathrm{MeV}, && \text{measured } 0.5110\ \mathrm{MeV},\\
m_\mu &= 105.22\ [98.5,\ 112.4]\ \mathrm{MeV}, && \text{measured } 105.66\ \mathrm{MeV},\\
m_\tau &= 1.7695\ [1.657,\ 1.890]\ \mathrm{GeV}, && \text{measured } 1.7769\ \mathrm{GeV},
\end{aligned}$$

with central values within about 0.4% of measurement and the physical triple inside every interval. The physical on-shell anchor requirement lies inside the certified gap interval. The solve also inverts exactly at the witness: the anchor-gap value $g = 0.6379$ closes the branch on the measured triple, and its distance $+0.0070$ from the standard on-shell reference deficit 0.631 is the scheme correction required by the bridge. A source-emitted bridge value is a sharp falsification target: a value at 0.6379 within the payload width closes the lepton branch on the witness, and a value outside the certified interval refutes the decomposition. The interval width reduces to the published payload uncertainty and the anchor gap. A source derivation requires the hadron backend and the scheme bridge.

The certified anchor-gap interval is the exact affine image of the hadronic payload interval, $g(X) = A^{-1}(0)(1 - X) - A_{\mathrm{lep}} - a_0$, so the rectangle solve above carries the one payload uncertainty in two anti-correlated slots. On the payload-coherent reading, where the gap is evaluated at the same payload value as the hadronic term, the payload cancels from the decomposition and the surviving width is the higher-order leptonic remainder and the kernel truncation. The coherent central coincides with the rectangle central, and the certified width contracts by a further factor of 3.8:

$$\begin{aligned} m_e &= 0.5089\ [0.5001,\ 0.5178]\ \text{MeV}, \quad \text{measured } 0.5110\ \text{MeV}, \\ m_\mu &= 105.22\ [103.40,\ 107.06]\ \text{MeV}, \quad \text{measured } 105.66\ \text{MeV}, \\ m_\tau &= 1.7695\ [1.7390,\ 1.8004]\ \text{GeV}, \quad \text{measured } 1.7769\ \text{GeV}, \end{aligned}$$

with the measured triple inside every coherent interval. The coherent logarithmic half-width is 0.01732, corresponding to multiplicative one-sided widths of -1.72% and $+1.75\%$. The rectangle interval remains the independent-gap statement for every bridge value inside its certified range; the coherent row is conditional on the declared payload-coherent anchor-gap premise. Both intervals require the same hadron backend and scheme bridge for source closure. The surviving coherent width is a premise floor rather than a budget slack: the higher-order band matches the published per-order structure of the leptonic running with negligible κ-sensitivity across the certified interval, so the certified floor is the scheme-bridge ambiguity itself, and a tighter certified width requires the source bridge rather than a smaller budget.

The determinant prescription is a consistency test on this scale. It reproduces the physical leptonic packet at $|\kappa| \approx 0.006$. Its runtime has no continuous fit parameter, while its architecture and charged ratios retain target ancestry. The construction adds no axiom. The interval branch carries absolute masses only on the empirical-closure branch; the source-only charged no-go is in force.

9.2 Conditional response candidate and symmetry-breaking boundary

A conditional package defines the charged response by stipulating a typed, incidence-complete response architecture on an oriented icosahedral flag: eight finite registers of dimensions $50, 31, 10, 512, 77, 21, 27, 5$, eight primitive path classes with normalized rank-one trace weights, a public-block amplitude, an oriented process phase, and a $\mathbb{Z}_6$ determinant character. Conditional on that architecture, the three-channel response map is a Banach contraction with contraction constant 1.5×10^{-3}; its unique fixed point, the regular-C_3 shape functional, and the determinant character emit one dimensionless charged triple on the declared-architecture branch with zero runtime charged-mass input:

$$\frac{(m_e, m_\mu, m_\tau)}{E_\star} = (4.18511 \times 10^{-23},\ 8.65348 \times 10^{-21},\ 1.45532 \times 10^{-19}),$$

with mass ratios $m_\mu/m_e = 206.7683$, $m_\tau/m_e = 3477.3655$, $m_\tau/m_\mu = 16.8177$, and unclosed-clock displays (0.51096, 105.649, 1776.78) MeV. A segregated compare-only calculation places this coherent branch about 84 ppm below the comparison triple; the candidate computation has no dependency on the comparison file.

The register dimensions, path table, signs, amplitude, phase, and determinant exponent are declared model inputs, so the triple is a declared-model coordinate. Those architecture choices are target-informed and stipulated with knowledge of the charged-family target. Their executable evaluation is runtime-reference-free, and the candidate is neither blind nor source-only. The 84 ppm comparison is diagnostic. The Koide subsection derives the finite public-block amplitude $1/\sqrt{2}$ from the connected M_6 response register. The candidate supplies no recoverable attachment between that event and a physical chiral mass channel, so its mass claim is conditional. The arithmetic identity $(N_c + 1)/(2N_cN_g) = 2/9$ supplies no oriented phase; a holonomy reading requires link variables, a loop, and a nonlocal readout map. A product of fourteen normalized rank-one traces is neither a $\mathbb{Z}_6$ character nor a determinant, and the canonical determinant norm carries the kinetic factor $\det \mathcal{M}_L = (v/\sqrt{2})^3 \,|\det Y_e|/\sqrt{\det Z_L \det Z_E}$; the declared 6^{-14} weight is a candidate positive path weight.

The source-side problem is a finite symmetry-breaking analysis on the icosahedral carrier. The vertex permutation module decomposes as $\mathbb{R}^{12} \cong W_1 \oplus W_3 \oplus W_3' \oplus W_5$; the traceless quadrupole map $Q(w) = \sum_i w_i(p_i p_i^T - I/3)$ satisfies $Q = QP_5$ and $Q^*Q = \frac{8}{5}P_5$, so it sees exactly W_5, and the two adjacent gaps of its spectrum span the centered family plane. A unique invariant MaxEnt state has zero expectation on every nontrivial irreducible module, and the homogeneous twelve-port branch has the unique uniform minimizer, so charged family shape requires a source-selected W_5 orbit. Because $\mathrm{Sym}^2_0(\mathbf{3}) \cong \mathrm{Sym}^2_0(\mathbf{3}') \cong W_5$, the family attachment is multiplicity-one up to one sign once the chiral family fibers carry a three-dimensional A_5 representation.

Residual symmetry alone does not select the mass ratios. Realize W_5 as the traceless symmetric real 3×3 matrices. Invariance under a threefold or fivefold rotation about n forces

$$A = \alpha(nn^T - I/3),$$

with spectrum $(2\alpha/3, -\alpha/3, -\alpha/3)$. Those fixed points have a double eigenvalue. The twofold fixed locus is

$$A = \begin{pmatrix} u & v & 0 \\ v & w & 0 \\ 0 & 0 & -u-w \end{pmatrix},$$

which is three-dimensional, hence two-dimensional after overall scaling, and admits simple spectrum. It leaves exactly enough continuous freedom to carry the two independent mass ratios without predicting them. The source-side family problem is therefore selection by a specific screen-derived A_5-invariant potential. One coefficient-free candidate is excluded: the quartic truncation of the relative-entropy expansion restricted to W_5 at fixed band amplitude has exactly two extremal orbit branches, the degenerate vertex-axis prolate orbit and a simple-spectrum twofold-fixed orbit with centered eigenvalues proportional to $(-(\varphi^4 - 1), -1, \varphi^4)$ and sorted-gap ratio exactly the golden ratio φ; the observed centered log-mass gap ratio 1.889 differs from φ by 16.7 percent in the closest orientation, so the entropy packet selects neither branch as the charged shape. No source mechanism beyond that truncation is supplied.

The finite theorem does not supply an effective action on W_5 with a unique simple-spectrum minimizing orbit. The invariant algebra has one quadratic, two cubic, and two quartic coefficients, so the W_5-restricted Hessian scalar h_5 at the homogeneous state is a necessary local discriminator. A physical charged shape additionally requires the physical A_5 family lift, normed determinant-line descent with kinetic factors, an interacting charged kernel, an operational scale, and a source packet fixed before comparison.

The affine scale has a second admissible closure through electromagnetic transport. With the exact one-loop Ward kernel $I(z) = \int_0^1 x(1-x)\log(1 + zx(1-x))\,\mathrm{d}x$ in closed form, the charged response $\mathcal{W}_Q(\mu;\ell) = \frac{2}{\pi}\sum_i I(q^2 e^{-2(\mu+\ell_i)})$ is a smooth, strictly decreasing bijection of the common log-scale μ onto $(0,\infty)$ at fixed centered shape ℓ, with high-energy slope $-2/\pi$. A source-complete spacelike Ward endpoint pair with a source-complete nonleptonic subtraction therefore fixes a unique μ_{ch}, hence the determinant line $|\det(M_e/E_\star)| = e^{3\mu_{\mathrm{ch}}}$ and the electron ratio $m_e c^2/E_\star$ consumed by the cesium clock packet. The common-shift no-go of the preceding subsections holds on the reduct without mass-dependent electromagnetic transport; a Ward endpoint response and a determinant-line basepoint are the two admissible closures of that orbit. The Ward inversion, its interval enclosure, and its fail-closed source-packet schema are encoded in the Ward determinant-line receipt. A source result requires the endpoint pair, nonleptonic subtraction, higher-order monotonicity certificate, and atomic transport factor.

10 Neutrino Family Derivation

The neutrino branch contains one exact no-go and one failed comparison candidate. The intrinsic isotropic branch is excluded by its spectral cap. The weighted-cycle construction descends from two hand-written family-transport matrices and declared label, orientation, cycle, and exponent choices selected with target information. Its bridge and absolute attachment are compare-only.

For the isotropic matrix $M = aI + \rho C$ with unit-modulus off-diagonal entries, the general Gershgorin estimate applied to $M^\dagger M$ gives

$$\max_{i,j} |\Delta m^2_{ij}| \leq 8a\rho + 4\rho^2.$$

The certified bound is $1.52304 \times 10^{-6}\,\mathrm{eV}^2$, about three orders of magnitude below the atmospheric scale.

The weighted-cycle comparison candidate gives

$$\begin{aligned}\theta_{12} &= 34.225904631810025^\circ,\\ \theta_{23} &= 49.72282845058266^\circ,\\ \theta_{13} &= 8.686355527700156^\circ,\end{aligned}$$

$$\begin{aligned}\delta_{\mathrm{PMNS}} &= 305.58061231449796^\circ,\\ J &= -0.02753115613565372,\end{aligned}$$

and the dimensionless hierarchy invariant

$$\frac{\Delta m^2_{21}}{\Delta m^2_{32}} = 0.030721110097966534.$$

The official NuFIT 6.1 normal-ordering profile gives

$$\begin{aligned}\Delta\chi^2_{23,\delta} &= 20.11955 \quad \text{with the tabulated atmospheric likelihood,}\\ \Delta\chi^2_{23,\delta} &= 18.43528 \quad \text{without it,}\end{aligned}$$

at $(\sin^2\theta_{23}, \delta_{\mathrm{CP}}) = (0.582056, -54.419^\circ)$ [17]. Both values exceed the two-parameter 3σ contour value 11.83. Separate marginal intervals do not test this correlation. The published profiles overlap and are not summed; their maximum is a lower bound on the unavailable full fixed-candidate $\Delta\chi^2$.

The declared shared-basis construction sets

$$M_{\mathrm{shared}} = U^*_{e,\mathrm{left}}\, M_{\mathrm{wc}}\, U^\dagger_{e,\mathrm{left}}, \qquad U_{\nu,\mathrm{shared}} = U_{e,\mathrm{left}}\, U_{\mathrm{wc}},$$

the identity

$$U^\dagger_{e,\mathrm{left}} U_{\nu,\mathrm{shared}} = U_{\mathrm{wc}}.$$

This cancellation is a tautology because $U_{\nu,\mathrm{shared}}$ was defined as $U_{e,\mathrm{left}} U_{\mathrm{wc}}$. It does not independently identify the physical charged-lepton basis or validate the PMNS point. No theorem in the combined theorems places the f-labelled weighted-cycle operator in the charged-lepton mass basis.

The charged-basis construction is open, and its three normalized shape singular values differ only at about the 10^{-12} relative level, so its left singular vectors do not define a stable physical family basis. The charged-lepton continuation is support-extension gated and does not supply a replacement labelled basis. At source level, the physical PMNS matrix is therefore unformed.

Taking the stored basis declarations literally gives a sharply different diagnostic. Reordering $M_{\rm wc}$ from $[f3, f1, f2]$ to the independently declared shared basis $[f1, f2, f3]$, computing its Takagi matrix $U_\nu^{(f)}$, and then forming $U_{e,\rm left}^\dagger U_\nu^{(f)}$ yields

$$\theta_{12} = 38.6812^\circ, \qquad \theta_{23} = 76.2759^\circ, \qquad \theta_{13} = 44.8283^\circ,$$

with $\delta = 324.2140^\circ$ and $J = -0.0233164$. In particular, $\sin^2\theta_{13} = 0.4970$ and $\sin^2\theta_{23} = 0.9437$, both outside the corresponding NuFIT 6.1 profile grids. This is a diagnostic conditional on the stored matrices and label order. It is not a source-closed prediction because the charged basis and family kernel also descend from template-level inputs.

The transported Takagi identity is

$$U_{\nu,\rm shared}^T M_{\rm shared} U_{\nu,\rm shared} = \mathrm{diag}(m_i) \in \mathbb{R}_{>0}.$$

Here $U_{\rm wc}$ denotes the canonical phase-fixed Takagi unitary, not the raw arbitrary-phase eigensystem of $M_{\rm wc}^\dagger M_{\rm wc}$. Equivalently, $U_{\rm wc}^T M_{\rm wc} U_{\rm wc}$ is positive diagonal in its declared basis. Canonical Takagi readout under the declared charged-basis assumption and the electron-row gauge $U_{e1} \in \mathbb{R}_{>0}$ gives the comparison-only Majorana pair

$$\alpha_{21}^{(\rm Maj)} = 153.6185177794357^\circ, \qquad \alpha_{31}^{(\rm Maj)} = 257.0032408220805^\circ.$$

The declared exponent law uses the positive transport-load segment between $\chi = 1+\epsilon$ and $1+\gamma_{1/2}$: on a one-dimensional affine segment, the balanced selector and the least-distortion selector for any positive translation-invariant quadratic form coincide at the midpoint, so

$$D_\nu = \frac{\chi + (1+\gamma_{1/2})}{2}, \qquad p = 1 + \gamma + \frac{\epsilon}{D_\nu}.$$

The midpoint fact does not derive the segment endpoints, the exponent law, the cycle topology, or their application to neutrino transport. The formula is target-informed and is not a source-side consequence of OPH. On the declared positive selector segment there is a stronger compare-only two-parameter adapter: under the declared normal-ordering hypothesis, solving τ_ν against the representative central ratio and then solving λ_ν against Δm_{32}^2 gives

$$(m_1, m_2, m_3) = (0.01745663295,\ 0.01948419960,\ 0.05308139066)\ \mathrm{eV},$$

$$\begin{aligned}\Delta m_{21}^2 &= 7.49 \times 10^{-5}\ \mathrm{eV}^2,\\ \Delta m_{31}^2 &= 2.5129 \times 10^{-3}\ \mathrm{eV}^2,\\ \Delta m_{32}^2 &= 2.438 \times 10^{-3}\ \mathrm{eV}^2.\end{aligned}$$

These exact central numbers are compare-only. The reduced invariant contains target feedback, so the following values are diagnostic coordinates:

$$\begin{aligned}C_\nu &= 0.9994295999075177,\\ P_\nu &= 6.699825740519345,\\ B_\nu &= P_\nu C_\nu = 6.696004159297337,\end{aligned}$$

and therefore

$$\lambda_\nu = \frac{m_{\star,\rm eV}}{q_{\rm mean}^{p_\nu}} P_\nu C_\nu = 1.7237014208357415,$$

$$m_i = \lambda_\nu \widehat{m}_i, \qquad \Delta m_{ij}^2 = \lambda_\nu^2 \widehat{\Delta}_{ij}.$$

The normalized same-label overlap-defect calculation is exact after the template and declared selectors are supplied. The positive-segment adapter, bridge corridor, and bridge-coordinate calculations are diagnostic. In particular, the following quantity is only a diagnostic coordinate:

$$C_\nu := \frac{B_\nu}{I_\nu^{1/2}\,\widehat{\text{ratio}}^{1/2}\,\text{sum\_defect}^{-1}}$$

It is defined from the declared proxy and normalizer. The construction also factors exactly through $q_e = q_{\text{mean}} qbar_e$, so a $qbar_e$-only collapse law for the bridge factor does not follow from the conditional algebraic surface stated here. The best algebraically complete conditional local object beneath that bridge is the defect-weighted same-label edge family $q_e = \sqrt{g_e d_e}$ together with the induced μ_e family. That family sits below C_ν and the induced amplitude B_ν; it does not replace them. No source-only PMNS, hierarchy, Majorana-phase, or absolute-mass prediction survives these gates.

Cosmological-neutrino pressure surface. The compare-only absolute attachment displays

$$\sum_i m_{\nu_i} = 0.09001192964464505 \text{ eV}.$$

Under the declared normal-ordering hypothesis, standard relic-neutrino inheritance, and absent an explicitly declared extra OPH relativistic coherence channel, the diagnostic cosmological-neutrino-background branch uses

$$N_{\text{eff}}^{\text{OPH}} = 3.044,$$

with the usual broad, low-amplitude normal-ordering free-streaming imprint. This is a diagnostic cosmological pressure surface for the rejected candidate. The displayed mass sum sits below the Planck+BAO 0.12 eV bound [15] but is exposed to strict DESI DR2 ΛCDM-style bounds near 0.0642 eV under their model assumptions [16]. A cosmological upper bound below the displayed compare-only value would exclude the displayed absolute-mass continuation under a model class that also respects oscillation lower limits and the OPH background assumptions.

On the same-label neutrino-only branch, the centered eta-class is exactly S_3-isotropic, so the same-label data are edge-constant and the solar $1-2$ split cannot open there by itself. The first solar mover therefore has to come from realized flavor-side same-label gap/defect readback, not from another neutrino-only selector tweak.

10.1 Intrinsic neutrino eta-chain

The proof-facing input is smaller than the raw same-label matrix payload. It compresses to the same-label scalar certificate

$$\big(g_e,\ \omega_e\big)_{e\in\{12,23,31\}}, \qquad \omega_e = \text{same-label overlap}_e^2, \qquad d_e = 1-\omega_e,$$

modulo one common scale. From that certificate one forms

$$q_e = \sqrt{g_e d_e}, \qquad \eta_e = \log q_e - \frac{1}{3}\sum_f \log q_f, \qquad e\in\{12,23,31\},$$

the intrinsic selector depends only on the centered class

$$[\eta_e] \in \mathbb{R}^3/\langle(1,1,1)\rangle.$$

Equivalently one may work with any positive normalized family

$$\mu_e = \frac{e^{\eta_e}}{\frac{1}{3}\sum_f e^{\eta_f}}.$$

Common scaling cancels identically, so the intrinsic selector is determined by the centered eta-class alone.

This factorization result is conditional on the scalar inputs; it is not a source-closure result. The scalar values are numerically complete. Their gap data inherit the template family-transport kernel, and their overlap data inherit a candidate-only line lift. Neither input is source-closed, so the intrinsic spectrum is a conditional diagnostic.

On the isotropic neutrino-only branch one has

$$(\eta_{12}, \eta_{23}, \eta_{31}) = (0, 0, 0),$$

which gives the conditional intrinsic singular values

$$(s_0, s_1, s_2) = (2.3986448447627196,\ 2.3986448447627196,\ 2.590074050773907) \times 10^{-12}\ \text{GeV},$$

with ascending squared-mass gaps

$$\begin{aligned} s_1^2 - s_0^2 &= 0, \\ s_2^2 - s_0^2 &= 9.549864971855843 \times 10^{-25}\ \text{GeV}^2. \end{aligned}$$

This is the exact neutrino-only isotropy obstruction behind the solar-splitting boundary.

Theorem 10.1 (Fixed isotropic reference plus same-label scalars suffice for the intrinsic deformation)**.** *Fix the isotropic reference M_0, equivalently its parameters (a, ρ, Ω). Assume the same-label gap and overlap scalars are supplied on all realized arrows. Then the full intrinsic neutrino mass-eigenstate bundle factors through the scalar certificate*

$$(g_e, \omega_e)_{e\in\{12,23,31\}}$$

or equivalently through the centered eta-class $[\eta_e]$. No additional raw matrix payload is needed to form the intrinsic selector, the depressed-cubic spectrum, the ascending gaps, or the intrinsic spectral subspaces. When the singular spectrum is simple, canonical Takagi congruence fixes an ordered column basis up to column signs. At a degeneracy, including the isotropic $s_0 = s_1$ point, only the degenerate spectral subspace and its projector are fixed. Ascending singular-value sorting does not select the physical normal or inverted mass-eigenstate labels.

Theorem 10.2 (Exact principal-branch selector from the centered eta-class)**.** *Fix the cycle sum*

$$\Omega = \psi_{12} + \psi_{23} + \psi_{31},$$

and a positive weight family μ_e. Define

$$\mu_{\min} = \min_e \mu_e, \qquad F_{\max} = \sum_e \arcsin\left(\frac{\mu_{\min}}{\mu_e}\right).$$

If $|\Omega| < F_{\max}$, then on the principal branch $\psi_e \in (-\pi/2, \pi/2)$, the affine energy

$$A(\psi) = \sum_e \mu_e(1 - \cos\psi_e)$$

has a unique minimizer. It is given by

$$\psi_e^\star = \arcsin(\lambda/\mu_e),$$

where λ *is the unique solution of*

$$\sum_e \arcsin(\lambda/\mu_e) = \Omega, \qquad \lambda \in (-\min_e \mu_e, \min_e \mu_e).$$

Proof. The Euler–Lagrange equations are $\mu_e \sin\psi_e = \lambda$. On the principal branch the scalar function

$$F(\lambda) = \sum_e \arcsin(\lambda/\mu_e)$$

is strictly increasing because

$$F'(\lambda) = \sum_e \frac{1}{\sqrt{\mu_e^2 - \lambda^2}} > 0.$$

Its range on $(-\mu_{\min}, \mu_{\min})$ is $(-F_{\max}, F_{\max})$, so $F(\lambda) = \Omega$ has a unique solution under the stated condition. Strict convexity follows because the Hessian of A is

$$\nabla^2 A = \operatorname{diag}(\mu_e \cos\psi_e),$$

which is positive definite on the principal branch and is positive definite after restriction to the affine plane $\sum_e \psi_e = \Omega$. □

For the normalized family

$$(\mu_{12}, \mu_{23}, \mu_{31}) = (1.4497003,\ 0.9968526,\ 0.5534471),$$

one has $F_{\max} = 2.5511001$, $|\Omega| = 1.7561443$, and positive domain margin 0.7949558. The numerical selector lies inside the theorem domain.

Let

$$a = m_\star = \frac{v^2}{\mu_u}, \qquad \rho = |(M_0)_{12}|.$$

Then the intrinsic Majorana matrix is

$$M(\eta) = \begin{pmatrix} a & \rho e^{i\psi_{12}} & \rho e^{i\psi_{31}} \\ \rho e^{i\psi_{12}} & a & \rho e^{i\psi_{23}} \\ \rho e^{i\psi_{31}} & \rho e^{i\psi_{23}} & a \end{pmatrix}.$$

Theorem 10.3 (Exact intrinsic spectral cubic)**.** *Let* $H = M^\dagger M$. *Then*

$$H = dI + T, \qquad d = a^2 + 2\rho^2,$$

where T *has zero diagonal and off-diagonals*

$$x_{12} = 2a\rho\cos\psi_{12} + \rho^2 e^{i(\psi_{23}-\psi_{31})},$$

$$x_{23} = 2a\rho\cos\psi_{23} + \rho^2 e^{i(\psi_{31}-\psi_{12})},$$

$$x_{13} = 2a\rho\cos\psi_{31} + \rho^2 e^{i(\psi_{23}-\psi_{12})}.$$

Define

$$P = |x_{12}|^2 + |x_{23}|^2 + |x_{13}|^2, \qquad Q = \Re(x_{12}x_{23}\overline{x_{13}}).$$

Then the eigenvalues λ_k of T are exactly the three real roots of

$$\lambda^3 - P\lambda - 2Q = 0,$$

and the intrinsic squared masses are

$$m_k^2 = d + \lambda_k.$$

Equivalently,

$$\lambda_k = 2\sqrt{\frac{P}{3}}\cos\left(\frac{1}{3}\arccos\left(\frac{3\sqrt{3}\,Q}{P^{3/2}}\right) - \frac{2\pi k}{3}\right).$$

Corollary 10.4 (Intrinsic eta-chain spectral closure)**.** *Once the centered eta-class is emitted at the flavor boundary, the intrinsic neutrino branch emits*

$$s_0 \le s_1 \le s_2, \qquad s_1^2 - s_0^2,\ s_2^2 - s_0^2,\ s_2^2 - s_1^2,$$

and the corresponding intrinsic spectral subspaces with no PMNS import and no flavor-label leakage. A columnwise Takagi basis is emitted only on the simple-spectrum locus. The physical normal-ordering assignment is $(\nu_1, \nu_2, \nu_3) = (s_0, s_1, s_2)$, while the inverted-ordering assignment is $(\nu_3, \nu_1, \nu_2) = (s_0, s_1, s_2)$. The source does not choose between them, so physical ordering is open.

The matrix used in this corollary is Majorana, so its physical column matrix is the Takagi matrix U_ν defined by

$$U_\nu^T M U_\nu = \mathrm{diag}(s_0, s_1, s_2) \in \mathbb{R}_{\ge 0}.$$

The Takagi matrix, whose columns diagonalize $M^\dagger M$, gives maximum congruence off-diagonal residual 5.4×10^{-28} GeV on the anisotropic matrix. On the stored charged matrix, the intrinsic combination gives

$$(\theta_{12}, \theta_{23}, \theta_{13}, \delta) = (45.0137^\circ,\ 2.50489^\circ,\ 2.53597^\circ,\ 210.6251^\circ).$$

That diagnostic is strongly incompatible with the NuFIT angle surface, but it is not an OPH prediction because the charged basis is open and template-derived.

10.2 Perturbative laws around the isotropic point

Write

$$\psi_e = \varphi + \delta_e, \qquad \delta_{12} + \delta_{23} + \delta_{31} = 0.$$

At first order in the centered eta-class,

$$\delta_e = -\tan\varphi\,\eta_e + O(\eta^2).$$

Define

$$\sigma^2 = \frac{2}{3}\sum_e \delta_e^2.$$

Assume $2a\cos\varphi + \rho \ne 0$ and that the collective state is separated from the isotropic doublet. Then the two ascending doublet eigenvalues satisfy

$$s_{0,1}^2 = m_d^2 \mp 2a\rho|\sin\varphi|\,\sigma + O(\delta^2),$$

so

$$s_1^2 - s_0^2 = 4a\rho|\sin\varphi|\,\sigma + O(\delta^2) = 4a\rho\frac{\sin^2\varphi}{|\cos\varphi|}\sqrt{\frac{2}{3}\sum_e \eta_e^2} + O(\eta^2).$$

Under the declared normal-ordering hypothesis, the ascending atmospheric gaps obey

$$\Delta m^2_{31} = \Delta_{\text{atm,iso}} + \frac{1}{2}\Delta m^2_{21} + O(\delta^2), \qquad \Delta m^2_{32} = \Delta_{\text{atm,iso}} - \frac{1}{2}\Delta m^2_{21} + O(\delta^2),$$

so the first-order invariant atmospheric object is the collective-to-doublet centroid gap

$$\Delta_{\text{cent}} = s_2^2 - \frac{1}{2}(s_0^2 + s_1^2) = \Delta_{\text{atm,iso}} + O(\delta^2).$$

Its quadratic shift is

$$\delta\Delta_{\text{cent}} = -a\rho\,\sigma^2 \frac{a(4\cos^2\varphi - 1) + 6\rho\cos\varphi}{2a\cos\varphi + \rho} + O(\delta^3).$$

The projective largest-mass right-singular direction, in the phase gauge that approaches the real democratic vector at the isotropic point, obeys the first-order deformation law

$$u_3 = u + \kappa \begin{pmatrix} \delta_{23} \\ \delta_{31} \\ \delta_{12} \end{pmatrix} + O(\delta^2), \qquad u = \frac{1}{\sqrt{3}}(1,1,1)^T, \qquad \kappa = \frac{\sqrt{3}\,(2a\sin\varphi + 3i\rho)}{9(2a\cos\varphi + \rho)}.$$

This equation fixes only the complex line. If v_3 denotes its normalized right-hand side, the canonical Takagi representative used by the executable construction is

$$u_3^{\text{Tak}} = e^{-\frac{i}{2}\arg(v_3^T M_\nu v_3)} v_3, \qquad (u_3^{\text{Tak}})^T M_\nu u_3^{\text{Tak}} = s_3 > 0,$$

up to the unavoidable column sign. In particular, at the isotropic point the real democratic vector is generally outside positive Takagi gauge. Equivalently,

$$u_3 = u - \tan\varphi\,\kappa \begin{pmatrix} \eta_{23} \\ \eta_{31} \\ \eta_{12} \end{pmatrix} + O(\eta^2).$$

One exact demonstration uses centered eta-class

$$(\eta_{12}, \eta_{23}, \eta_{31}) = \begin{array}{l} (0.13631072512014578,\ -0.18362987264261327, \\ \qquad\qquad\qquad 0.0473191475224675), \end{array}$$

and returns the ascending intrinsic singular values

$$(s_0, s_1, s_2) = (2.3929601069646055,\ 2.4048200109774875,\ 2.589606227283229) \times 10^{-12}\ \text{GeV},$$

with

$$s_1^2 - s_0^2 = 5.690121167370743 \times 10^{-26}\ \text{GeV}^2,$$
$$s_2^2 - s_0^2 = 9.798023388600230 \times 10^{-25}\ \text{GeV}^2.$$

These are exact outputs of the intrinsic eta-chain once the fixed isotropic reference and centered eta-class are supplied. They are not flavor-labeled OPH rows, and ascending sorting does not pick a physical ordering. Under the declared normal-ordering hypothesis $(\nu_1, \nu_2, \nu_3) = (s_0, s_1, s_2)$; under the inverted-ordering hypothesis $(\nu_3, \nu_1, \nu_2) = (s_0, s_1, s_2)$. The source-side label rule is absent.

10.3 Weighted-cycle bridge rigidity and absolute attachment

The weighted-cycle route supplies a frozen PMNS/hierarchy candidate. The reduced invariant C_ν was selected on a compare-only correction surface, with $B_\nu = P_\nu C_\nu$ retained as a diagnostic amplitude parameterization:

$$\begin{aligned} C_\nu &= G^2 Q S^{-1/2} = 0.9994295999075177, \\ P_\nu &= 6.699825740519345, \\ B_\nu &= P_\nu C_\nu = 6.696004159297337. \end{aligned}$$

The blocked absolute attachment displays

$$\begin{aligned} \lambda_\nu &= \frac{m_{\star,\mathrm{eV}}}{q_{\mathrm{mean}}^{p_\nu}} P_\nu C_\nu = 1.7237014208357415, \\ m_i &= \lambda_\nu \hat{m}_i, \\ \Delta m_{ij}^2 &= \lambda_\nu^2 \widehat{\Delta m_{ij}^2}. \end{aligned}$$

The basis permutation, holonomy orientation, and CP sign lack source-side selection. Exhaustive enumeration of both stored orientations and all row and mass-column relabelings of the stored candidate matrix finds no normal-ordering-consistent relabeling that passes both NuFIT 6.1 correlated 3σ gates. This excludes a convention-only rescue of that matrix; it neither derives nor exhausts possible source-side physical charged-basis placements. The one-parameter atmospheric-anchor slice and the two-parameter positive-segment adapter are compare-only continuations.

11 Hadrons, QCD, and the Emergence of Ordinary Matter

The hadron derivation is the most operationally demanding part of the particle construction. The source-only artifact is not a fitted hadronic residual and not a bare scalar ρ_{had}. It is a source-derived hadronic spectral backend: a QCD quotient ensemble, source QCD parameter map, Euclidean slab/vacuum-transfer construction, hadronic Hilbert quotient, Ward-normalized electromagnetic current accounting, positive two-current spectral export $d\rho_Q^{(2)}$, higher-point and transition spectral exports, same-scheme remainder Ξ_Q, and systematics accounting. A source-only hadron mass row requires a working OPH hadron backend, such as GLORB/Echosahedron, that can emit these objects with verifiable provenance and production systematics. Small local surrogates cannot provide the required quantum chromodynamics. The two-current spectral measure is the marginal needed for running-α and HVP transport; it is insufficient for HLbL and rare-decay long-distance amplitudes without the four-current and transition sectors. The source-backend boundary has an empirical comparison contract. The empirical comparison uses a separate $e^+e^- \to$ hadrons payload class for the hadronic spectral contribution and stays separate from source-only OPH rows. The source-only derivation nevertheless keeps the mathematical execution bridge, deterministic sampling law, evaluator, and surrogate validation as an execution contract for that backend.

11.1 Seeded $2+1$ family and unquenched measure

Let

$$m_l := \frac{m_u + m_d}{2}, \qquad \rho_l := \frac{m_u + m_d}{2\,\Lambda_{\overline{\mathrm{MS}}}^{(3)}}, \qquad \rho_s := \frac{m_s}{\Lambda_{\overline{\mathrm{MS}}}^{(3)}}.$$

The seeded fixed-physics family is

$$a\Lambda_n = a\Lambda_{\text{seed}}\, 2^{-n}, \qquad a\Lambda_{\text{seed}} = \sqrt{\rho_l \rho_s},$$

$$am_l^{(n)} = a\Lambda_n \rho_l, \qquad am_s^{(n)} = a\Lambda_n \rho_s,$$

$$\beta_n = 6 + \frac{9}{2\pi^2} \log \frac{1}{a\Lambda_n}, \qquad L_n = \left\lceil \frac{\lambda_L^{\text{target}}}{a\Lambda_n} \right\rceil, \qquad T_n = \left\lceil \frac{\lambda_T^{\text{target}}}{a\Lambda_n} \right\rceil.$$

The unquenched ensemble measure is

$$d\mu_n(U) = Z_n^{-1} \exp[-S_g(U;\beta_n)] \det D_l(U; am_l^{(n)})^2 \det D_s(U; am_s^{(n)})\, dU.$$

On this branch, $N_f = 2+1$ and QED is off.

11.2 Deterministic configuration and source contract

The emitted execution law is

$$U_{n,c} = K_n^{N_{\text{therm}} + cN_{\text{sep}}}(U_{\text{cold}}; \text{seed}_{n,c}),$$

with stop-time formula

$$t_{\text{stop}}(n,c) = N_{\text{therm}} + cN_{\text{sep}}.$$

The deterministic cfg seed law is

$$\text{seed}_{n,c} = \text{Decode}\big(h_{n,c}\big),$$

with

$$h_{n,c} = \text{SHA256}\Big(\text{Serialize}\big(n, \beta_n, L_n, T_n, am_l^{(n)}, am_s^{(n)}, c\big)\Big).$$

The source set is fixed as

$$S_{n,c} = \left\{(0,0,0,0), \left(\left\lfloor \frac{L_n}{2} \right\rfloor, \left\lfloor \frac{L_n}{2} \right\rfloor, \left\lfloor \frac{L_n}{2} \right\rfloor, \left\lfloor \frac{T_n}{2} \right\rfloor\right)\right\}.$$

The surrogate execution uses

$$N_{\text{therm}} = 2048, \qquad N_{\text{sep}} = 512.$$

These values are surrogate execution inputs only; they are not claimed as theorem outputs.

The production geometry summary makes the runtime boundary concrete. On the emitted seeded $2+1$ family there are three ensembles and six configurations total. If one stores all four links at every site as full double-complex 3×3 matrices, the naive raw gauge storage estimate is

$$576 \text{ bytes/site},$$

which gives total naive raw gauge storage

$$2.80071464105088 \times 10^{14} \text{ bytes}$$

over the production schedule. The normalized backend correlator dump needed by the analysis layer is tiny:

$$195264 \text{ bytes}$$

for the full π_{iso}, $N_{\text{iso,dir}}$, and $N_{\text{iso,ex}}$ payload on the production schedule. The hadron requirement is the backend export bundle that would feed the normalized dump from real production execution.

11.3 Operational reading of the required computation

The lightweight analysis layer is separate from the physical backend. The seeded ensemble family, deterministic configuration and source contract, jackknife evaluator, forward-window selector, and uncertainty schema are explicit. A physical result requires Ward-projected hadronic spectral data from a working OPH hadron backend.

Technically, the required production computation is the standard lattice-QCD stable-channel workflow on the emitted family: run unquenched rational or standard hybrid Monte Carlo for the three seeded ensembles; for each realized configuration and fixed source solve the clover-improved Wilson light/strange systems; construct the zero-momentum π_{iso}, $N_{\mathrm{iso,dir}}$, and $N_{\mathrm{iso,ex}}$ two-point sequences; write the normalized backend bundle; then apply the jackknife and forward-window estimators to obtain $am_{X,\mathrm{ground}}$, R_X, $m_X[\mathrm{GeV}]$, and the published σ_{stat}, δ_{cont}, δ_{vol}, and δ_χ fields. This gives a complete source-only execution contract and a separate empirical display policy.

The engineering burden is asymmetric. Local execution suffices for surrogate validation and downstream readout tests because those calculations use a tiny correlator payload. The physical branch needs backend hardware and execution semantics. This paper treats GLORB/Echosahedron-class OPH hardware, or an equivalent working OPH hadron backend, as the source-only backend requirement. Production hadron masses enter the particle construction only with a backend-emitted Ward-projected spectral measure and its uncertainty budget. This is a source-backend scope boundary.

11.4 Stable-channel correlators, effective masses, and forward windows

The pion stable channel is

$$p_\pi^{(n,c,s)}(t) = \sum_x \Re\,\mathrm{tr}_{c,\mathrm{spin}}\left[\gamma_5 S_l(x;s)\gamma_5 S_l(s;x)\right].$$

The nucleon stable channel is

$$p_{N,\mathrm{dir}}^{(n,c,s)}(t) = \sum_x G_d, \qquad p_{N,\mathrm{ex}}^{(n,c,s)}(t) = \sum_x G_x,$$

$$p_N^{(n,c,s)}(t) = p_{N,\mathrm{dir}}^{(n,c,s)}(t) - p_{N,\mathrm{ex}}^{(n,c,s)}(t).$$

Cfg/source and ensemble averaging are

$$\bar{p}_X^{(n,c)}(t) = \frac{1}{|S_{n,c}|}\sum_{s\in S_{n,c}} p_X^{(n,c,s)}(t),$$

$$C_X^{(n)}(t) = \frac{1}{|C_n|}\sum_{c\in C_n} \bar{p}_X^{(n,c)}(t).$$

The effective-mass laws are

$$am_{\mathrm{eff},\pi}(t) = \log\frac{C_\pi(t)}{C_\pi(t+1)}, \qquad am_{\mathrm{eff},N}(t) = \log\frac{|C_N(t)|}{|C_N(t+1)|}.$$

The forward window is

$$W_n = \{t : 1 \le t+1 < \lfloor T_n/2\rfloor\}.$$

The evaluator monitors log-convexity,

$$R_{\log\mathrm{conv},\pi}(t) = C_\pi(t)^2 - C_\pi(t-1)C_\pi(t+1),$$

$$R_{\mathrm{log\,conv},N}(t) = |C_N(t)|^2 - |C_N(t-1)|\,|C_N(t+1)|,$$

tail-drop,

$$D_X(t) = am_{\mathrm{eff},X}(t) - am_{\mathrm{eff},X}(t+1),$$

and mirror suppression,

$$M_X(t) = \exp[-am_{\mathrm{eff},X}(t)(T_n - 2t)].$$

The selected forward window is the longest contiguous run satisfying finite effective masses, non-negative log-convexity up to tolerance, nonnegative tail-drop up to tolerance, mirror suppression below threshold, and local plateau flatness. The candidate ground-state mass is then the weighted window average

$$am_{X,\mathrm{ground}} = \frac{\sum_{t\in W_X^{\mathrm{sel}}} w_t\, am_{\mathrm{eff},X}(t)}{\sum_{t\in W_X^{\mathrm{sel}}} w_t}, \qquad w_t = \frac{1}{\max(\sigma_t^2, \varepsilon)}.$$

11.5 Statistics, systematics, and dimensional readout

Delete-1 jackknife is performed over the cfg axis after source averaging inside each cfg. With n_{cfg} configurations and integrated autocorrelation time $\tau_{\mathrm{int,cfg}}$, the effective cfg count is

$$n_{\mathrm{eff,cfg}} = \frac{n_{\mathrm{cfg}}}{2\,\tau_{\mathrm{int,cfg}}}.$$

The published statistical error is

$$\sigma_{\mathrm{stat},X} = \mathrm{JKstderr}(am_{X,\mathrm{ground}}).$$

The machine-readable systematics field uses

$$\sigma_{\mathrm{sys},X} = \sqrt{\delta^2_{\mathrm{cont},X} + \delta^2_{\mathrm{vol},X} + \delta^2_{\chi,X}}.$$

Continuum, volume, and chiral proxies are encoded as

$$R_X^{(n)} = \frac{am_X^{(n)}}{a\Lambda_n}, \qquad R_X^{(n)} \approx R_X(0) + c_X (a\Lambda_n)^2,$$

$$\delta^{(n)}_{\mathrm{cont},X} = a\Lambda_n \left| R_X^{(n)} - R_X(0) \right|,$$

$$\delta^{(n)}_{\mathrm{vol},\pi} = am_\pi^{(n)} \frac{e^{-am_\pi^{(n)} L_n}}{\max(am_\pi^{(n)} L_n, 1)},$$

$$\delta^{(n)}_{\mathrm{vol},N} = am_N^{(n)} \frac{e^{-am_\pi^{(n)} L_n}}{\max(am_\pi^{(n)} L_n, 1)},$$

$$\delta^{(n)}_{\chi,\pi} = a\Lambda_n \left| R_\pi^{(n)} - \langle R_\pi \rangle_n \right|,$$

$$Q_N^{(n)} = \frac{R_N^{(n)}}{R_\pi^{(n)}}, \qquad \delta^{(n)}_{\chi,N} = a\Lambda_n \langle R_\pi \rangle_n \left| Q_N^{(n)} - \langle Q_N \rangle_n \right|.$$

These are surrogate publication proxies, not production physical systematics. Given a ground-state candidate,

$$R_X = \frac{am_{X,\mathrm{ground}}}{a\Lambda^{(3)}_{\overline{\mathrm{MS}}}}, \qquad m_X[\mathrm{GeV}] = R_X\, \Lambda^{(3)}_{\overline{\mathrm{MS}}}[\mathrm{GeV}],$$

with

$$\Lambda^{(3)}_{\overline{\mathrm{MS}}} = 0.3344017073 \text{ GeV}.$$

11.6 Surrogate execution bundle and frontier

The surrogate execution evolves a latent state $z \in \mathbb{R}^d$ with Hamiltonian

$$H(z,p) = S(z) + \frac{1}{2}\sum_i p_i^2,$$

where

$$S(z) = \frac{1}{2}\sum_i \omega_i^2 z_i^2 + \lambda_4 \sum_i z_i^4 + \kappa \sum_i (z_{i+1} - z_i)^2 + 2\alpha_l \sum_i \log(\mu_l^2 + z_i^2) + \alpha_s \sum_i \log(\mu_s^2 + \tfrac{1}{2} z_i^2).$$

Leapfrog integration plus Metropolis accept/reject gives the surrogate HMC update

$$(z,p) \mapsto (z', p'), \qquad P_{\rm acc} = \min(1, e^{-\Delta H}).$$

This kernel is a deterministic executable surrogate honoring the emitted receipt and seed law, with no physical lattice-QCD RHMC/HMC claim.

For validation of the execution bridge, the surrogate locks the ground-state masses to the hadron comparison proxy values

$$m_{\pi,\rm proxy} = 0.13497682776768472 \text{ GeV},$$
$$m_{N,\rm iso,proxy} = \frac{m_p + m_n}{2} = 0.93891875434 \text{ GeV}.$$

Thus

$$am^{(n)}_{\pi,\rm proxy} = \frac{m_{\pi,\rm proxy}}{\Lambda^{(3)}_{\overline{\rm MS}}[\text{GeV}]}\, a\Lambda_n,$$
$$am^{(n)}_{N,\rm iso,proxy} = \frac{m_{N,\rm iso,proxy}}{\Lambda^{(3)}_{\overline{\rm MS}}[\text{GeV}]}\, a\Lambda_n.$$

The surrogate correlator is

$$C_X^{\rm sur}(t) = A_{X,0} e^{-am_X t} + A_{X,1} e^{-am_{X,\rm ex} t} + A_{X,\rm mir} e^{-am_X (T_n - t)}$$

up to a multiplicative correlated noise factor. Direct and exchange nucleon pieces are written as

$$C^{\rm sur}_{N,\rm dir}(t) = f_{\rm dir}\, C_N^{\rm sur}(t), \qquad C^{\rm sur}_{N,\rm ex}(t) = (f_{\rm dir} - 1)\, C_N^{\rm sur}(t),$$

so that

$$C_N^{\rm sur}(t) = C^{\rm sur}_{N,\rm dir}(t) - C^{\rm sur}_{N,\rm ex}(t).$$

On the finest surrogate ensemble, the resulting diagnostic candidates are

$$m^{\rm sur}_{\pi,\rm iso} = 0.135039383836 \text{ GeV}, \qquad m^{\rm sur}_{N,\rm iso} = 0.938960210578 \text{ GeV},$$

with worst absolute error approximately 6.26×10^{-5} GeV across the surrogate stable-channel family. These numbers validate the emitted execution bridge. They are not promotable production hadron predictions.

So the hadron derivation closes the full

receipt → execution → writeback → evaluation → budgets → forward-window summary

path on executed surrogate data, while the physical closure requires production unquenched RHMC/HMC, real Dirac solves and baryon contractions, production autocorrelation studies, and production continuum / finite-volume / chiral systematics.

12 Observer-Centric Particle Ontology and Measurement

OPH is observer-centric at the level of its basic ontology, so particle physics also requires an account of what a particle *is* and how a measurement turns an excitation into an observed state. The fixed-cutoff record and measurement theorems give that account precise mathematical content.

12.1 Observer patches, overlap consistency, and particle data

Ref. [3] formulates the basic kinematic picture in patch-net language: each observer patch carries a local state, neighboring patches compare a shared interface alphabet, and a global state is physically admissible exactly when neighboring projections agree on the overlap. The algebraic OPH form of the same statement is the first and second axioms: local physical data are carried by patch algebras $\mathcal{A}(P)$, and those local states must agree on shared subalgebras whenever patches overlap.

The particle interpretation starts from patch-local algebras, overlap-visible observables, edge sectors, and transport data rather than a single absolute global particle basis. A particle row in the reported derivation is a readout claim about a stable or candidate-stable excitation structure visible to a family of observer patches and consistent on their overlaps.

For this record-algebra discussion, use the support-local algebra-state-record reduct

$$O_{\text{red}} = (P, \mathcal{A}(P), \rho, R),$$

where P is the support patch, $\mathcal{A}(P)$ its local algebra, ρ the local state, and R the record algebra. The full operational observer additionally carries overlap interface algebras and restriction maps, allowed update and repair instruments, and checkpoint data used for continuation. For the particle derivation, R belongs to the quantum observer surface itself. On the exact fixed-cutoff measurement surface it is generated by central record projectors, and practical readout may use approximately commuting projectors that are close to that central reference algebra. That is what makes the observer-facing record surface shareable across overlapping observer descriptions without violating the usual no-cloning constraints on generic quantum states.

12.2 Record algebras and definite outcomes

The integrated measurement appendices and Ref. [4] make the measurement interface precise at fixed cutoff. On the declared operational surface, the completed write/verify slice carries a finite commutative central record algebra. Practical readout may instead use projectors Q_a on the same declared slots with commuting central reference projectors $\widehat{Q}_a$, where

$$\delta_{\text{rec}} := \max_a \|Q_a - \widehat{Q}_a\|.$$

Then

$$\|[Q_a, Q_b]\| \le 4\,\delta_{\text{rec}},$$

and, whenever $\|\widetilde{\rho} - \rho\|_1 \le \varepsilon$,

$$\left|\mathrm{Tr}(\widetilde{\rho} Q_a) - \mathrm{Tr}(\rho \widehat{Q}_a)\right| \le \varepsilon + \delta_{\text{rec}}.$$

On the explicit fixed-cutoff screen architecture, the exact record algebra after a completed write/verify cycle is

$$\mathcal{Z}_{\text{rec}}(t) = \mathrm{Alg}\Big(\{P_{m_I^{(J)}(t)}\}_{I\neq J}, \{P_{r_I^{\text{bulk}}(t)}\}_I, \{\Pi_\alpha^{(IJ)}(t)\}_{I<J,\alpha}\Big).$$

Ref. [4] proves that, on the declared operational measurement surface, $\mathcal{Z}_{\mathrm{rec}}(t)$ is commutative and central for the readout instrument being used.

This fixed-cutoff centrality result is what supports definite outcomes without adding a second ontology. The integrated measurement construction phrases the same idea through edge-center decomposition: on a fixed collar with exact Markov structure aligned with the edge split (the Standard Model gauge paper's Markov-split alignment hypothesis), the state splits into superselection blocks,

$$\rho_{ABC} = \bigoplus_j q_j \, \rho^{(j)}_{Ab_L^{(j)}} \otimes \rho^{(j)}_{b_R^{(j)}C},$$

and the label j is classical center data. The supplement then goes one step further: in the physical algebra there are no interference observables between different sector blocks. So once a particle detection event has been recorded in the observer-accessible center/record algebra, the accessible physics is organized as a classical mixture over those blocks instead of as a superposed measurement record.

Accordingly, a measured particle family or excitation channel should be thought of as an observer-accessible sector or record value carried by the central measurement surface, not as a mysterious absolute collapse of the full universe-state into a preferred basis chosen by hand.

12.3 Born probabilities and post-measurement states

The fixed-cutoff measurement package is also explicit about probabilities and updates. For every event E in the sigma algebra generated by $\mathcal{Z}_{\mathrm{rec}}(t)$, the measurement probability is

$$\mathbb{P}_t(E) = \mathrm{Tr}(\rho_t P_E),$$

where $P_E \in \mathcal{Z}_{\mathrm{rec}}(t)$ is the projector for that event. Conditioning on the event then produces the post-measurement state

$$\rho_t|_E = \frac{P_E \rho_t P_E}{\mathrm{Tr}(\rho_t P_E)}.$$

In the supplement, the same statement appears as the Lüders update on the record algebra. In the synthesis paper *Observers Are All You Need* [2], this fixed-cutoff Born/Lüders package and its Bell/CHSH extension are imported as theorem-bearing, not non-theorem commentary.

This directly answers the particle-state question in the present framework. Measurements give particles actual states by conditioning the observer-accessible state on the central record algebra. A detector click, a stable overlap-sector readout, or a pointer value does more than announce a pre-existing classical label; it defines the conditioned state for subsequent observer-accessible physics. If the same event is re-read without any accepted repair move touching its support, the result in Ref. [4] shows that the re-read probability is 1. So the measurement interface is well defined and operationally auditable.

12.4 Particle identity as transport-stable structure

Once one asks "which event happened?" and then "which particle family is this?", the flavor derivation becomes essential. The active flavor derivation does not start with the names electron, muon, tau lepton, or up quark built into the fundamental observable. It starts with transport kernels, generation-bundle data, same-label eigenline transport, overlap-edge cocycles, and a persistent flavor observable carrying intrinsic labels $f1, f2, f3$. This is an important discipline condition in the paper. At the deepest flavor surface used here, family identity is transport-stable intrinsic structure first and named experimental family assignment second.

The shared Yukawa/excitation dictionary carries substantial weight even though it is not itself the leading missing object. The flavor construction gives an invariant base: projectors, spectral gaps, pair suppressions, cycle phases, and common sector-response objects. No map from that intrinsic base to the named low-energy fermion families is derived. The quark calculation supports direct numerical comparison, while the charged-lepton theorem emits no physical mass. The neutrino weighted-cycle construction is a target-informed template candidate without a physical flavor-labelled attachment.

Accordingly, particle identity is a refinement-stable observer-accessible transport pattern whose named interpretation is inherited from the available dictionary and declared comparison domain. Without that dictionary, no named physical identity follows.

13 Affine Event Records and Cross-Boundary Token Stitching

The sector-identity analysis concerns which transport-stable excitation type is being read. A different question is whether two located record tokens, seen near a chart or partition boundary, are the same continuing observer-visible token. This section closes that second question at the finite-certificate level. It does not add a particle species, mass, charge, or scattering theorem. It says when a bounded patch system is allowed to stitch affine event records into nonbranching record-worldlines, and when it must report ambiguity.

13.1 Event chart, frame, and clock atlas

The compact cap-normal theorem supplies the Lorentz-natural frame hyperboloid. It identifies sky directions with $q(\Omega) = (1, \Omega)$, represents oriented round caps by $n_C = (\cot\alpha, \csc\alpha\,\mathbf{c})$, and maps each cap to a geodesic half-space in H^3. The compact event-manifold packet separately supplies affine event charts under its population, separation, chart, cone, and reachability receipts. This section consumes event supports, observer clocks, chart transitions, gauge transport, and worldline stitching. Frame estimation, event location, overlap descent, and worldline stitching are separate certificates.

Fix a curvature radius $R_H > 0$ and write

$$H^3_{R_H} = \{X \in \mathbb{R}^{1,3} : \langle X, X\rangle_L = -R_H^2,\ X^0 > 0\}, \qquad \langle X, Y\rangle_L = -X^0Y^0 + X^1Y^1 + X^2Y^2 + X^3Y^3.$$

The hyperbolic distance used for conditioned frame comparison is

$$d_H(X, Y) = R_H \operatorname{arcosh}\left(-\frac{\langle X, Y\rangle_L}{R_H^2}\right).$$

An event-record atlas is a tuple

$$\mathcal{A}_H = (R_H, \{U_i, \chi_i, u_i, J_i, \vartheta_i\}, \{G_{ji}, F_{ji}\}, E, \nabla, \mathsf{Prov}_{R_H}, \mathsf{Prov}_u)\,.$$

Here χ_i are affine event charts, $u_i \in H^3_{R_H}$ is the selected local observer-frame anchor when such an anchor is claimed, J_i are event-chart domains for record support, and $\vartheta_i : J_i \to T$ are observer-clock maps into a common comparison-time line T. Prov_{R_H} records whether R_H is a unit convention, an imported physical scale, or an independently derived OPH scale. The unit-chart convention alone supplies no independent OPH scale. Prov_u records which observer, tetrad, or clock object selected the anchors u_i. The Lorentz part $G_{ji} \in \mathrm{SO}^+(1,3)$ transports frames and tetrads; the full Poincaré transition transports affine event coordinates. The map F_{ji} is the sector or gauge transport functor on the record payload. These two holonomies are deliberately separate: atlas holonomy tests geometry, and gauge holonomy tests transported sector data. When two charts describe the same transported observer frame, the atlas additionally requires

$$G_{ji}u_i = u_j.$$

Theorem 13.1 (H3 geometric naturality)**.** *For every $A \in \mathrm{SO}^+(1,3)$, the map $X \mapsto AX$ preserves $H^3_{R_H}$, d_H, geodesics, exponential and logarithm maps, parallel transport, Hausdorff distances between compact frame supports, and Fréchet means whenever the mean is unique.*

Proof. The defining equation of $H^3_{R_H}$ and the formula for d_H depend only on the Lorentz inner product and the future sheet. Elements of $\mathrm{SO}^+(1,3)$ preserve both. The Levi-Civita connection, geodesic exponential, logarithm, and parallel transport are functorial for isometries, so the listed constructions commute with A. Hausdorff distance and unique Fréchet means are metric constructions, hence are preserved by any isometry. □

13.2 Cap-response frame balls and affine event supports

The Standard Model gauge paper supplies the conditional inverse theorem that turns calibrated modular cap responses into conditioned observer-frame supports. For every descended record token i and clock slice t, the frame layer consumes a certificate

$$R_i(C,t,O) = \omega_{i,O}\Big(\sigma_t^{C,O}(M_{C,0,O})\Big), \qquad y_{i,j} = F_j(X_i(t)) + e_{i,j},$$

on a compact frame domain $\Omega \subset H^3_{R_H}$, with quantitative observability

$$\alpha\bar{d}(X,Y) \le |F(X) - F(Y)|_W \le L\bar{d}(X,Y), \qquad |e_i|_W \le \sigma_i(t).$$

Given an ε-net and residual tolerance $\tau_i(t)$, the imported theorem emits

$$S_i(t) = B_H(\widehat{X}_i(t), r_i(t)), \qquad r_i(t) = R_H\left[\frac{L}{\alpha}\varepsilon + \frac{2}{\alpha}\sigma_i(t) + \frac{1}{\alpha}\tau_i(t)\right].$$

A frame estimate $\widehat{X}_i(t)$ is a unique finite output only when $\Delta_{\mathrm{frame},i}(t) > 0$. If this gap fails, the output is a frame ambiguity set or support ball. This ball cannot serve as an event position. Event stitching consumes an affine event support $A_i(t)$ produced by the event-manifold chart and ancestry receipts. Distinct event supports at a shared clock slice require a certified separation in the affine event metric. A conditioned-frame separation may also be recorded:

$$d_H(\widehat{X}_i, \widehat{X}_k) > r_i + r_k + m_{\mathrm{sep}}$$

for a declared positive frame margin m_{sep}. It does not replace event separation.

13.3 Proto-record extraction and descent

A raw local proto-record is a connected component of an invariant detector scalar above a frozen threshold, together with its certified affine event support, conditioned frame ball, local clock interval, sector payload, and uncertainty collar. The detector scalar is built from record-algebra data, not from implementation object IDs. Extraction is chart-natural when transporting the raw field by G_{ji} and F_{ji} carries extracted components in chart i to the extracted components in chart j, up to the declared support collar.

Before temporal stitching, overlapping charts are descended. The overlap-descent relation joins two proto-records only when their transported affine event supports lie in the same connected component on a real chart overlap and the same-component margin dominates the support collar. If distinct component separation does not dominate the collar, the descent emits an ambiguity certificate instead of choosing a label. On component-connected covers, this descent produces canonical global descended records by the usual sheaf gluing argument: cocycle-compatible local components with positive separation margins have a unique global component, and the construction is natural under atlas restriction.

13.4 Boundary-crossing germs and transported residuals

A cross-boundary candidate edge is admissible only when the two descended records meet a real patch interface in adjacent observer-clock intervals and share a common comparison chart. The interface contact must be oriented and transverse: the signed-distance interval brackets the interface, and the normal velocity has a positive lower bound. Tangencies, grazing contacts, and file or shard-name boundaries without a physical interface are rejected. If the local slab contains an interaction marker, the record is routed to the interaction solver instead of stitched by the free-propagation rule.

For an admissible pair, the certificate transports the left record into the right comparison chart and evaluates frozen residuals

$$r_t, \quad r_\partial, \quad r_x, \quad r_v, \quad r_{\mathrm{int}},$$

for clock adjacency, boundary contact, affine event position, transported velocity or finite difference, and interaction absence. The cost function is declared before seeing the proposed matching. Record IDs, global IDs, shard-local IDs, and stitch keys are forbidden in admissibility, costs, and tie-breaking. All affine event-position and velocity residuals are evaluated as certified interval or uncertainty-box distances. A point-estimate residual may be printed as a diagnostic; it cannot replace the event-support radius in an admissibility or matching certificate. Conditioned H^3 frame residuals are a separate tetrad-consistency check.

Lemma 13.2 (Candidate-edge naturality)**.** *The admissible candidate-edge set and its residual intervals are invariant under common* $\mathrm{SO}^+(1,3)$ *chart changes and under gauge-bundle trivialization changes.*

Proof. Every event term is computed from the affine overlap cocycle, oriented interface data, and the common clock line. Frame terms use d_H and Theorem 13.1. Sector and gauge residuals compare transported payloads after applying the declared connector, so changing a local trivialization conjugates both sides of the comparison. The residual intervals and the resulting admissibility predicate are therefore unchanged. □

13.5 Assignment gap, nonbranching paths, and ambiguity

Let L and R be the descended record sets on two adjacent time slabs. A stitch matching is a partial one-to-one assignment between L and R, augmented by declared appearance and disappearance interval costs. Its certified gap is

$$\Delta_{\mathrm{cert}} = \min_{M \neq M_*} \underline{C}(M) - \overline{C}(M_*),$$

where M_* is the proposed matching, $\overline{C}(M_*)$ is its upper cost bound, and $\underline{C}(M)$ is the lower cost bound for every competing matching. The event-location gap Δ_{event} and stitch-assignment gap

Δ_{cert} are distinct receipts: the first certifies a unique event support inside one clock slice, while the second certifies a unique temporal assignment across slices.

Theorem 13.3 (Stable ID-independent stitch assignment). *If the candidate-edge set is natural, the costs are frozen and ID-independent, the proposed matching is one-to-one, and $\Delta_{\text{cert}} > 0$, then M_* is the unique certified minimum-cost stitch. The certified matching is natural under chart changes, gauge trivializations, and relabeling of implementation record IDs.*

Proof. The strict gap says every competing matching has lower cost bound above the proposed matching's upper cost bound, so no competitor can tie or beat it under any realization inside the certified intervals. Lemma 13.2 makes the candidate graph and cost intervals invariant under chart and gauge presentation changes. Because IDs are excluded from admissibility and costs, relabeling them changes only bookkeeping fields, not the optimum. □

Theorem 13.4 (Unavoidable ambiguity). *If a nontrivial permutation of the visible descended records preserves the atlas, clocks, interfaces, sector/gauge payloads, candidate graph, and cost intervals, then no natural ID-independent finite procedure can certify a unique stitch across that slab.*

Proof. A natural procedure must commute with the permutation. A unique certified output would therefore be fixed by the permutation. But the permutation sends one equally admissible stitch to a distinct equally admissible stitch. Choosing between them would introduce a label or presentation dependence. The only natural finite output is an ambiguity label. □

The certified stitch graph is then the graph obtained by chaining the unique matchings across adjacent slabs. Since each matching is partial one-to-one, every connected component is a nonbranching path, ray, or isolated record. These are the affine event-record worldlines of this section.

13.6 Refinement naturality and dynamical lift

For a refinement $s \to r$, let Q_{sr} contract fine descended records to coarse records and let η_{sr} bound the cost distortion between the fine and coarse stitch problems. If the coarse certified gap obeys $\Delta_r > 2\eta_{sr}$, then the contracted fine optimum is isomorphic to the coarse optimum. This is the refinement-naturality gate: a cross-boundary stitch is not paper-grade until the coarse/fine contraction agrees within the declared margin.

The stitch certificate lives on the conditional Lorentzian event manifold. A geodesic or free-particle interpretation requires a separate dynamical receipt with a timelike speed margin, connection compatibility, and an action or propagation law. The stitch theorem proves token continuation in event charts; it does not prove geodesic motion.

Theorem 13.5 (Certified affine event-record stitching). *Given a valid affine event atlas satisfying The Standard Model gauge paper's event-manifold receipts, a common operational clock atlas, affine event supports for every descended record token and clock slice, and separately typed conditioned-frame certificates where observer frames are used, chart-natural proto-record extraction, overlap descent with positive event-support separation margins, real transverse interface-crossing germs, sector and gauge transport continuity, a frozen ID-independent one-to-one assignment with $\Delta_{\text{cert}} > 0$, no interaction marker in the free-propagation slab, and a passing coarse/fine contraction gate, the emitted stitch graph is a unique natural nonbranching affine event-record graph. If any positive event-location gap, separation margin, or assignment gap is absent, the natural output is* AMBIGUOUS*; if an interaction marker is present, the output is* INTERACTION REQUIRED*; if atlas, interface, event-location, or transport gates fail, the stitch is rejected.*

Proof. The atlas, event-location, and extraction gates make local proto-records presentation-independent. The positive Δ_{event} gate licenses event-position outputs; otherwise the procedure carries event boxes or ambiguity sets forward. Descent turns overlap-compatible local components into descended records before any temporal decision is made. The interface and transport gates define a natural candidate-edge graph by Lemma 13.2. The strict assignment gap gives a unique ID-independent matching by Theorem 13.3. Chaining partial one-to-one matchings gives nonbranching components, and the refinement gate makes the result stable under the declared coarse/fine contraction. The alternative outcomes are exactly the negations of these conditions, with Theorem 13.4 forcing ambiguity when the visible data do not separate a unique continuation. □

13.7 Scope boundary of the measurement chapter

The fixed-cutoff measurement theorem gives the central record algebra, Born probabilities, Lüders conditioning, repeated-read stability, and the stated two-wing Bell/CHSH result. It does not prove global observer continuation or strange-loop closure. Those metaphysical proposals do not enter the technical argument.

14 Discussion: Matter, Antimatter, Supersymmetry, and Physical Boundaries

The OPH theorems distinguish structural results, conditional quantitative branches, and target-conditioned comparisons. Matter versus antimatter, supersymmetry, and several adjacent questions lie on that boundary. The quark calculation is an obstruction and target-conditioned surface rather than a set of source-only numeric predictions. Its S_3 table is a target-anchored diagnostic.

14.1 Matter and antimatter

The Standard Model structural branch fixes the chiral gauge architecture within which matter and antimatter are defined. Once a chiral fermion representation is present, its conjugate representation gives the corresponding antiparticle content. In that limited sense, OPH explains why matter and antimatter both exist: they are paired by the realized gauge and chiral structure of the low-energy branch.

The cosmological asymmetry requires a separate finite source object. At regulator r, write

$$\mathfrak{B}_r = (Q_r, C_r, E_r, \omega_{R,r}, L_{r,T}, p_{r,i}, \tau_r, T_r, \rho_{R,r}).$$

Here Q_r is the physical quotient, C_r is its CP involution, $\omega_{R,r}$ is a CP-odd integral winding cocycle, $L_{r,T}$ is the physical quotient repair generator, $p_{r,i}$ is the initial law, τ_r and $T_r(\tau)$ are the physical clock and temperature map, and $\rho_{R,r}$ assigns record charges r_ψ to chiral matter.

For a record phase acting as $\psi \mapsto e^{ir_\psi \Theta_R}\psi$, the finite-quotient anomaly-and-current theorem gives

$$\boxed{k_R = \sum_{\psi\,\text{LH}} r_\psi \, 2T_2(R_\psi)}$$

and

$$\boxed{\dot{\Theta}_R(T) = \theta_0 \sum_{q,q' \in Q_r} p_r(q,T) L_{r,T}(q,q') \omega_{R,r}(q,q')}, \qquad \theta_0 = \frac{2\pi}{m_R}.$$

The first equation is the mixed electroweak anomaly index. The second is an oriented probability current in proper time. Quotient settlement supplies no rate matrix, and a repair iteration count supplies no physical clock.

The theorem also fixes the sign boundary. If the initial law and generator are CP symmetric,

$$p_{r,i}(C_r q) = p_{r,i}(q), \qquad L_{r,T}(C_r q, C_r q') = L_{r,T}(q, q'),$$

then every transition cancels its CP image and

$$\boxed{\dot{\Theta}_R(T) = Y_B = 0.}$$

An oriented register and a normal-form map therefore select no cosmological matter sign. A nonzero sign requires a quotient-intrinsic CP-odd boundary condition, action term, or transition affinity.

The realized $\mathbb{Z}_6$ determinant/deck winding supplies an integral phase coordinate with inversion under orientation reversal. Its natural Standard Model attachment is the central hypercharge direction. The associated coefficient vanishes generation by generation:

$$\boxed{k_R^{YWW} = N_g(3Y_Q + Y_L) = 3\left(3 \cdot \frac{1}{6} - \frac{1}{2}\right) = 0.}$$

This is the mixed $SU(2)_L^2 U(1)_Y$ anomaly-cancellation condition. The gauge/deck phase cannot directly supply the required $\Theta_R W\widetilde{W}$ coupling. Quotient periodicity and the twelve electroweak zero modes do not select $k_R = 1$ or any other nonzero record attachment.

A distinct global record attachment can carry a nonzero index. On the three-generation branch,

$$k_R(Y) = k_R(B-L) = 0, \quad k_R(B) = k_R(L) = 3, \quad k_R(B+L) = 6.$$

A quotient-visible gauge-singlet $B+L$ record phase is therefore a viable conditional attachment. Its derivation from OPH repair data is open. On that branch, the freeze-out transport functional requires

$$\left\langle \frac{\dot{\Theta}_R}{T} \right\rangle_{\mathrm{fo}} = (4.463 \pm 0.028) \times 10^{-9}.$$

This is a source-generator target. It may not be consumed to select $L_{r,T}$, the attachment, or the CP-odd affinity.

The exact source theorem and gauge/deck no-go are closed at finite quotient. The physical baryon abundance is work in progress. Its open source packet consists of a distinct anomalous record phase, source-only transition rates, a proper-time and temperature calibration, a CP-odd initial or boundary law, sign-domain coherence, and the resulting full source history. Transport, washout, and freeze-out must then be evaluated on that emitted history.

14.2 Why the declared branch does not require supersymmetry

The heat-kernel appendix below contains the gauge-coupling calibration. The declared branch needs no supersymmetric partner spectrum to produce its unification-like running benchmark. This does not rule out supersymmetry in a different physical completion.

At one loop, the appendix records the familiar fact that Standard Model beta-function coefficients do not produce successful naive unification, whereas minimal supersymmetric Standard Model coefficients do. The conditional calibration hypothesis is that edge-sector heat-kernel weights shift the effective beta-function coefficients in that direction through geometric or entropic sector multiplicity instead of through an actual low-energy superpartner spectrum. In that reading, "unification-like behavior" and "a supersymmetric particle zoo" come apart.

This calibration is not a derived unification theorem. The appendix records one calibration branch in which Peter–Weyl multiplicities, the printed one-loop running frame, the threshold conventions, and an additional fermionic-grading restriction together produce an supersymmetric benchmark shift. It does not prove that those effective loop multiplicities, statistics restrictions, and decoupling conventions are uniquely forced by OPH edge sectors alone.

For this paper, the correct conclusion is therefore scope-limited and modest:

1. the conditional derivation uses the declared Standard Model branch without having to introduce a supersymmetric partner for every known field;
2. the heat-kernel appendix contains a conditional calibration in which unification-style running features could arise from edge-sector structure instead of supersymmetric particle content;
3. the beta-shift calculation uses explicit calibration assumptions rather than a closed edge-sector theorem;
4. none of this is a universal theorem that supersymmetry is impossible.

So if the paper asks "why is there no supersymmetry in the derived spectrum?", the best technical answer is: because the declared branch stops at the Standard Model content, and the existing unification discussion is trying to reproduce the relevant running behavior through a separate heat-kernel calibration, without extending the particle content to a full supersymmetric multiplet structure.

14.3 Three Generations And Flavor Closure

The target-free structural matter packet leaves the three values in $3 \le N_g \le 5$ indistinguishable. On the strengthened screen interface, the operational Laplacian cost selects the canonical rank-three band uniquely among the single complete faithful candidates. The declared unitary simulator recovers the same band at its lowest positive generator frequency. This supplies an exact finite family candidate. The fifteen-state charge table has exact anomaly cancellation, diagonal $\mathbb{Z}_6$ action, and a nondegenerate chirality grading. Tensoring the rank-three response band with that table gives a conditional complex rank-45 candidate. On a separate finite local domain, the receipt checks the declared scalar signed operator tensored with I_{45}; its zero kernel and exact positive dimensionless gap are inherited with multiplicity 45. The source does not select this matter action. The twelve-port Spin packet and the 8,662-node operator domain have no certified source, domain, or transport bridge. Matter-pole identification, the continuum Spin/locality limit, physical seam selection, the missing persistence leg, and laboratory current identification have no receipts [5]. The full flavor dictionary, excitation map, and CKM closure are not derived. The charged source landing from P to physical charged data is a source-domain no-go, and the neutrino branch has no source-closed flavor prediction. Its weighted-cycle candidate fails the NuFIT 6.1 correlated profile [17]. The flavor and family chapters therefore spend substantial space on constructive object boundaries as well as final numbers.

14.4 Ordinary matter and the hadron backend boundary

A second distinction belongs in discussion form instead of in the results sections: the difference between "elementary rows exist" and "ordinary matter is fully derived." Ordinary visible matter is dominated by hadrons, especially protons and neutrons. The OPH derivation has a hadron construction, with source-only hadron masses gated by a production backend export. The paper emits the eligible elementary rows and retains the weighted-cycle coordinates only as a rejected comparison record, while the full observed matter spectrum is not emitted from a source-only OPH backend.

14.5 Claim boundary for conditional branches

The reader should therefore interpret this whole discussion section with the same rule used by *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency and Deriving Standard Model Gauge Structure from Observer Overlap Consistency* [6, 7]. The structural gauge and gravity theorems do not depend on a quark source-spread selector, charged-lepton source landing, a source-derived flavor-labelled neutrino kernel, or source-only hadron masses. The independently supported bosonic rows have the same separation.

15 Conclusion

Under the explicit inverse-port response and rank-15 matter contracts, anomaly balance and tensor descent imply the Standard Model Lie type, hypercharge lattice, three-color carrier, and $\mathbb{Z}_6$ global form from the stated premises alone. The transportable-sector/Tannaka chain remains a separate conditional classification. Physical source binding, laboratory current identification, the two routes' current identity, matter-pole and continuum identification of the conditional rank-45 candidate, selection of the local matter action, cross-domain Spin transport, scalar multiplicity/dynamics, and continuum QFT are open. The strengthened finite screen interface selects its rank-three candidate under the stated premises. Exclusion of extra light sectors is a declared completion.

On the declared quantitative branch, one local coordinate P organizes the electromagnetic, electroweak, hierarchy, and Higgs readouts. The common-load premise gives a dimensionless hierarchy and $\epsilon_H = 0$. The weak-boson and Higgs numbers are chart coordinates. The conditional perturbative implication from a complete renormalized packet to strict charged and neutral complex-pole coefficients is specified. The OPH source does not produce that packet: its source-selected action, complete Yukawas, Fleischer–Jegerlehner coordinate [26, 27], target-clean running/matching, two independent engines, executable ST/Nielsen/current-pole receipts, source law/covariance, physical reference transition, and SI clock binding are work in progress. The finite local-domain Hamiltonian has an exact positive gap for its signed seam operator. This finite graph operator is distinct from the compact-gauge repair generator and the continuum Yang–Mills Hamiltonian. It supplies no physical clock or mass scale. A nonperturbative chiral gauge construction would additionally face the known lattice chiral-measure program [22, 23, 24] and the γ_5-scheme restoration bookkeeping [25]. A finite chiral quantum object and a nonperturbative observable reconstruction are not prerequisites for the bounded imported-Standard-Model perturbative validation. Both are work in progress for their stronger claims.

The flavor analysis produces exact algebraic statements and explicit negative results. The positive-chamber face-circulant identity reduces Koide balance to $|b|/a = 1/\sqrt{2}$, and equal rank-two event blocks give that modulus in the conditional finite tracial-GNS packet. Physical chiral-mass attachment, phase, and numerical ratios are open. The analysis proves a two-modulus quark non-identifiability result, and rejects the tested register-Clebsch, reciprocal-quark, and weighted-cycle neutrino routes. The Clebsch rejection leaves a conditional channel-pairing boundary and a target-free unordered weight set under declared constraints; neither supplies Yukawa-coefficient equality or a source-derived family order. The exact real-axis enumeration excludes direct identification of the Cabibbo angle with an acute angle between two of the 31 real three-dimensional icosahedral residual axes, while leaving spinorial, higher-order, and dynamical routes outside its scope. The W_5 stabilizer theorem shows why residual symmetry cannot finish the charged-family problem: threefold and fivefold fixed points are degenerate, and the twofold fixed locus retains two parameters after scaling. Absolute charged-lepton masses require a source-derived clock and determinant map. Source-only hadron masses require a working hadronic production backend. Empirical hadronic transport remains identified as empirical input.

The relation between the particle hierarchy coordinate and cosmic capacity is conditional on two constructions in the companion papers: the direct public-record fixed point and the identification of its carrier with the screen/electroweak load. The particle results are stated separately as theorems, conditional implications, comparisons, rejected candidates, and requirements for physical identification.

A Heat-Kernel Calibration Supplement

This appendix gives the compact-group heat-kernel calculation used by the particle calibration. The fixed-cutoff same-overlap thermal/Casimir law is proved in Ref. [4]. The calculation below supplies the compact-group Peter–Weyl lift and beta-shift bookkeeping. It does not derive the physical one-loop beta coefficients from edge sectors alone.

A.1 $\mathcal{R}_U$ Unified-Coupling Certificate

The $\mathcal{R}_U$ certificate supplies the unified diffusion coupling used by the hierarchy discussion in the synthesis paper and by the electroweak transmutation factor

$$\frac{v}{E_\star} = P^{-1/2} \exp\left[-\frac{2\pi}{4\alpha_U(P)}\right].$$

It does not certify the full clock branch

$$\mathcal{R}_\gamma = \mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\mathrm{abs}} + \mathcal{R}_{\mathrm{QCD/nuc}}^{\mathrm{133Cs}} + \mathcal{R}_{\mathrm{atom}}^{\mathrm{133Cs}}.$$

The complete $\mathcal{R}_\gamma$ construction requires a source-only electromagnetic endpoint, charged-lepton absolute scale, cesium nuclear data, and atomic spectral enclosures. The numerical package is a witness for $\mathcal{R}_U$. Its mathematical conclusion requires the stated outward-rounded interval arithmetic.

Declared source packet. The public-pixel branch uses

$$P_C = \begin{array}{l}\texttt{1.63096820940395932487927984778264894133598285162792506066 61}\\ \texttt{507533907793398933432}\end{array}.$$

The source-only branch uses the certified coordinate

$$P_{\mathrm{fwd}} = 1.630972095858897\ldots.$$

It is interval-certified unique on its interval and, by the domain-global certificate, the only fixed point of its readout map on the declared physical domain $\alpha^{-1} \in [100, 200]$ (the interval contraction certificate of 2026-07-14, whose closure row is a certified source-root row). The finite heat-kernel cutoffs are

$$N_2 = 128, \qquad N_3 = 64.$$

The one-loop running packet is

$$(b_1, b_2, b_3) = \left(\frac{33}{5}, 1, -3\right), \qquad N_c = 3, \qquad \beta_{\mathrm{EW}} = N_c + 1 = 4.$$

This packet is an explicit premise of the certificate. This appendix does not derive it from the recovered OPH core.

Definition A.1 (No-hidden-free-variable rule for $\mathcal{R}_U$)**.** *The following objects must be fixed before comparison with electroweak or gravity data:*

$$(b_1, b_2, b_3), \quad N_2, \quad N_3, \quad M_U(P), \quad E_{\mathrm{cell}}(P),$$
$$\beta_{\mathrm{EW}}, \quad \textit{Casimir normalization}, \quad \textit{matching scheme}.$$

Changing any of them after comparison with M_W, M_Z, $\alpha_s(M_Z)$, low-energy electroweak couplings, or G_{exp} demotes the row to calibration.

Dimensionless source map. Set $E_\star = 1$ for the dimensionless calculation. For a trial coupling a, define

$$M_U(P) = e^{-2\pi} P^{1/6}, \qquad E_{\mathrm{cell}}(P) = P^{-1/2},$$
$$v(P, a) = E_{\mathrm{cell}}(P) \exp\left[-\frac{2\pi}{4a}\right].$$

For $i = 1, 2, 3$,

$$\alpha_i^{-1}(\mu; P, a) = a^{-1} + \frac{b_i}{2\pi} \log\left(\frac{M_U(P)}{\mu}\right), \qquad \alpha_Y(\mu; P, a) = \frac{3}{5}\alpha_1(\mu; P, a).$$

The source Z-scale is the positive solution of

$$\mu = \frac{v(P, a)}{2} \sqrt{4\pi\alpha_2(\mu; P, a) + 4\pi\alpha_Y(\mu; P, a)}.$$

For SU(2), with $j = n/2$ and $0 \le n \le N_2$,

$$d_j = 2j + 1, \qquad C_2(j) = j(j + 1).$$

For SU(3), with $0 \le p, q \le N_3$,

$$d_{p,q} = \frac{(p + 1)(q + 1)(p + q + 2)}{2}, \qquad C_2(p, q) = \frac{p^2 + q^2 + pq + 3p + 3q}{3}.$$

The group sums are

$$Z_G(t) = \sum_R d_R e^{-tC_2(R)}, \qquad \bar{\ell}_G(t) = \frac{1}{Z_G(t)} \sum_R d_R e^{-tC_2(R)} \log d_R.$$

Define

$$t_2(P, a) = 4\pi^2 \alpha_2(\mu_Z(P, a); P, a), \qquad t_3(P, a) = 4\pi^2 \alpha_3(\mu_Z(P, a); P, a),$$

and

$$\Phi_U(P, a) = \bar{\ell}_{\mathrm{SU}(2)}(t_2(P, a)) + \bar{\ell}_{\mathrm{SU}(3)}(t_3(P, a)) - \frac{P}{4}.$$

The certificate value is the zero $\Phi_U(P, \alpha_U(P)) = 0$.

Numerical witness. For $P = P_C$, the checker emits

$$\begin{aligned}\alpha_U(P_C) &= 0.041124336195630495,\\ \alpha_U(P_C)^{-1} &= 24.316501918546496.\end{aligned}$$

The corresponding dimensionless hierarchy data are

$$\begin{aligned}\frac{M_U(P_C)}{E_\star} &= 0.0020260720012100037,\\ \frac{v(P_C)}{E_\star} &= 2.0199803239725553 \times 10^{-17},\\ \frac{\mu_Z(P_C)}{E_\star} &= 7.502403238986022 \times 10^{-18}.\end{aligned}$$

The emitted couplings at μ_Z are

$$\begin{aligned}\alpha_1 &= 0.016885708038988166,\\ \alpha_Y &= 0.010131424823392899,\\ \alpha_2 &= 0.033777889448908055,\\ \alpha_3 &= 0.11833602747445048.\end{aligned}$$

The heat-kernel parameters and entropy readouts are

$$\begin{aligned}t_2 &= 1.3334976254578108,\\ t_3 &= 4.6717191102770705,\end{aligned}$$

$$\begin{aligned}\bar{\ell}_{\mathrm{SU}(2)} &= 0.39488144681077636,\\ \bar{\ell}_{\mathrm{SU}(3)} &= 0.012860605540213493.\end{aligned}$$

Thus

$$\begin{aligned}\bar{\ell}_{\mathrm{SU}(2)} + \bar{\ell}_{\mathrm{SU}(3)} &= 0.40774205235098987,\\ P_C/4 &= 0.4077420523509898,\end{aligned}$$

and

$$\Phi_U(P_C, 0.041124336195630495) = 5.55 \times 10^{-17}.$$

A centered derivative check gives

$$\partial_a \Phi_U(P_C, a) \simeq -10.990524683118785.$$

With $\delta = 10^{-6}$, the bracket signs are

$$\Phi_U(P_C, \alpha_U - \delta) = 1.0990748697592423 \times 10^{-5},$$

$$\Phi_U(P_C, \alpha_U + \delta) = -1.0990300680135956 \times 10^{-5}.$$

For the certified source-only pixel,

$$\begin{aligned}P_{\mathrm{fwd}} &= 1.630972095858897\ldots,\\ \alpha_U(P_{\mathrm{fwd}}) &= 0.041124247441816685\ldots,\\ \frac{v(P_{\mathrm{fwd}})}{E_\star} &= 2.0198114078576331 \times 10^{-17}.\end{aligned}$$

This branch is the one to cite when avoiding the measured Thomson endpoint in the upstream pixel solve.

Theorem A.2 (Electroweak unified-coupling numerical witness)**.** *Assume the declared source map, running packet, heat-kernel cutoff policy, and no-hidden-free-variable rule above. Let*

$$I_U = [0.041123336195630494,\ 0.041125336195630496].$$

If outward-rounded interval arithmetic verifies

$$0 \in \Phi_U(P_C, I_U), \qquad 0 \notin \partial_a \Phi_U(P_C, I_U),$$

then there is a unique $\alpha_U(P_C) \in I_U$ satisfying

$$\Phi_U(P_C, \alpha_U(P_C)) = 0.$$

The dependency graph of this zero contains no measured G, no Planck unit formed using measured G, no measured Λ, no measured M_Z, no measured M_W, and no measured low-energy gauge coupling.

Proof. Continuity follows because the running couplings, self-consistent Z-scale solution, and finite heat-kernel sums are continuous on the declared positive interval. The endpoint sign change gives existence by the intermediate value theorem. The derivative exclusion gives uniqueness. The dependency statement follows by inspection of the declared source map: the residual is built only from P_C, the declared running packet, the self-consistent Z-scale equation, and finite compact-group representation sums. □

Precision policy. The exponential map makes downstream gravity displays sensitive to α_U:

$$\frac{\partial \log G}{\partial \alpha_U} \simeq \frac{\pi}{\alpha_U^2} \approx 1.86 \times 10^3.$$

Therefore the number of digits printed for any downstream G or $\varepsilon_{\mathrm{Cs}}$ row may not exceed the interval precision certified by

$$\mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\mathrm{abs}} + \mathcal{R}_{\mathrm{QCD/nuc}}^{133\mathrm{Cs}} + \mathcal{R}_{\mathrm{atom}}^{133\mathrm{Cs}}.$$

The witness supports an electroweak hierarchy certificate. It does not support a 52-digit source-only gravity prediction.

A.2 One-Loop Calibration Frame

At one loop, if couplings unify at (M_U, α_U),

$$\alpha_i^{-1}(M_Z) = \alpha_U^{-1} + \frac{b_i}{2\pi} \ln \frac{M_U}{M_Z}.$$

On the forward calibration branch these are the values $(M_U(P), \alpha_U(P))$ emitted by the source solve. They are not inferred by inverse readback from measured low-energy couplings. Writing $A_i := \alpha_i^{-1}(M_Z)$ and $L := \ln(M_U/M_Z)$, one finds

$$L = \frac{2\pi}{b_1 - b_2}(A_1 - A_2),$$

and the corresponding consistency relation for this calibration is

$$A_3^{\mathrm{pred}} = \frac{b_3 - b_2}{b_1 - b_2} A_1 + \frac{b_1 - b_3}{b_1 - b_2} A_2.$$

This appendix keeps that algebra on the page as a calibration relation, not as a theorem that OPH derives a supersymmetric UV spectrum.

A.3 Benchmark coefficient comparison

For the standard one-loop benchmark coefficients,

Model	(b_1, b_2, b_3)	$\alpha_s(M_Z)^{\text{pred}}$
Standard Model	$(41/10, -19/6, -7)$	≈ 0.071
minimal supersymmetric Standard Model	$(33/5, 1, -3)$	≈ 0.116
Observed		0.1179 ± 0.0010

At one loop, the printed minimal supersymmetric Standard Model (MSSM) coefficients reproduce the observed closure far better than the plain Standard Model coefficients. The forward calibration matches this benchmark. It does not imply that OPH derives a supersymmetric ultraviolet spectrum.

A.4 Heat-Kernel Sector Law

Read this subsection as the compact-group merge boundary above the fixed-cutoff microphysics theorem. MaxEnt plus the bi-invariant compact-group package yield the familiar $d_R e^{-tC_2(R)}$ law once the Peter–Weyl lift is supplied; they are not a second, independent proof of the finite-cutoff same-overlap Casimir branch.

Theorem A.3 (Heat-kernel edge-sector weights)**.** *Under MaxEnt with bi-invariant constraints on a compact Lie group G, the sector probabilities take heat-kernel form:*

$$p_R(t) \propto d_R \, e^{-tC_2(R)},$$

where $d_R = \dim R$ and $C_2(R)$ is the quadratic Casimir.

Lemma A.4 (Bi-invariant operators)**.** *If $G = \prod_i G_i$ is a compact semisimple Lie group written as a product of compact simple factors, then any bi-invariant second-order differential operator on G has the form*

$$D = c_0 \mathbf{1} - \sum_i c_i \Delta_{G_i},$$

where Δ_{G_i} is the Laplace–Beltrami operator on the factor G_i.

Remark A.5. *Bi-invariance is equivalent to $D \in Z(U(\mathfrak{g}))$. Linear terms vanish because there are no invariant vectors in the adjoint representation, and the quadratic part is proportional to the Killing form on each simple factor, giving one independent coefficient per factor. The Casimir element then acts as $-\Delta_G$ in the regular representation.*

A.5 Peter–Weyl Multiplicity and Beta Shifts

By Peter–Weyl decomposition, the effective refinement-limit edge representation space is

$$L^2(G) \cong \bigoplus_R V_R \otimes V_R^*.$$

Entanglement traces over one factor give multiplicity d_R in p_R, but loops see both factors, so the effective multiplicity entering the calibration branch is

$$N_{\text{eff}}(R) = d_R \, p_R.$$

The one-loop beta shift from edge sectors is then

$$\Delta b_a = \sum_R p_R \, d_R \, T_a(R),$$

where $T_a(R)$ is the Dynkin index for gauge factor a. This is the exact mathematical point where edge-sector heat-kernel weights feed the calibration branch. Matching to MSSM-like one-loop running is a comparison benchmark, not a theorem-level MSSM spectrum claim.

The branch boundary for theorem-level unification closure is explicit. One requires a derived refinement-limit identification of which transportable sectors actually contribute to the running carrier, a derived statistics/grading rule rather than an imposed fermionic-grading restriction, controlled threshold and decoupling conventions on that same carrier, a derived representation content/truncation rule for the sectors retained in the comparison, and a proof that the effective loop multiplicities entering Δb_a are exactly the ones used in this calibration package rather than nearby alternatives. The declared runtime surface does not emit the full RG/matching/threshold/scheme packet: scheme lock, threshold map, beta provenance, and interval composition are separate certificate requirements.

At the branch unification diffusion parameter $t_U(P) \approx 1.64$ selected by the forward source solve, the benchmark shift is

$$\Delta b \approx (2.49, 4.38, 3.97) \qquad \text{vs MSSM} \qquad (2.50, 4.17, 4.00).$$

With the additional calibration assumption of a fermionic-grading restriction to half-integer SU(2) sectors, this sharpens to

$$\Delta b \approx (2.50, 4.17, 3.97),$$

which matches the MSSM-style one-loop benchmark at the percent level under that declared calibration package. Equivalently, the benchmark ratio

$$\frac{\Delta b_3}{\Delta b_2} \approx 0.91$$

sits close to the MSSM comparison value 0.96.

B H3 Worldline-Stitch Certificate

This appendix records the finite evidence-facing certificate for Theorem 13.5. It is separate from the heat-kernel calibration package above. A passing certificate is a record-continuation statement in a declared H^3 observer chart, not a particle-species, mass, charge, or scattering-amplitude theorem.

The certificate payload has the following required blocks.

1. **Atlas.** It declares the hyperboloid model $H^3_{R_H} \subset \mathbb{R}^{1,3}$, the Lorentz signature $(-+++)$, the curvature radius R_H, chart domains, and chart transitions $G_{ji} \in \mathrm{SO}^+(1,3)$. Transition residuals must certify Lorentz inner-product preservation, orientation, and future-sheet preservation.

2. **Clock map.** Each local record interval is mapped to a common comparison-time line. The observer-time adjacency margin must dominate the declared clock uncertainty.

3. **Chart-natural extraction.** Raw local records are connected components of a frozen detector scalar on the H^3 chart. The detector scalar, thresholds, support collars, and chart-naturality residuals are included. Implementation record IDs are not inputs.

4. **Overlap descent.** Records are first descended across genuine chart overlaps. The payload records same-component join margins, distinct-component separation margins, support errors, and triple-overlap cocycle residuals.

5. **Interface crossing.** A candidate stitch must cross a real interface in adjacent clock intervals. It includes oriented signed-distance margins and a positive lower bound on normal velocity. Tangential, grazing, and file-boundary-only contacts are rejected.

6. **Transport.** The candidate edge is evaluated in a common chart after sector and gauge transport. The atlas holonomy comparison is separate from the gauge-holonomy comparison.

7. **Assignment.** The matching is one-to-one with appearance/disappearance interval costs. The proposed winner has an upper cost bound, every competitor has a lower cost bound, and

$$\Delta_{\text{cert}} = \min_{M \neq M_*} \underline{C}(M) - \overline{C}(M_*) > 0.$$

   Record IDs, global IDs, stitch keys, and shard-local IDs are forbidden in admissibility, costs, and tie-breaking.

8. **Refinement.** A coarse/fine pair supplies Q_{sr}, a distortion bound η_{sr}, and a contracted-graph isomorphism check. The coarse gap must satisfy $\Delta_r > 2\eta_{sr}$.

9. **Interaction firewall.** A free-propagation slab is required. If an interaction marker is present, the terminal label is H3 interaction required and the event is handed to the interaction solver.

The allowed terminal outcomes are

$$\text{H3 stitch certified}, \quad \text{H3 stitch ambiguous}, \quad \text{H3 stitch rejected},$$
$$\text{H3 interaction required}, \quad \text{H3 atlas invalid}, \quad \text{H3 certificate incomplete}.$$

H3 stitch certified is emitted only when every block above passes and the stable assignment gap is positive. Equal-cost matchings, preserved nontrivial permutations of the visible data, or coarse/fine gaps below margin force H3 stitch ambiguous. Missing fields force H3 certificate incomplete. Failed atlas, interface, metric, or transport gates force rejection or atlas invalidity. In particular, Euclidean distances on the stored coordinate triples are not certificate distances; the certificate distance is the hyperboloid geodesic d_H from Section 13.

References

[1] B. Müller, D. Matscheko, and J. Hill, *Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems*, 2026.
Available at https://github.com/FloatingPragma/observer-patch-holography/blob/main/extra/observable_normal_forms.pdf.

[2] B. Müller, A. Osika, M. Poneder, K. Xue, B. Cassie, P. Nguyen, M. A. Visser, K. A. Anirudha, D. Matscheko, and J. Hill, *Observers Are All You Need*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/observers_are_all_you_need.pdf

[3] B. Müller, K. Xue, K. A. Anirudha, D. Matscheko, and J. Hill, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/reality_as_consensus_protocol.pdf

[4] B. Müller, A. Osika, K. Xue, B. Cassie, M. A. Visser, and D. Matscheko, *Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/screen_microphysics_and_observer_synchronization.pdf

[5] B. Müller, *OPH-FPE: finite simulator and receipt engine for Observer-Patch Holography physics experiments*, 2026. https://github.com/muellerberndt/oph-physics-sim

[6] B. Müller, A. Osika, M. Poneder, K. Xue, P. Nguyen, M. A. Visser, and D. Matscheko, *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/recovering_observer_spacetime_and_einstein_dynamics_from_overlap_consistency.pdf

[7] B. Müller, A. Osika, M. Poneder, K. Xue, P. Nguyen, M. A. Visser, and D. Matscheko, *Deriving Standard Model Gauge Structure from Observer Overlap Consistency*. GitHub PDF: https://github.com/FloatingPragma/observer-patch-holography/blob/main/paper/deriving_standard_model_gauge_structure_from_observer_overlap_consistency.pdf

[8] National Institute of Standards and Technology, *2022 CODATA Recommended Values of the Fundamental Constants of Physics and Chemistry*, NIST SP 959, May 2024. https://physics.nist.gov/cuu/pdf/wallet_2022.pdf

[9] Y. Koide, *New view of quark and lepton mass hierarchy*, Physical Review D **28**, 252 (1983).

[10] C. A. Brannen, *The Lepton Masses*, preprint, 2006. http://brannenworks.com/MASSES2.pdf

[11] S. Antusch, K. Hinze, and S. Saad, *Updated Running Quark and Lepton Parameters at Various Scales*, arXiv:2510.01312, 2025. https://arxiv.org/abs/2510.01312

[12] Y. Aoki et al. (Flavour Lattice Averaging Group), *FLAG Review 2024*, arXiv:2411.04268, 2024, Tables 11–12. https://arxiv.org/abs/2411.04268

[13] S. Navas et al. (Particle Data Group), *CKM Quark-Mixing Matrix*, Review of Particle Physics 2024, Sec. 12.2.2, revised May 31, 2024. https://pdg.lbl.gov/2024/reviews/rpp2024-rev-ckm-matrix.pdf

[14] Particle Data Group, *Review of Particle Physics: 2026 particle listings*, 2026. https://pdg.lbl.gov/2026/listings/particle_properties.html

[15] Planck Collaboration, *Planck 2018 results. VI. Cosmological parameters*, Astronomy & Astrophysics 641, A6, 2020. https://arxiv.org/abs/1807.06209

[16] DESI Collaboration, *Constraints on Neutrino Physics from DESI DR2 BAO and DR1 Full Shape*, arXiv:2503.14744, 2025. https://arxiv.org/abs/2503.14744

[17] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro, and T. Schwetz, *NuFit-6.0: Updated global analysis of three-flavor neutrino oscillations*, JHEP **12** (2024) 216, arXiv:2410.05380, with the NuFIT 6.1 (2025) profile-table release at https://www.nu-fit.org/?q=node/309.

[18] B. A. Kniehl, *All-order renormalization of the propagator matrix for fermionic systems with flavor mixing*, Physical Review D **89**, 096005 (2014), arXiv:1308.3140.

[19] P. Gambino and P. A. Grassi, *The Nielsen identities of the Standard Model and the definition of mass*, Physical Review D **62**, 076002 (2000), arXiv:hep-ph/9907254.

[20] D. Buchholz, *Gauss' law and the infraparticle problem*, Physics Letters B **174**, 331–334 (1986).

[21] P. Duch and W. Dybalski, *Infrared problem in quantum electrodynamics*, arXiv:2307.06114.

[22] M. Lüscher, *Abelian chiral gauge theories on the lattice with exact gauge invariance*, Nuclear Physics B **549**, 295–334 (1999), arXiv:hep-lat/9811032.

[23] M. Lüscher, *Lattice regularization of chiral gauge theories to all orders of perturbation theory*, Journal of High Energy Physics **06** (2000) 028, arXiv:hep-lat/0006014.

[24] D. Kadoh and Y. Kikukawa, *A simple construction of fermion measure term in $U(1)$ chiral lattice gauge theories with exact gauge invariance*, Journal of High Energy Physics **02** (2008) 063, arXiv:0709.3658.

[25] H. Bélusca-Maïto, A. Ilakovac, M. Mađor-Božinović, and D. Stöckinger, *Dimensional regularization and Breitenlohner–Maison/'t Hooft–Veltman scheme for γ_5 applied to chiral Yang–Mills theory*, Journal of High Energy Physics **08** (2020) 024, arXiv:2004.14398.

[26] V. Dūdėnas and M. Löschner, *Vacuum expectation value renormalization in the Standard Model and beyond*, Physical Review D **103**, 013003 (2021), arXiv:2010.15076.

[27] S. Dittmaier and H. Rzehak, *Electroweak renormalization based on gauge-invariant vacuum expectation values of non-linear Higgs representations*, Journal of High Energy Physics **05** (2022) 125, arXiv:2203.07236.

[28] P. A. Grassi, B. A. Kniehl, and A. Sirlin, *Width and partial widths of unstable particles in the light of the Nielsen identities*, Physical Review D **65**, 085001 (2002), arXiv:hep-ph/0109228.

Chain	OPH output	Role
P-closure and fine structure	certified incomplete-map root $136.994835177413\ldots$ (certified source-root row); mixed diagnostic $137.0359595136\ldots$; certified gauge-width fixed point $137.035660136946577\ldots$; empirical endpoint $136.3827548175 \in [136.3670480603, 136.3984651934]$; measured 137.035999177(21)	distinct support classes: incomplete declared-map output, mixed diagnostic excluded from physical output, empirical hadron closure with anchor gap $[0.6198609041, 0.6505569679]$, and measurement
Classical carrier modes	two transverse Maxwell modes; $2\dim G$ perturbative pure-Yang–Mills modes; two pure-Einstein TT modes	conditional quadratic-action results; quantum particles require separate Hilbert-space, pole, and phase premises
Electroweak W/Z	source-only running/chart coordinates (80.330, 91.119) GeV; candidate value-law coordinates (80.3770000154, 91.1879780779) GeV; no OPH-native pole	these coordinates are not commensurate with the PDG mass-dependent-width Breit–Wigner masses or with complex-pole masses. The exact convention map gives energy-pole masses $M_W = 80.3411410$ GeV and $M_Z = 91.1623040$ GeV; the distinct coordinates $\sqrt{\mathrm{Re}\, s}$ are 80.3340218 GeV and 91.1537725 GeV. The finite quotient-path certificate, source-law selection, matching prescription, theory covariance, and physical pole readout are open; numerical proximity is not mass evidence
	$v/E_\star$ on each named pixel branch	
Electroweak hierarchy/naturality	$N_{\mathrm{EW}} \simeq 3.53235 \times 10^{122}$ $\mathcal{B}_{\mathrm{EW}} = 0,\ \epsilon_H = 0$	selected dimensionless bridge; identifying N_{EW} with direct cosmic capacity requires construction of the public-record producer and the common screen/electroweak load-carrier map. The Planck-Λ central capacity is $N_\Lambda \simeq 3.313 \times 10^{122}$, a 6.6-percent mismatch; the horizon–record identification and $E_\star$ are separate gates
Higgs/top relation	double-criticality branch $(m_H, m_t) = (125.77, 172.63)$ GeV at the declared boundary-scale candidate $E_\star e^{-\pi} P^{-1/6}$ (two loops), with $m_H = 125.72$ GeV on the fit-free curve at the measured top; target-anchored declared-surface fit (125.1995304097, 172.3523553288) GeV kept separate; no source-only physical mass emitted	the criticality family has zero continuous parameters over the boundary scale. Its scale selection assumes that the boundary record has exactly two equal-depth anchor registers, one geometric and one transmutational, together with the declared Gaussian/affine record and port-additive repair premises. The target-anchored fit carries no predictive force. This branch does not supply the source root, physical scale, rigidity, provenance, uncertainty enclosure, or complex pole
Charged leptons	target-anchored witness values withheld; finite eight-path digital carrier and empirical transport intervals retained as diagnostics	the model proves schema satisfiability and a central record dilation, not physical source selection; the intervals use target-anchored ratios, measured α, and empirical transport
Selected-frame quarks	public numeric rows withheld; reciprocal-ray product 0.835323 and held-out error 21.56% at M_Z	common-scale data reject the reciprocal-ray candidate; the generic interface has six scalars and requires a source-derived flavor-orbit selector

Sector	Theorem scope	Derivation	Output(s)	Caveat / theorem boundary
Carrier roles and modes	structural group/content theorem plus conditional action theorem	explicit transverse/TT quadratic kernels with no derived particle mass	action, background, and phase are hypotheses; quantization, positive-residue pole, and asymptotic/deconfinement conditions are separate	
Electroweak bosons	running/chart calculation, conditional quotient transport, and comparison adapter	source-only zero-selector law; selected carrier $\to$ two-coordinate chart; finite quotient-path certificate $\to$ conditional value law	no OPH-native pole; source-only coordinates (80.330, 91.119) GeV and candidate value-law coordinates (80.3770000154, 91.1879780779) GeV are chart outputs	the chart coordinates are not commensurate with PDG mass-dependent-width Breit–Wigner masses or complex-pole masses. Source-law selection, matching, scale, threshold, tadpole, and pole derivations are open
Electroweak hierarchy/naturality	selected-branch dimensionless theorem	local $P \to \alpha_U \to v/E_\star$ hierarchy branch $\to$ exterior weak multiplicity $3+1=4$ $\to$ mathematical capacity bridge and RG/Higgs square; conditional screen branch: 12 ports and an oriented 24-slot register	$N_{\mathrm{EW}} = 3.5323546226929906511\ldots \times 10^{122}$, $\mathcal{B}_{\mathrm{EW}} = 0$, $\epsilon_H = 0$	Supports the dimensionless hierarchy and naturality identities on the named branch. The 24-slot register does not produce four weak loads. Equality with cosmic capacity requires the direct public-record fixed point and the common screen/electroweak load-carrier map; the Planck-Λ central value is lower by about 6.6 percent. An independent physical $E_\star$, the public Thomson endpoint, theorem-level W/Z, and the other listed mass rows are separate surfaces.
Higgs/top stage	double-criticality family from the gauge sector plus a target-anchored fit	criticality law $\lambda = 0,\ \beta_\lambda = 0$ at one source scale $\to$ fully constrained (m_H, m_t) family $\to$ declared boundary-scale candidate	no source-only mass; double-criticality branch $(m_H, m_t) =$ (125.77, 172.63) GeV, with $m_H = 125.72$ GeV on the fit-free curve at the measured top; target-anchored fit (125.1995304097, 172.3523553288) GeV	A physical mass requires the declared Gaussian/affine record and port-additive repair premises, the two equal-depth boundary-record anchors, a source root and physical scale, finite quotient-path selection, same-scheme running, target-independent rigidity, provenance, uncertainty enclosure, and a complex pole. The companion top coordinate is not a separate public top-mass prediction.
Quark family	common-scale rejection plus restricted source non-identifiability theorem	one-scheme dimensionless Yukawas $\to$ reciprocal-ray test $\to$ six-scalar generic interface and flavor-selector boundary	no public numeric quark row; at M_Z, $\rho_u\rho_d = 0.835323$ and the endpoint-granted held-out error is 21.56%	The reciprocal-ray candidate is physically rejected across the tested running scales. The $(\mathbb{R}_{>0})^2$ theorem is a restricted lower-bound obstruction. Mixed-convention template and target-conditioned cumulant residuals are diagnostics. A flavor-orbit selector, quark–Higgs carrier, and common-scale source transport are absent
Charged leptons	conditional construction plus exact target-anchored witness	shared excitation dictionary $\to$ ordered charged carrier $\to$ exact centered readback $\to$ determinant-line lift on theorem-level physical charged data	exact centered readback; same-family target-anchored charged triple withheld from public prediction tables	public charged masses are not emitted from P. The theorem branch does not emit a theorem-level sector-isolated charged determinant exponent vector. It does not attach a source-side determinant character to the physical charged determinant line. The determinant-line lift and algebraic mass readout apply only on theorem-level physical charged data
Neutrinos	target-informed conditional candidate plus auxiliary checks	template family transport $\to$ same-label scalar certificate $\to$ weighted-cycle candidate $\to$ compare-only bridge and absolute attachment $\to$ shared-basis identity	declared scale-free candidate and PMNS/Majorana comparison rows; absolute mass attachments withheld from public prediction tables	exact linear algebra conditional on the declared template and selectors. The candidate fails the NuFIT 6.1 correlated $(\sin^2\theta_{23}, \delta_{\mathrm{CP}})$ profile, and neither the scale-free point nor the bridge invariant is a prediction or theorem [17]
Hadrons	strong-binding construction absent; empirical comparison contract	source-derived hadronic spectral backend contract, including quotient ensemble, source QCD parameter map, Ward current accounting, spectral exports, and empirical $e^+e^- \to$ hadrons payload schema	no first-principles hadron masses; empirical comparison rows are separate from first-principles OPH results	First-principles hadron prediction requires a working OPH hadron construction, such as GLORB/Echosahedron, with the Ward-projected two-current spectral measure, higher-point/transition spectral sectors, same-scheme remainder, and systematics. The empirical comparison uses separate $e^+e^- \to$ hadrons input and cannot establish the first-principles theorem
Affine event-record stitching	conditional certificate theorem	event-manifold affine chart and observer clock atlas $\to$ affine event support for each descended token and clock slice $\to$ separately typed conditioned-frame ball where required $\to$ positive event-location gap $\to$ overlap descent $\to$ real transverse interface-crossing germs $\to$ common-chart sector/gauge transport $\to$ ID-independent one-to-one assignment gap $\to$ coarse/fine contraction check	nonbranching event-record paths, rays, or isolated records when the location and stitch certificates pass; otherwise AMBIGUOUS, REJECTED, or INTERACTION REQUIRED	This conditional theorem concerns observer-visible records on the event-manifold branch. It does not derive particle species, masses, gauge charges, scattering amplitudes, or geodesic motion. Repeated implementation IDs, nearest-neighbor fits, frame coordinates used as event positions, and file-boundary coincidences are inadmissible evidence.

Result class	Exact output(s)	Exact derivation	Conditions
Carrier roles and classical modes	no photon/gluon/graviton particle-mass output	axioms $\to$ realized electromagnetic/color/dynamical-metric roles; additional action/background/phase receipt $\to$ transverse/TT quadratic modes	quantum-particle receipt not supplied; confined color has no free asymptotic-gluon claim
Electroweak chart and comparison	no nonzero source-only mass output	source tuple $\to$ exact two-coordinate chart; finite quotient-path certificate $\Rightarrow$ candidate value law; inverse adapter separate	all numerical coordinates belong to the comparison discussion; the finite quotient-path certificate is an open premise and no pole pair follows
Hierarchy/ naturality bridge	exact dimensionless bridge residual and zero Higgs naturality defect	axioms $\to$ P-fixed source branch $\to$ exact exterior weak multiplicity four $\to$ mathematical N_{EW} bridge map $\to$ source-to-Higgs settled-form square; conditional screen branch: 12 ports and an oriented 24-slot register	selected-branch theorem for $v/E_\star$ and the naturality identities; the register count does not construct the weak load; equality with cosmic capacity requires the direct public-record fixed point and the common screen/electroweak load-carrier map; it does not supply a physical $E_\star$ or derive the mass rows
Higgs/top inverse slice	no nonzero source-only mass output	gauge core $\to$ declared Jacobian $\to$ exact inverse slice	target-conditioned benchmark only; the conditional coordinates inherit every open physical premise
Charged exact witness	exact same-family charged triple withheld	axioms $\to$ shared excitation dictionary $\to$ ordered charged carrier $\to$ exact centered readback $\to$ closed quadratic readout theorem $\to$ same-family exact witness	same-family target-anchored comparison witness only; theorem branch carries exact centered readback plus the closed common-shift no-go. The theorem does not provide a $\widehat{C}_e$ lift that emits the physical scalar $\mu_{\mathrm{phys}}(Y_e)$
Quark physical boundary	numeric predictions withheld; common-scale reciprocal-ray candidate rejected; mixed-convention formula residuals are target-anchored	six dimensionless Yukawas in one scheme and scale $\to$ reciprocal-ray rejection $\to$ six-scalar interface and selector no-go	at M_Z, $\rho_u = 1.110889$, $\rho_d = 0.751941$, product 0.835323; even with four endpoints granted, the held-out miss is 21.56%. The two-spread theorem is a restricted non-identifiability result
Neutrino candidate	declared scale-free comparison point; absolute mass attachments withheld	template family transport $\to$ same-label scalar certificate $\to$ target-informed weighted-cycle law $\to$ compare-only bridge and absolute attachment $\to$ shared-basis identity	rejected by the NuFIT 6.1 correlated profile; target dependence, source closure, basis orientation, and absolute normalization preclude a prediction [17]

Carrier role	Theorem scope	Action-level output	Physical degrees of freedom	Quantum-particle requirements
Electro-magnetic connection	conditional Maxwell mode	transverse $k^2 = 0$ kernel	two transverse classical modes	photon requires quantization and a positive-residue physical pole
Color connection	conditional perturbative pure-Yang–Mills mode	one transverse $k^2 = 0$ kernel per generator	$2 \dim G$ perturbative modes	deconfined BRST/LSZ receipt required; no free-gluon claim in confined QCD
Metric perturbation	conditional pure-Einstein mode	$D^{\mathrm{TT}} \propto i\Pi^{\mathrm{TT}}/(k^2 + i0)$	two classical TT modes	graviton requires metric quantization, physical Hilbert space, and pole receipt

Particle	Calculation	Theorem scope	Value
W boson	electroweak calibration	no source-only physical mass	no OPH-native pole
Z boson	electroweak calibration	no source-only physical mass	no OPH-native pole
Higgs boson	Higgs/top criticality	no source-only physical mass	no OPH-native pole
Top quark	Higgs/top criticality	companion coordinate on declared surface; direct-top auxiliary codomain no-go	not a separate source-only prediction

Quark	Theorem scope	Public value	Mass convention
Up quark	source spread non-identifiable	withheld	$\overline{\mathrm{MS}}$, $\mu = 2\,\mathrm{GeV}$, comparison chart
Down quark	source spread non-identifiable	withheld	$\overline{\mathrm{MS}}$, $\mu = 2\,\mathrm{GeV}$, comparison chart
Strange quark	source spread non-identifiable	withheld	$\overline{\mathrm{MS}}$, $\mu = 2\,\mathrm{GeV}$, comparison chart
Charm quark	source spread non-identifiable	withheld	$\overline{\mathrm{MS}}$, $\mu = m_c(\mu)$, comparison chart
Bottom quark	source spread non-identifiable	withheld	$\overline{\mathrm{MS}}$, $\mu = m_b(\mu)$, comparison chart
Top quark	separate extraction coordinate	withheld	cross-section pole-mass chart